POLITECNICO DI TORINO

MASTER's Degree in MECHANICAL ENGINEERING



MASTER's Degree Thesis

An Open-Source Framework for the hydroelastic analysis of composite hydrofoils

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MARCH 2026

Abstract

The increasing integration of composite materials into marine lifting surfaces, such as hydrofoils and turbine blades, is unlocking new opportunities for weight reduction and structural efficiency. However, this trend amplifies the need for reliable prediction of unsteady hydrodynamic behaviour and coupled hydro-elastic instabilities, with flutter remaining one of the most critical challenges. Most available numerical tools are either accurate but computationally expensive or rely on oversimplified modelling that fails to capture the complex coupling between hydrodynamics, structural anisotropy, and dynamic stability. This work introduces a comprehensive framework that directly addresses these limitations. The toolchain integrates four key components into a unified workflow: a finite-element pre-processor for generating fully coupled six degrees of freedom beam models; a dynamic beam solver for time-domain structural response; a Doublet Lattice Method implementation for unsteady hydrodynamic load prediction; and a robust P-K analysis module featuring automatic mode tracking. Together, these modules enable accurate coupling of structural and hydrodynamic phenomena, delivering a comprehensive toolset for frequency-domain response evaluation and instability prediction. The proposed framework, using reduced-order modelling, achieves high computational efficiency while maintaining accuracy. This balance between fidelity and cost-effectiveness makes the tool ideal for early-stage design, parametric studies, and optimisation tasks. The methodology and software architecture are described in detail and validated against experimental data for a hydrofoil with heterogeneous cross-sections. Results show very good agreement, confirming the tool's capability to capture coupled fluid–structure dynamics and accurately predict critical flutter conditions.

ACKNOWLEDGMENTS

a due persone speciali, a due persone che sanno gioire sempre, che mi insegnano ogni giorno, che mi guardano con occhi critici ma pieni di orgoglio, che faticano in silenzio e che mi vogliono bene. A mio papà Andrea e mia mamma Angelica.

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Nomenclature

$\mathbf{M}$	Mass matrix
$\mathbf{K}$	Stiffness matrix
$\mathbf{C}$	Damping matrix
$\mathbf{Z}$	Coupling matrix between structural and aerodynamic degrees of freedom
$\mathbf{M}_s$	Structural mass matrix
$\mathbf{K}_s$	Structural stiffness matrix
$\mathbf{A}_{\text{hydro}}$	Hydrodynamic added mass matrix
$\mathbf{B}_{\text{hydro}}$	Hydrodynamic damping matrix
$\mathbf{Q}(k)$	Aerodynamic Influence Coefficient matrix as a function of reduced frequency
$\mathbf{u}$	Displacement vector
$\mathbf{F}$	Force vector
$\mathbf{w}$	Downwash vector
$\mathbf{GAF}$	Generalised Aerodynamic Force
V_F	Flutter speed
Φ	Modal matrix, eigenmodes
ω	Angular frequency
ρ	Density
k	Reduced frequency
U	Flow velocity
ω_{dry}	Dry natural frequency
s	Laplace variable
p_j	Pole in Rational Function Approximation
$\text{AIC}(\omega)$	Aerodynamic Influence Coefficient matrix
$(\cdot)_s$	Subscript indicating structural or dry properties
$(\cdot)_{\text{wet}}$	Subscript indicating wet or coupled properties
DLM	Double Lattice Method
FSI	Fluid-Structure Interaction
FEM	Finite Element Method
CoG	Center of Mass
M_a	Mach number

Introduction

1.1 Context and motivation

In the last few years, the use of slender load-bearing profiles in hydrodynamic environments, such as hydrofoils for racing yachts, marine propulsion systems and renewable energy devices, has grown significantly. The adoption of lightweight structures, often made of composite materials, allows for improving overall system efficiency, but it makes such components more sensitive to aero-hydro-elastic undesired phenomena, particularly flutter.

Flutter represents a form of self-excited dynamic instability, where the interaction between structural rigidity, inertias and fluid-dynamic loads leads to increasing amplitude oscillations. For a hydrofoil, the onset of flutter can lead to loss of performance, structural damage or collapse, with possible relevant safety consequences, reliability impacts and operational costs.

It thus becomes essential to develop analysis instruments able to prevent, during the design phase, the onset of these instabilities, guiding geometrical, material and mass distribution choices.

High-fidelity approaches based on computational fluid dynamics coupled with complex structural models guarantee high accuracy of the phenomenon, but they are often computationally expensive and difficult to integrate into iterative design cycles. On the other hand, aeroelastic reduced-order methods, which combine beam models with unsteady potential aerodynamics, offer an advantageous compromise between accuracy and computational cost. Naturally, the workflow chain must be defined coherently, verified and reproducible.

This thesis fits into this context, aiming to develop a modular numerical procedure for analyzing flutter of slender wings, with particular attention to hydrofoils.

1.2 Current challenges

Extensive literature covers aeroelasticity of wings in aerodynamic environments and, increasingly, of hydrofoils in hydrodynamic flows. However, several recurring critical gaps are observed:

- absence of an integrated workflow spanning from realistic section definition (materials, layup, shear center and mass center locations) to flutter prediction in a coherent manner;
- limited availability of readily integrable tools combining:
  - flexible parametrization of geometry and material properties,
  - generation of stiffness and mass properties for composite sections,
  - a dedicated flexible beam solver,
  - a consolidated, unsteady panel aerodynamic model,
  - a flutter solver based on standard industrial methods;
- fragmented documentation, hindering result reuse and adoption in preliminary design workflows.

These gaps motivate the development of a transparent computational environment, verified against known benchmarks and readily extensible to diverse wing and hydrofoil configurations.

1.3 Objectives of this thesis

The overall objective of this thesis is to define, implement, and validate a modular numerical procedure for analyzing flutter of slender wing structures in fluid environments, with particular reference to hydrofoils, based on the seamless coupling of structural, aerodynamic, and hydrodynamic analysis modules.

In particular, the work aims to:

1. define a **standardized input model** for describing geometry, materials and layup of wing sections,
2. build a **modular computational chain** using the consolidated **SONATA/ANBA4** tool to generate cross-sectional structural matrices and the associated one-dimensional Euler-Bernoulli beam model,
3. implement and verify a **custom FEM beam solver** for static and modal analysis of equivalent beams, comparing results with literature solutions and reference cases (Timoshenko beams, cantilevers, and modal verification),
4. develop a comprehensive **aerodynamic and hydrodynamic modeling framework** integrating the **Doublet Lattice Method** (via **PanelAero**) for unsteady forces, **Rational Function Approximation** (RFA) for frequency-dependent aerodynamic matrices, and a **hybrid DLM-BEM approach** combining DLM with **Capytaine BEM** to capture three-dimensional added-mass effects in water,
5. define a coherent **coupling strategy** between the aerodynamic/hydrodynamic grid and structural finite-element nodes, with explicit treatment of both aeroelastic (air) and hydroelastic (water) environments,
6. implement a **flutter solver** based on the industrial-standard **p–k method**, including automatic mode tracking and eigenvalue convergence criteria,
7. **integrate all modules** into a coherent, reproducible, and openly documented computational workflow,
8. **verify and validate** each module and the complete procedure against established benchmarks (Goland wing for aeroelasticity, NACA0003 for hydroelasticity, literature data),
9. apply the **end-to-end procedure** to realistic **hydrofoil case studies** (wing01 flexible wing and tns_multibody assembly), evaluating the effect of structural and modeling parameters on dynamic behavior and flutter boundary conditions.

To ensure the validity of the work, each custom-built module and the entire workflow are validated both analytically and with reference benchmark cases.

The unified workflow for aeroelastic and hydroelastic flutter analysis is represented in Figure 1.1. The scheme illustrates the hybrid DLM-BEM approach tailored for hydrofoil design:

- The **left branch** develops structural properties: geometry and material definitions are processed through SONATA/ANBA4 to generate cross-sectional matrices, which are then assembled into a one-dimensional finite-element beam model. A "dry" modal analysis (without fluid loading) is performed to extract the structural eigenvalues and eigenvectors serving as the baseline for aeroelastic coupling.
- The **right branch** develops aerodynamic matrices: the aerodynamic grid is discretized for the Doublet Lattice Method (PanelAero), the Aerodynamic Influence Coefficient (AIC) matrix $\mathbf{Q}(k)$ is computed as a function of reduced frequency, and Rational Function Approximation is applied to obtain frequency-independent aerodynamic coefficients suitable for flutter analysis.
- A **decision point** routes the analysis to either the **aeroelastic path** (air environment,

using DLM exclusively) or the **hydroelastic path** (water environment, combining DLM with Capytaine BEM corrections). The hybrid approach eliminates the thin-wing approximation error inherent in DLM by replacing the asymptotic added-mass term with Capytaine's three-dimensional boundary-element calculation $\mathbf{A}_{hydro}(\infty)$.

- Both paths converge at the **coupling matrix Z**, where structural and aerodynamic/hydrodynamic terms are assembled into a unified FSI system with mass **M**, damping **C**, and stiffness **K** matrices.
- The **p-k method** solves for the stability characteristics (eigenvalues and eigenvectors) across a sweep of velocities, identifying the flutter boundary where damping transitions from positive (stable) to negative (unstable).
- The output is the complete flutter envelope: "wet" natural frequencies and damping ratios plotted as functions of flow velocity.

1.4 Document structure

The remainder of this document is organized as follows:

Chapter 2 (*Structural Modeling and FEM Analysis*) introduces the theoretical foundations of beam structural modeling for slender structures, the development of cross-sectional properties, assembly of finite-element matrices, and the concept of "dry" modal analysis without fluid coupling.

Chapter 3 (*Aeroelastic and Hydroelastic Modeling*) establishes the distinction between aeroelastic (air) and hydroelastic (water) environments, describes the aerodynamic and hydrodynamic force models based on the Doublet Lattice Method (DLM) and Boundary Element Method (BEM), introduces Rational Function Approximation (RFA) for frequency-dependent aerodynamic matrices, and defines the coupling strategy between structural and fluid domains. A hybrid DLM-BEM approach is presented for hydrofoil applications to capture three-dimensional added-mass effects while maintaining computational efficiency.

Chapter 4 (*Flutter Analysis: The p-k Method*) details the mathematical formulation and numerical implementation of the p–k method, including the nested outer loop (velocity sweep, altitude, Mach number) and inner loop (mode-by-mode iterative solution with reduced frequency updating). The chapter includes detailed flowcharts, convergence criteria, and mode-tracking algorithms essential for automated flutter identification.

Chapter 5 (*Verification and Validation*) presents systematic validation of the aeroelastic and hydroelastic frameworks through sensitivity studies, grid convergence analyses, and benchmark comparisons. The Goland wing flutter test case is used as the primary aeroelastic reference, while the NACA0003 hydrofoil provides hydroelastic validation. Each stage of the pipeline is verified against established analytical and numerical results.

Chapter 6 (*Hydrofoil Case Study*) demonstrates the complete end-to-end application of the developed workflow to realistic hydrofoil structures. The chapter covers two primary applications: the `wing01` flexible composite hydrofoil (examining isolated wing dynamics) and the `tns_multibody` assembly (representing a practical racing-yacht foil system). Results include dry and wet modal characteristics, flutter boundaries, and parametric sensitivity studies.

Chapter 7 (*Discussion, Conclusion and Further Work*) synthesizes the results, provides critical assessment of the developed framework, discusses model limitations and design implications, and proposes future developments including complete Rational Function Approximation, viscous effects modeling, cavitation analysis, and experimental validation.

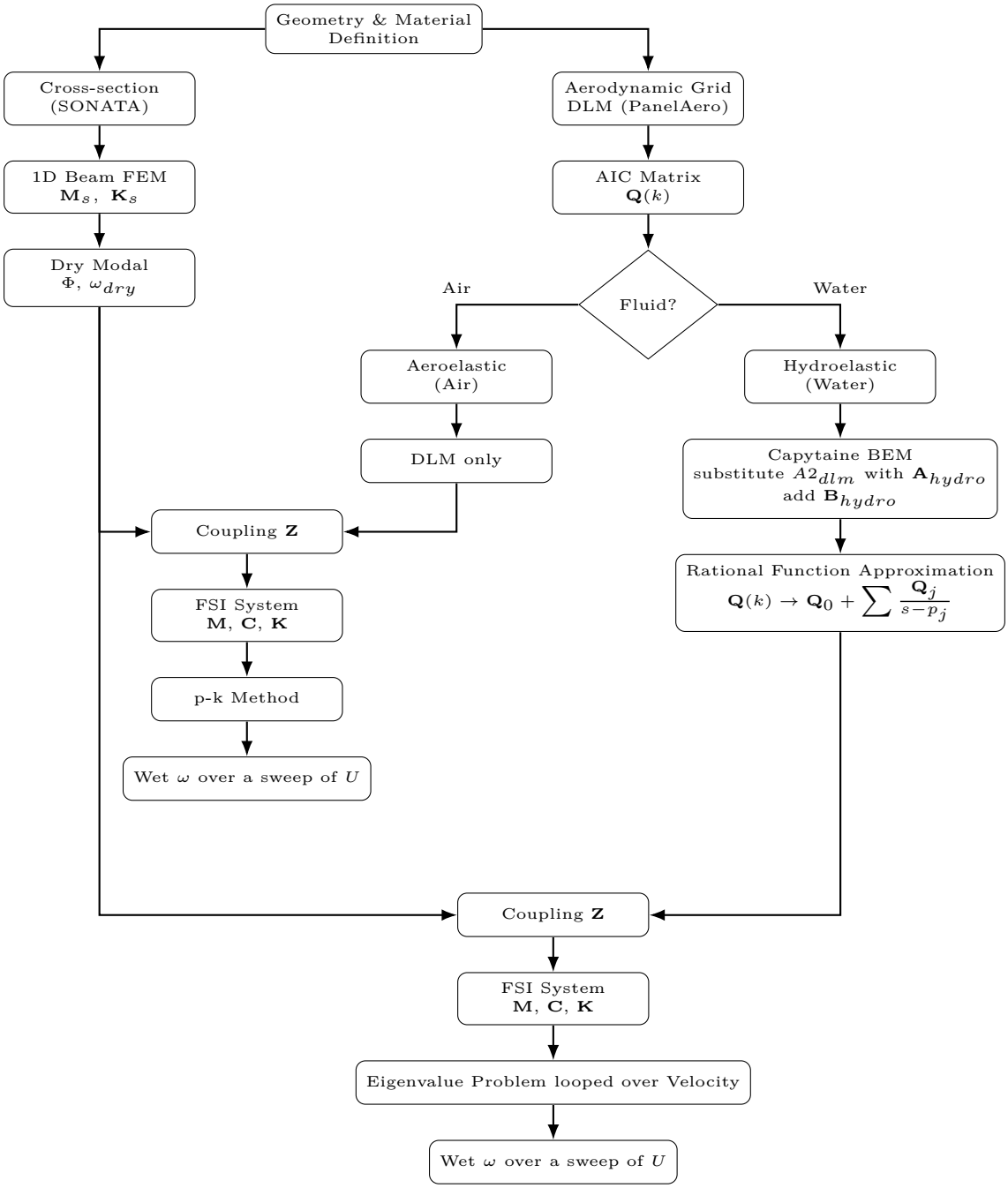


Figure 1.1: Unified workflow for aeroelastic and hydroelastic flutter analysis.

Structural Modeling and FEM Analysis

2.1 Description of the physical problem

This work is dedicated to the study of dynamic behavior of load-bearing slender structures immersed in a fluid flow, with particular reference to wings and hydrofoils. Such structures are characterized by high geometric slenderness, comparable flexural and torsional stiffness and marked interaction with the surrounding fluid.

When a deformable structure is immersed in a flow, the aerodynamic or hydrodynamic forces acting on it depend not only on the instantaneous geometric configuration, but also on the motion of the structure itself. Displacements and rotations modify the velocity field of the fluid, which in turn generates pressure variations responsible for additional loads on the structure. This feedback mechanism constitutes the foundation of fluid-structure interaction.

Under certain operating conditions, the interaction between elastic, inertial and fluid forces can lead to dynamic instability phenomena. Among these, flutter represents one of the most relevant issues from a design perspective, as it manifests as a self-excited oscillation that can rapidly lead to response levels incompatible with structural integrity.

The objective of aeroelastic analysis, therefore, is the prediction of conditions in which such instabilities can arise, starting from a physically coherent description of the structure, fluid and their coupling. In the present work, the problem is addressed in the context of linear dynamics, assuming small displacements and linear elastic behavior of the structure.

2.2 General equation of aeroelastic motion

The dynamic behavior of a deformable structure immersed in a flow can be described by a set of differential equations that express the equilibrium between inertial, elastic, dissipative and fluid forces. Introducing a vector of generalized coordinates $\mathbf{q}(t)$, which represents the dynamic state of the structure, the equation of motion can be written in general form:

$$\mathbf{M_s}\ddot{\mathbf{q}}(t) + \mathbf{C_s}\dot{\mathbf{q}}(t) + \mathbf{K_s}\mathbf{q}(t) = \mathbf{F_{aero}}(\mathbf{q(t)}, \dot{\mathbf{q}}(\mathbf{t}), \mathbf{U}_\infty(\mathbf{t})) \tag{2.1}$$

where:

- $\mathbf{M_s}$ is the structural mass matrix;
- $\mathbf{C_s}$ is the structural damping matrix;
- $\mathbf{K_s}$ is the elastic stiffness matrix;
- $\mathbf{F_{aero}}$ represents the vector of unsteady aerodynamic or hydrodynamic forces;
- $\mathbf{U}_\infty$ is the undisturbed flow velocity vector.

Equation 2.1 highlights how the aeroelastic problem is intrinsically coupled: the fluid load term depends on the motion of the structure and, at the same time, contributes to modifying its dynamic response. Unlike classical structural problems, in which external forces are prescribed, in this context they constitute a function of the dynamic state of the system.

In the present work, attention is directed toward small-amplitude harmonic motion regimes, for which it is possible to linearize the fluid term with respect to generalized coordinates and their

derivatives. This assumption allows us to reduce the problem to the search for exponential solutions and to analyze their stability through eigenvalue techniques.

The equation of motion represents the heart of aeroelastic analysis and serves as the conceptual reference for the entire thesis. In subsequent chapters, each term will be defined, modeled and physically interpreted, always maintaining explicit the link between theoretical formulation and the physical reality of the phenomenon.

2.3 Reference system and kinematic assumptions

To describe the aeroelastic problem coherently, it is necessary to uniquely establish the reference system and kinematic assumptions adopted.

The structure is modeled using an equivalent beam, which coincides with the elastic axis of the wing. A Cartesian reference system is introduced:

- the y-axis is oriented along the span, positive from root to tip, aligned with the elastic axis of the beam;
- the x-axis is oriented along the chord, positive from leading edge to trailing edge;
- the z-axis is oriented vertically, normal to the mean plane of the structure.

For models with multiple wings, a local reference system is introduced for each beam, with the same axis orientation convention. All nodal degrees of freedom, section properties and sectional forces are defined with respect to these local reference systems, which are fixed to the beam. To build the global model, it is necessary to define a transformation that allows expressing all quantities with respect to a common global reference system.

2.4 Definition of structural matrices

In the equation of motion 2.1, the matrices $\mathbf{M_s}$, $\mathbf{C_s}$ and $\mathbf{K_s}$ represent respectively the mass, damping and stiffness of the structure. These matrices are not introduced directly in global form, but derive from a hierarchical construction that reflects the physical nature of the problem. In the present work, structural modeling follows a three-level procedure:

- **definition of section properties**, expressed through sectional matrices in a local reference system;
- **construction of beam elemental matrices** between consecutive nodes;
- **assembly of global matrices** of the structural system.
- if multiple wings are considered, transformation of structural matrices with respect to the common global reference system.

The first level of structural modeling is the **definition of cross-sectional properties**. Since the structure is modeled as an equivalent beam, whose nodes coincide with the elastic axis, each section is described through sectional stiffness properties $\mathbf{K_{sec}}$ and sectional mass properties $\mathbf{M_{sec}}$, defined in a local reference system fixed to the section.

This system is defined in the same way as the global system, with the care to correctly orient the y-axis along the direction tangent to the elastic axis.

The sectional matrices derive from **three-dimensional beam theory**, which assumes that:

- cross-sections remain plane and normal to the elastic axis during deformation (Bernoulli-Euler hypothesis);
- deformations are small compared to section dimensions;

In general form, these are expressed with respect to deformations and generalized velocities of the three-dimensional beam. It is therefore important to emphasize that these matrices are not defined directly with respect to nodal degrees of freedom, but with respect to local kinematic quantities that describe the deformation state of the section.

Since the beam axis coincides with the y-direction (spanwise direction), the vector of generalized sectional deformations is defined as:

$$\epsilon_{sec} = \begin{bmatrix} \kappa_x & \epsilon_y & \kappa_z & \gamma_x & \tau_y & \gamma_z \end{bmatrix}^T \tag{2.2}$$

where:

- ϵ_y is the axial deformation along the y-axis;
- κ_x and κ_z are the flexural curvatures about the x and z axes;
- τ_y is the torsional deformation about the y-axis;
- γ_x and γ_z are the shear deformations in the xy and yz planes.

The generalized sectional forces that are conjugate are collected in the vector:

$$f_{sec} = \begin{bmatrix} M_x & N_y & M_z & Q_x & T_y & Q_z \end{bmatrix}^T = K_{sec} \cdot \epsilon_{sec} \tag{2.3}$$

In the above equation, we see for the first time the sectional stiffness matrix $\mathbf{K_{sec}}$, which links sectional deformations to sectional forces. In the general case, the sectional stiffness matrix is full, including all possible couplings between the different deformation modes of the section, and assumes the form:

$$\mathbf{K_{sec}} = \begin{bmatrix} EI_x & K_{12} & K_{13} & K_{14} & K_{15} & K_{16} \\ K_{21} & EA & K_{23} & K_{24} & K_{25} & K_{26} \\ K_{31} & K_{32} & EI_z & K_{34} & K_{35} & K_{36} \\ K_{41} & K_{42} & K_{43} & GA_x & K_{45} & K_{46} \\ K_{51} & K_{52} & K_{53} & K_{54} & GJ & K_{56} \\ K_{61} & K_{62} & K_{63} & K_{64} & K_{65} & GA_z \end{bmatrix} \tag{2.4}$$

In this matrix, the diagonal terms represent axial, flexural, torsional and shear stiffness; while off-diagonal terms represent possible elastic couplings such as bending-torsion, bending-shear and axial-bending coupling.

For symmetric sections and isotropic materials, many off-diagonal terms vanish, reducing the matrix to an approximately diagonal form. However, in the present work, the general formulation is maintained to allow analysis of geometrically complex composite structures.

Similarly, the sectional mass matrix $\mathbf{M_{sec}}$ is defined with respect to generalized velocities of the section, as it calculates the sectional momentum. The velocities are collected in the vector:

$$\dot{\eta_{sec}} = \begin{bmatrix} \dot{u_x} & \dot{u_y} & \dot{u_z} & \dot{\theta_x} & \dot{\theta_y} & \dot{\theta_z} \end{bmatrix}^T \tag{2.5}$$

The generalized momentum is therefore expressed as:

$$P_{sec} = M_{sec} \cdot \dot{\eta_{sec}} \tag{2.6}$$

where the sectional mass matrix $\mathbf{M_{sec}}$ assumes the general form:

$$\mathbf{M_{sec}} = \begin{bmatrix} m & 0 & 0 & 0 & mz_c & 0 \\ 0 & m & I_{xz} & I_{xy} & 0 & 0 \\ 0 & I_{zx} & I_{zz} & I_{zy} & mx_c & 0 \\ 0 & I_{yx} & I_{yz} & I_{yy} & 0 & 0 \\ mz_c & 0 & mx_c & 0 & M & 0 \\ 0 & 0 & 0 & 0 & 0 & m \end{bmatrix} \tag{2.7}$$

where:

- m is the mass per unit length of the section;
- I_{xx}, I_{yy}, I_{zz} are the principal moments of inertia of the section;
- I_{xy}, I_{xz}, I_{yz} are the products of inertia of the section;
- x_c and z_c are the coordinates of the section center of mass with respect to the elastic axis.

Off-diagonal terms represent the coupling between different motion modes of the section.

Once section properties are defined, the second level of modeling consists of the construction of **beam elemental matrices** associated with each element between two consecutive nodes.

Since the sectional matrices are expressed with respect to generalized sectional deformations and velocities, it is necessary to introduce a transformation that allows expressing sectional deformations and velocities as functions of nodal displacements and rotations of the beam element.

Consider a beam element between two consecutive nodes i and j, of length L_e. The vector of nodal degrees of freedom of the element is defined as:

$$\mathbf{q_e} = \begin{bmatrix} q_i & q_j \end{bmatrix}^T \tag{2.8}$$

The vector of nodal degrees of freedom of each node can be expanded as:

$$\mathbf{q_e} = \begin{bmatrix} u_{x_i} & u_{y_i} & u_{z_i} & \theta_{x_i} & \theta_{y_i} & \theta_{z_i} \end{bmatrix}^T \tag{2.9}$$

Sectional deformations and velocities can therefore be expressed as:

$$\epsilon_{sec}(y) = \mathbf{B}(y) \cdot \mathbf{q_e} \tag{2.10}$$

$$\dot{\eta_{sec}}(y) = \mathbf{N}(y) \cdot \dot{\mathbf{q}}_\mathbf{e} \tag{2.11}$$

where $\mathbf{B}(y)$ and $\mathbf{N}(y)$ are the shape function matrices associated with generalized sectional deformations and velocities.

Using the kinematic relations introduced, the elemental stiffness and mass matrices can be expressed in compact form from sectional properties, integrating along the element length:

$$\mathbf{K_e} = \int_0^{L_e} \mathbf{B}^T(y) \cdot \mathbf{K_{sec}}(y) \cdot \mathbf{B}(y)\, dy \tag{2.12}$$

$$\mathbf{M_e} = \int_0^{L_e} \mathbf{N}^T(y) \cdot \mathbf{M_{sec}}(y) \cdot \mathbf{N}(y)\, dy \tag{2.13}$$

In the present work, sectional properties are defined at beam nodes and assumed constant within each element through interpolation of nodal properties. This assumption is consistent with sufficiently fine discretization of the beam and is commonly adopted in FEM analyses of slender beams.

Finally, the third level of structural modeling consists of the **assembly of global matrices** of the structural system. Assembly occurs by summing contributions of elemental matrices in shared

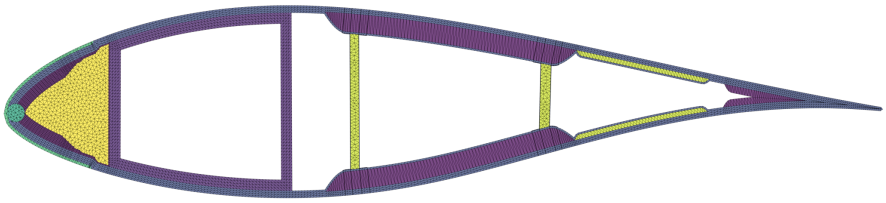


Figure 2.1: Example of a complex composite cross section handled by SONATA, representing the internal architecture of a 15-MW wind turbine blade. The figure is taken from [1].

global degrees of freedom, according to the nodal connectivity of the beam. The final result is the global mass $\mathbf{M_s}$, stiffness $\mathbf{K_s}$ and damping $\mathbf{C_s}$ matrices, which represent the structure as a whole and are ready to be used in the equation of motion 2.1.

Structural damping is introduced in proportional form (Rayleigh damping), combining global mass and stiffness matrices.

The entire procedure for defining structural matrices is implemented in a modular way, allowing the use of constant or variable properties along the span and integration of section properties derived from external tools such as SONATA.

2.5 Cross-Sectional Analysis with SONATA

2.5.1 Motivation and Role in the Workflow

Slender lifting structures such as hydrofoils are typically characterised by complex composite laminates whose through-thickness arrangement of plies determines the structural response in a highly non-trivial way. Modelling the full three-dimensional geometry in a finite element solver would be computationally prohibitive in an iterative design or aeroelastic context. The standard engineering approach instead reduces the blade to an equivalent one-dimensional (1-D) beam, characterised at each spanwise station by a 6×6 stiffness matrix and a 6×6 inertia matrix. The translation from the full 3-D composite definition to these sectionwise matrices requires a dedicated cross-sectional structural analysis.

In the present work, this step is carried out with **SONATA** (Structural Optimization and Aeroelastic Analysis), an open-source Python framework developed at the National Renewable Energy Laboratory (NREL) [1]. SONATA was originally conceived for rotorcraft and wind turbine blade design; its application here to a hydrofoil cross section is a natural extension, given the geometrical and material similarity between rotor blades and slender marine lifting surfaces. The framework closes the gap between 1-D beam finite element models and the full 3-D blade design by evaluating multiple 2-D cross sections of the slender composite structure, as illustrated schematically in the SONATA architecture of [1].

A clear example of the type of lamination that can be handled by SONATA is shown in the cross section at figure 2.1, which represents the internal architecture of a turbine blade. Section considered in this work as a template.

The way SONATA works is described in detail by Feil et al. [1]; the following sections summarise the main features of the framework and its application to the hydrofoil cross section. To improve understanding and aid discussions of the results, particular attention is paid to the coordinate system convention and the structural solver (ANBA4) adopted in this work.

2.5.2 Framework Architecture

SONATA comprises five main components that are invoked sequentially:

1. **Parametrisation and surface generation**: the outer shape of the section and all lamination data are loaded from a YAML input file. Airfoil profiles (in the present case, NACA 0012 and NACA 0015 sections) are defined at each radial station together with twist, chord, and the twist-reference-axis location.
2. **Topology generation**: the internal composite architecture (shell layers, spar caps, webs, foam cores) is described by specifying, for each layer, its thickness, its start and end positions along the contour coordinate $s \in [0, 1]$ (measured counter-clockwise from the trailing edge), the fibre orientation angle Θ_{11}, and the material assignment. B-spline curves represent layer boundaries, enabling smooth offset operations and the automatic detection of overlapping layer intervals through an interval-tree data structure.
3. **Mesh discretisation**: once the topology is established, each layer is meshed from the inside outward by projecting nodes orthogonally between adjacent B-spline boundaries. Corner cells are resolved through a library of predefined corner-style rules; remaining cavities (e.g., foam-filled bays) are triangulated with the Shewchuk algorithm [2].
4. **Structural solving**: the mesh and material data are passed to the structural solver—either the commercial VABS [3] or the fully open-source ANBA4 [4]—which returns the 6×6 Timoshenko stiffness matrix $\mathbf{S}$ and the sectional inertia matrix.
5. **Postprocessing**: cross-sectional outputs (shear centre, neutral axis, mass centre, stiffness and inertia matrices) are exported in CSV format for downstream use; three-dimensional lofted geometries and 2-D mesh plots can be generated for quality inspection.

2.5.3 Coordinate System Convention

SONATA adopts a local cross-section frame (superscript L) whose x_1^L-axis is tangent to the beam geometric curve (the elastic axis), while x_2^L points toward the leading edge parallel to the chord and x_3^L completes the right-handed triad. The ply orientation angle Θ_{11} is defined as a rotation of the material frame about x_1^L, and the material orientation angle Θ_3 is a subsequent rotation about the out-of-plane ply axis. This two-level rotation convention allows the framework to handle arbitrary lamination angles, including off-axis plies that introduce extension–torsion or bending–torsion coupling in the stiffness matrix.

2.5.4 The ANBA4 Structural Solver

In the present work, the open-source solver ANBA4 (ANisotropic Beam Analysis, version 4) is adopted as the structural engine, making the entire cross-sectional analysis chain fully open-source [4, 5].

ANBA4 starts from the weak form of the linear equilibrium equations over the beam volume:

$$\int_V \delta\boldsymbol{\varepsilon} : \boldsymbol{\sigma}\, dV = \delta L_e, \tag{2.14}$$

where $\boldsymbol{\varepsilon} = \frac{1}{2}(\nabla\mathbf{u}^T + \nabla\mathbf{u})$ is the small strain tensor, $\boldsymbol{\sigma} = \mathbf{E} : \boldsymbol{\varepsilon}$ the Cauchy stress, and δL_e the virtual work of external loads applied at the beam extremities.

Integration by parts along the local x_1^L-axis and the subsequent finite element discretisation of the cross-sectional displacement field lead to a second-order homogeneous differential equation in the

nodal displacements $\tilde{\mathbf{u}}$:

$$\mathbf{M}\,\frac{\partial^2 \tilde{\mathbf{u}}}{\partial (x_1^L)^2} - \mathbf{H}\,\frac{\partial \tilde{\mathbf{u}}}{\partial x_1^L} - \mathbf{E}\,\check{\mathbf{u}} = \mathbf{0}. \tag{2.15}$$

This equation possesses a Hamiltonian structure characterised by twelve null eigenvalues, corresponding to six rigid-body motions and six polynomial deformation modes (traction, torsion, two bending directions, and two shear-bending directions). Once these polynomial solutions are known, the 6×6 Timoshenko stiffness matrix is obtained by enforcing equivalence between the internal virtual work per unit length and the virtual work of the sectional forces and moments [4].

The stiffness matrix $\mathbf{S}$ relates sectional forces $\mathbf{F}$ and moments $\mathbf{M}$ to strains $\boldsymbol{\epsilon}$ and curvatures $\boldsymbol{\kappa}$:

$$\begin{pmatrix} \mathbf{F} \\ \mathbf{M} \end{pmatrix} = \mathbf{S} \begin{pmatrix} \boldsymbol{\epsilon} \\ \boldsymbol{\kappa} \end{pmatrix}, \qquad \mathbf{S} = \begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} & S_{15} & S_{16} \\ S_{12} & S_{22} & S_{23} & S_{24} & S_{25} & S_{26} \\ S_{13} & S_{23} & S_{33} & S_{34} & S_{35} & S_{36} \\ S_{14} & S_{24} & S_{34} & S_{44} & S_{45} & S_{46} \\ S_{15} & S_{25} & S_{35} & S_{45} & S_{55} & S_{56} \\ S_{16} & S_{26} & S_{36} & S_{46} & S_{56} & S_{66} \end{bmatrix}, \tag{2.16}$$

where the diagonal entries represent the extensional ($S_{11} \equiv EA$), shear ($S_{22} \equiv GA_x$, $S_{33} \equiv GA_z$), torsional ($S_{44} \equiv GJ$), and bending ($S_{55} \equiv EI_x$, $S_{66} \equiv EI_z$) stiffnesses. The off-diagonal terms encode elastic coupling effects such as bending–torsion (S_{45}, S_{46}) or extension–torsion (S_{14}), which are particularly significant for composite structures with off-axis ply orientations and have a direct bearing on the flutter mechanism.

SONATA has been validated through an extensive code-to-code comparison with VABS and with NABSA results for composite box beams with various layup configurations, including off-axis plies that generate extension–torsion and shear–bending coupling terms [1]. The framework proved capable of accurately reproducing all off-diagonal stiffness entries, even for the most demanding layup configurations.

2.5.5 Application to the Hydrofoil Cross Section

The lamination of the hydrofoil studied in this work is defined in a YAML configuration file following the SONATA input convention. The structural model is analysed at multiple spanwise stations; at each station, the 2-D cross-sectional mesh is generated automatically by SONATA and passed to ANBA4, which returns the full 6×6 stiffness and inertia matrices. These are then exported in CSV format and loaded by the custom beam FEM assembler described in the preceding sections of this chapter.

An important feature of this approach is that SONATA resolves all off-diagonal coupling terms of the stiffness matrix, including any bending–torsion coupling arising from geometric twist or asymmetric ply stacking. Such terms have a direct influence on the flutter mechanism: their presence modifies both the flutter speed and the modal coalescence pattern, and their neglect would lead to predictions that may be unconservative.

Furthermore, the integration of SONATA in the workflow allows to represent complex geometries and laminates, enhancing powerfully this tool effectiveness.

Aeroelastic and Hydroelastic Modeling

3.1 Aeroelasticity and Hydroelasticity: Foundational Concepts

The fluid-structure interaction component appearing in the structural equation of motion (Eq. 2.1) is:

$$\mathbf{F_{fluid}} = \mathbf{F_{fluid}}(\mathbf{q(t)}, \dot{\mathbf{q}}\mathbf{(t)}, \mathbf{U}_\infty\mathbf{(t)}) \tag{3.1}$$

It is fundamental to describe the interaction between the structure and the surrounding fluid medium, which is defined more precisely here.

3.1.1 Aeroelasticity

Aeroelasticity is the science that studies the interaction between elastic structures and aerodynamic forces in *gaseous* environments (primarily air). It is concerned with analyzing how elastic, inertial, and fluid-dynamic forces interact on load-bearing structures immersed in an airflow.

In aeroelastic systems:

- The surrounding fluid is typically air at atmospheric conditions ($\rho_{\text{air}} \approx 1.225$ kg/m^3 at sea level).
- The flow is described by inviscid potential-flow theory, which is well-suited to slender aerodynamic surfaces where viscous effects are localized to thin boundary layers.
- Aerodynamic forces are dominated by *circulatory lift* (controlled by the Kutta-Joukowski theorem) and the development of a von Kármán wake.
- Added-mass effects (inertial reaction of the fluid) are typically small relative to structural inertia.

The structural response to loading cannot be interpreted separately from the actions of the fluid: displacements modify the flow field, which in turn alters pressures and generates additional loads on the structure. In the case of slender wing structures (wings, blades, aerodynamic foils), such interactions can lead to dynamic instability phenomena, including **flutter**, characterized by self-excited oscillations that develop beyond a critical airspeed and can lead to structural damage or catastrophic failure if not adequately predicted and mitigated.

3.1.2 Hydroelasticity

Hydroelasticity is the science that studies the interaction between elastic structures and hydrodynamic forces in *liquid* environments (primarily water). It encompasses the physical phenomena of aeroelasticity but with profound modifications arising from the properties of the fluid medium.

In hydroelastic systems:

- The surrounding fluid is water at ambient conditions ($\rho_{\text{water}} \approx 1000$ kg/m^3), approximately **800 times denser** than air.
- The fluid is incompressible, and viscous effects are more significant than in air.

- Hydrodynamic forces exhibit two distinct contributions: (i) *circulatory lift* from the lifting surface (via potential-flow methods), and (ii) *added mass and radiation damping* arising from the volume displacement of the elastic body and the pressure field it generates.
- Added-mass effects are **not negligible**: they often dominate the system inertia, requiring accurate three-dimensional calculation of pressure distributions over the entire wetted surface, not just the thin lifting-surface approximation.

The physical origin of the distinction lies in the density ratio. For a structure with mass m and reference dimension d, immersed in a fluid of density ρ:

- In air: the fluid-to-structure mass ratio $\mu_{\text{air}} = \rho_{\text{air}}\pi d^2 L/m$ is typically order 0.01–0.1 (very light coupling).
- In water: the fluid-to-structure mass ratio $\mu_{\text{water}} = \rho_{\text{water}}\pi d^2 L/m$ is typically order 1–10 (strong coupling, often greater than structural mass).

Consequently, the fluid-added-mass term, often negligible in aeroelastic design, becomes the dominant inertial effect in hydroelastic applications. A simplified 2D thin-wing aerodynamic model (where the wing is represented as a curve line) is inadequate in water; instead, the three-dimensional volume and surface curvature of the hydrofoil must be explicitly modeled to capture the full hydrodynamic inertial forces.

Aeroelastic Workflow (Air Environment) For aeroelastic analysis in air, the workflow simplifies to:

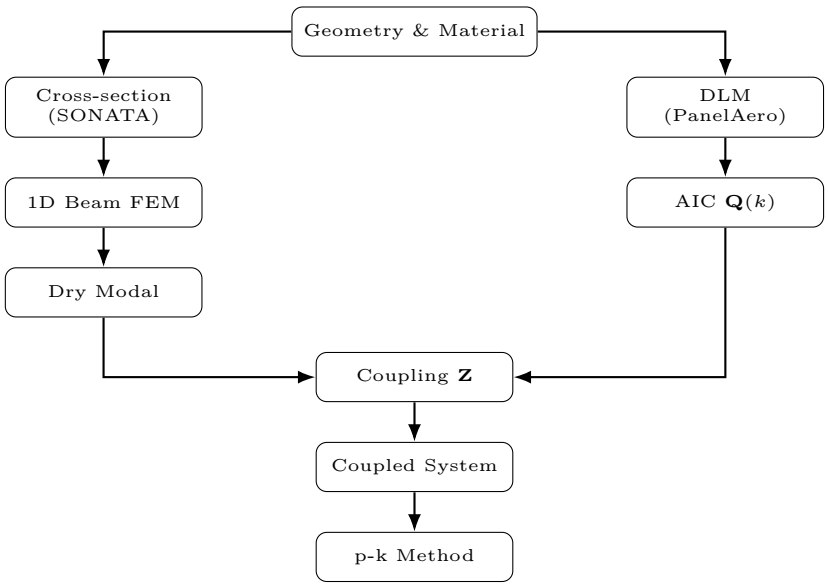


Figure 3.1: Aeroelastic analysis workflow: DLM alone is sufficient for air-based applications.

3.2 Structural and aerodynamic kinematic spaces

The structural model describes the motion of the structure through a vector of nodal degrees of freedom, defined on the nodes of the equivalent beam. These degrees of freedom include translations and rotations about the principal axes, and are organized in a vector $\mathbf{q}_{\mathbf{struct}}$ (Eq. 2.9).

The aerodynamic model, based on a panel discretization, provides fluid forces at corresponding control points. Each control point is associated with a panel, typically on the three-quarter chord line.

Since the two models are defined on different sets of points, it is necessary to introduce a kinematic relation that allows us to:

- express the motion of aerodynamic control points as a function of structural degrees of freedom;
- transfer aerodynamic forces calculated on panels back to the structural domain.

3.3 Kinematics of the aerodynamic control point

From a physical perspective, fluid-structure interaction is governed by the **normal component of relative velocity** between the lifting surface and the undisturbed flow. In other words, what matters to the aerodynamic model is the velocity at which the flow is deflected by the lifting surface due to both the structural motion of the wing and its geometry. This quantity is called **downwash** and represents the fundamental variable through which structural motion influences the aerodynamic pressure field.

In the context of panel discretization, downwash is calculated at the aerodynamic control points, located on each panel of the aerodynamic grid. Therefore, the coupling between the two domains requires expressing the motion of aerodynamic control points as a function of structural degrees of freedom, and projecting the aerodynamic forces calculated on panels into the structural domain.

Consider an aerodynamic control point p, belonging to a panel whose unit normal is denoted $\mathbf{n_p}$, and let $\mathbf{U}_\infty$ be the velocity of the undisturbed flow. Consequently, for coupling purposes, it is not necessary to preserve the entire displacement vector $\mathbf{V_p}$, but only its projection along the normal to the panel $\mathbf{n_p}$.

Downwash is defined as the normal component of relative velocity between surface and fluid:

$$w_p(t) \;=\; \mathbf{n}_p \cdot \big(\mathbf{V}_p(t) - \mathbf{U}_\infty\big). \tag{3.2}$$

In a regime of small deformations and for main flow aligned with the x-axis, the relevant component reduces to the scalar form

$$w(x,t) \;=\; \frac{\partial z(x,t)}{\partial t} + U_\infty \frac{\partial z(x,t)}{\partial x}, \tag{3.3}$$

where $z(x,t)$ is the deformation of the surface in the reference normal direction and U_∞ the environment velocity vector. The first term is the *local* contribution (surface velocity) and leads, in the harmonic domain, to aerodynamic damping effects; the second term is the *convective* / geometric contribution (slope) that generates quasi-steady lift.

Assuming harmonic solutions $\sim e^{pt}$ (with $p = \sigma + i\omega$), the formula becomes

$$\hat{w}(x) \;=\; p\,\hat{z}(x) + U_\infty \frac{d\hat{z}(x)}{dx}. \tag{3.4}$$

In the present work, the structural velocity of the control point is obtained by assuming that each cross-section of the beam behaves as a rigid body in its own plane. The motion of control point p is thus calculated as the sum of the **translational contribution** with the **rotational contribution** associated with each structural node.

3.4 Discretization and kinematic matrix

Let N be the nodes of structural discretization (with 6 DOF per node: $u_x, u_y, u_z, \theta_x, \theta_y, \theta_z$) and n_p be the aerodynamic panels (or control points). We introduce the kinematic matrix

$$\mathbf{Z} \in \mathbb{R}^{n_p \times 6N}, \tag{3.5}$$

such that the vector of dynamic-origin downwash is written as

$$\mathbf{w}_{vel} \;=\; \mathbf{Z}\,\mathbf{q}, \tag{3.6}$$

where $\mathbf{q}$ is the structural nodal vector. The matrix $\mathbf{Z}$ maps structural degrees of freedom to aerodynamic/hydrodynamic downwash, including both translational and rotational contributions.

Recall that an aerodynamic control point p, belonging to an aerodynamic panel, is located between two consecutive structural nodes of the equivalent beam. Once the two structural nodes 1 and 2 are identified, a local dimensionless coordinate $\epsilon \in [0, 1]$ is introduced that defines the position of the control point along the element. It is defined as:

$$\epsilon = \frac{(\mathbf{y_p} - \mathbf{y_1})}{(\mathbf{y_2} - \mathbf{y_1})} \tag{3.7}$$

where $\mathbf{y_p}$ is the coordinate of the control point along the span axis, and $\mathbf{y_1}$ and $\mathbf{y_2}$ are the coordinates of the structural nodes.

The linear shape functions of the beam element are thus:

$$\mathbf{N_1}(\epsilon) = 1 - \epsilon, \quad \mathbf{N_2}(\epsilon) = \epsilon \tag{3.8}$$

These functions allow continuous interpolation along the element of both nodal displacements and nodal rotations. We therefore introduce the kinematic matrix associated with the aerodynamic panel p, whose contributions to the corresponding row of matrix Z are:

$$\mathbf{Z_p} = \mathbf{N_1}(\epsilon) \begin{bmatrix} \mathbf{I} & \mathbf{S}(\mathbf{x_p} - \mathbf{x_1}) \end{bmatrix} + \mathbf{N_2}(\epsilon) \begin{bmatrix} \mathbf{I} & \mathbf{S}(\mathbf{x_p} - \mathbf{x_2}) \end{bmatrix} \tag{3.9}$$

where:

- the first part represents the translational contribution, with $\mathbf{I}$ being the 3×3 identity matrix
- the second part represents the rotational contribution, with $\mathbf{S}(\mathbf{r})$ being the skew-symmetric matrix associated with vector $\mathbf{r}$
- $\mathbf{x_p}$ is the position of the aerodynamic control point
- $\mathbf{x_1}$ and $\mathbf{x_2}$ are the positions of the adjacent structural nodes

The construction adopted for the Z matrix therefore presents some fundamental properties:

- **kinematic continuity**: the motion of control points varies continuously along the span
- **physical coherence**: bending and torsion contribute naturally to downwash
- **locality**: each panel is influenced only by adjacent structural nodes
- **energetic compatibility**: the matrix enables a force transfer formulation that preserves virtual work

It allows expression of the normal motion of control points as a function of FEM degrees of freedom and constitutes the first step for introducing aerodynamic forces into the equations of motion.

3.5 Convective term of downwash

In addition to the dynamic contribution of downwash, associated with the structural velocity of control points, it is also necessary to consider the convective contribution related to the geometry of the lifting surface. This term depends on the slope of the lifting surface relative to the incident flow and contributes to the generation of quasi-steady lift and corresponds to the term $U_\infty \dfrac{dz}{dx}$ of Eq. 3.4. The term is represented in discrete form using a matrix that maps nodal rotations to

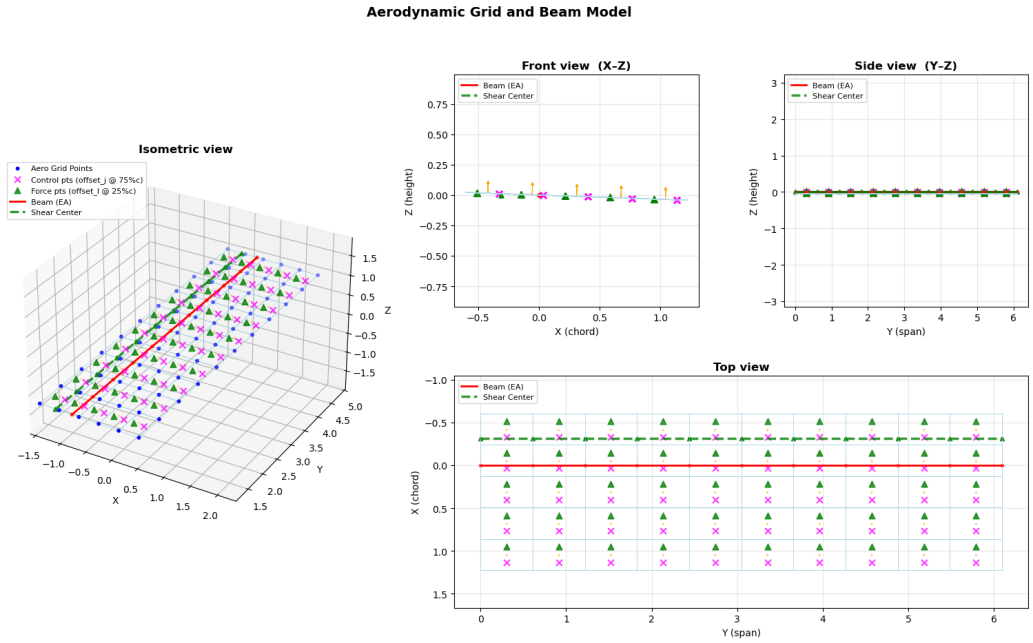


Figure 3.2: Aerodynamic grid and beam model showing the discretization and coupling schemes.

inclination changes seen by the flow. In the case of undisturbed flow oriented along x with unit vector $\mathbf{e}_\infty = [1, 0, 0]$ the useful projection is:

$$\mathbf{n}_i \times \mathbf{e}_\infty = \begin{bmatrix} 0 \\ n_{i,z} \\ -n_{i,y} \end{bmatrix}, \tag{3.10}$$

where $\mathbf{n}_i$ is the unit normal of panel i.

The quasi-steady downwash (normalized for U_∞) is approximately:

$$\frac{\mathbf{w}_{qs}}{U_\infty} \approx (\mathbf{n} \times \mathbf{e}_\infty)^\top \boldsymbol{\theta}. \tag{3.11}$$

Therefore, the matrix $\mathbf{Z}_{qs} \in \mathbb{R}^{n_p \times 6N}$ contains, for each panel i and node n, the entries

$$\begin{aligned} (\mathbf{Z}_{qs})_{i,\,3n+3} &= 0, \\ (\mathbf{Z}_{qs})_{i,\,3n+4} &= N_n(\xi)\; n_{i,z}, \\ (\mathbf{Z}_{qs})_{i,\,3n+5} &= -N_n(\xi)\; n_{i,y}, \end{aligned} \tag{3.12}$$

in accordance with the θ_y and θ_z components that determine the slope useful for generating lift.

Note: it is not correct to neglect the component associated with θ_z in general: the presence of normals not perfectly orthogonal to z or of non-flat wing geometry makes both relevant.

3.6 Non stationary Aerodynamics and matrices of influence

The **Doublet Lattice Method** (DLM) is a panel method based on potential and incompressible flow assumptions. It enables the calculation of unsteady aerodynamic forces generated by a wing or hydrofoil in motion, represented by a grid of lifting panels. The method is based on solving potential flow equations, imposing non-penetration boundary conditions on the lifting surface and Kutta conditions at trailing edges. Through this procedure, for each reduced frequency k and Mach number

Ma, an aerodynamic/hydrodynamic influence matrix $\mathbf{Q}(k) \in \mathbb{C}^{n_p \times n_p}$ is calculated that allows the calculation of fluid forces as a function of displacements/velocities of aerodynamic/hydrodynamic control points.

In general form, to calculate the vector of aerodynamic/hydrodynamic forces in the frequency domain, it can be written as:

$$\hat{\mathbf{f}}_{aero}(\omega) \;=\; \mathbf{Q}(k)\,\hat{\mathbf{w}}(\omega). \tag{3.13}$$

where we recall:

- $\mathbf{w}(\omega)$ is the downwash (Eq. 3.2)

More in particular, the DLM analysis result is a (complex) matrix $\mathbf{Q}(k)$ mapping panel circulations into pressure coefficients for each reduced frequency k and Mach number Ma. One matrix entry (row = receiving panel, column = source panel) represents the aerodynamic influence of a unit (harmonic) circulation on the normal downwash at the receiving panel. Physically: "If panel j has a unit circulation, how much downwash does it induce on panel i?"

For harmonic motion at frequency ω (reduced frequency $k = \frac{\omega b}{U_\infty}$), the influence is complex:

- the real part is the in-phase component of the aerodynamic force, contributing to the mass-like behavior of the system.
- the imaginary part is the out-of-phase component, contributing to the damping-like behavior of the system.

The influence decays with distance and depends strongly on relative geometry (spanwise offset, dihedral, sweep, local panel orientation).

3.7 Force transfer to structural space: $\mathbf{Z}_f$ matrix

The pressure on each panel i generates a force $\Delta\mathbf{F}_i = \Delta p_i\, A_i\, \mathbf{n}_i$. To transfer such forces to the space of structural degrees of freedom we define

$$\mathbf{Z}_f \in \mathbb{R}^{n_p \times 6N} \tag{3.14}$$

such that generalized nodal forces are obtained from

$$\mathbf{F}_{aero} \;=\; \mathbf{Z}_f^{\top}\,\mathbf{A}\,\Delta\mathbf{p}. \tag{3.15}$$

The entries of $\mathbf{Z}_f$ are analogous to those of $\mathbf{Z}$ but calculated at the *force point*, and include the moment contribution due to the force applied at a distance Δx_{fp} from the elastic axis:

$$(\mathbf{Z}_f)_{i,\,3n+2} = N_n(\xi)\,n_{i,z}, \qquad (\mathbf{Z}_f)_{i,\,3n+4} = -N_n(\xi)\,\Delta x_{fp}\,n_{i,z}, \tag{3.16}$$

where the sign of the moment term must be chosen consistently with the rotation convention (positive pitch, etc.). The quantity Δx_{fp} must be calculated with reference to the local elastic axis (or the reference chosen for rotation).

3.7.1 Physical location of calculation points in aerodynamic panels: offset_j and offset_l

A fundamental distinction in the aerodynamic model concerns the *physical location* where aerodynamic forces are calculated and applied. The Doublet Lattice Method (DLM) formulation provides two distinct sets of points on each panel:

- $\mathbf{Z}$ is calculated at the *control point* (or collocation point), positioned at **75**% chord ($\text{offset}_j = 0.75$), where the flow tangency boundary condition is applied. These points represent the *downwash positions*, where the method enforces that the normal velocity of the surface equals the normal component of the perturbed flow velocity.
- $\mathbf{Z}_f$ is calculated at the *force point*, positioned at **25**% chord ($\text{offset}_l = 0.25$), where aerodynamic forces are integrated and applied to the structure. This position represents approximately the center of pressure distribution on rigid wings.

The physical separation between these two sets of points – a distance of approximately 50% chord – has fundamental implications for aeroelastic coupling. When the aerodynamic panel is not aligned with the elastic axis of the structure (typical condition in real wings), this separation introduces a *significant moment arm*. Aerodynamic forces applied at the force point generate torsional and bending moments on the structure according to the relation $\mathbf{M} = \mathbf{r} \times \mathbf{F}$, where $\mathbf{r}$ is the position vector between the force application point and the elastic axis.

This moment offset is automatically accounted for in the structure of the $\mathbf{Z}_f$ matrix (see equation (3.16)), where the term Δx_{fp} represents the distance between the force application point and the structural reference. The accuracy of flutter instability prediction depends critically on correct representation of this geometry. In the context of numerical implementation (see Section 3.8), these offsets are defined and calculated in the aerodynamic grid construction routine and in the coupling matrix.

3.8 Final composition and insertion into the equation of motion

Composing (3.6), (3.12), (3.13) and (3.15), assuming harmonic solution e^{pt}:

$$\hat{\mathbf{w}} \;=\; \frac{p}{U_\infty}\,\mathbf{Z}\,\hat{\mathbf{q}} \;+\; \mathbf{Z}_{qs}\,\hat{\mathbf{q}}, \tag{3.17}$$

and therefore

$$\hat{\mathbf{F}}_{aero}(p) \;=\; \mathbf{Z}_f^{\top}\,\mathbf{A}\,\mathbf{Q}\big(k(p)\big)\left(\frac{p}{U_\infty}\mathbf{Z} + \mathbf{Z}_{qs}\right)\hat{\mathbf{q}}. \tag{3.18}$$

After introducing aerodynamic forces and coupling operators, it is possible to write the complete aeroelastic equation of motion in the complex frequency domain p, assuming harmonic solution $q(t) = \hat{q}e^{pt}$:

$$[Mp^2 + Cp + K]\hat{q} = \hat{F_{aero}}. \tag{3.19}$$

Following Rodden et al. (2017), the final aeroelastic equation for aeroelastic stability modal analysis is:

$$\mathbf{M}_{hh}p^2 + (\mathbf{B}_{hh} - \frac{1}{4}\rho c U_\infty \mathbf{Q}_{hh}^I/k)p + (\mathbf{K}_{hh} - \frac{1}{2}\rho U_\infty^2 \mathbf{Q}_{hh}^R) = 0, \tag{3.20}$$

where $\mathbf{Q}_{hh}^R$ and $\mathbf{Q}_{hh}^I$ are respectively the real and imaginary parts of the aerodynamic/hydrodynamic influence matrix projected in the structural modal space. where $\mathbf{B}_{hh}$, $\mathbf{M}_{hh}$ and $\mathbf{K}_{hh}$ are respectively the damping, mass and stiffness matrices projected in the structural modal space.

This represents the complex eigenvalue problem used in linear flutter analyses. Once the equation of motion projected onto a reduced modal basis is defined, the aeroelastic problem assumes the form of a nonlinear eigenvalue equation. This nonlinearity arises from the fact that unsteady aerodynamic matrices depend on the reduced frequency k, which in turn depends on the oscillation frequency ω that is one of the unknowns (the imaginary part of the eigenvalue) of the problem itself.

To solve this problem and find the flutter speed, the **p-k Method** proceeds through a rigorous iterative algorithm based on two nested loops and a post-processing phase.

3.9 Unsteady Aerodynamics with PanelAero

3.9.1 Overview of the PanelAero Library

PanelAero is an open-source Python library that implements both the Vortex Lattice Method (VLM) and the Doublet Lattice Method (DLM) [6]. It was originally developed as part of the Loads Kernel software at the DLR Institute of Aeroelasticity and is publicly available under a BSD 3-Clause licence at `https://github.com/DLR-AE/PanelAero`. The DLM implementation follows the formulations of Albano and Rodden [7], Rodden et al. [8, 9], and Blair [10], and has been validated against MSC.Nastran for both parabolic and quartic integration schemes across a variety of configurations including dihedral, sweep, and horizontal-tail/wing combinations at different Mach numbers and reduced frequencies [6].

The library operates in the frequency domain. For a given reduced frequency

$$k = \frac{\omega\, b}{U_\infty}, \tag{3.21}$$

where ω is the circular frequency of harmonic oscillation, b is the reference half-chord, and U_∞ is the free-stream speed, PanelAero returns the complex Aerodynamic Influence Coefficient (AIC) matrix

$$\mathbf{Q}(k, \mathrm{Ma}) \in \mathbb{C}^{n_p \times n_p}. \tag{3.22}$$

The entry (i, j) of $\mathbf{Q}$ represents the aerodynamic influence of a unit harmonic circulation on panel j (the *sending* panel) on the normal downwash at the control point of panel i (the *receiving* panel), for the prescribed reduced frequency k. As established in the preceding sections, the real part of $\mathbf{Q}$ contributes an in-phase (mass-like) aerodynamic stiffness, while the imaginary part contributes an out-of-phase (damping-like) aerodynamic damping.

3.9.2 Vortex Lattice Method: Steady Aerodynamic Baseline

The VLM implemented in PanelAero follows the horseshoe-vortex formulation of Katz and Plotkin [11]. A Prandtl–Glauert compressibility correction is applied by contracting the streamwise x-coordinates by the factor $\beta = \sqrt{1 - \mathrm{Ma}^2}$ prior to evaluating the induced velocity integrals. The AIC matrix of the VLM, $\mathbf{Q}_{\mathrm{VLM}}$, is obtained as the negative inverse of the downwash matrix, scaled by the Kutta–Joukowski relation to map panel circulations to pressure coefficients.

For vanishing reduced frequency ($k \to 0$), the DLM solution reduces formally to the VLM solution; however, because the DLM involves numerical integration and analytical approximations, the VLM is numerically more accurate in the quasi-steady limit. PanelAero exploits this fact explicitly by subtracting the steady DLM contribution from the full DLM result and subsequently replacing it with the VLM solution, so that the final composite AIC reads:

$$\mathbf{Q}(k) = \mathbf{Q}_{\mathrm{VLM}} + \big[\mathbf{Q}_{\mathrm{DLM}}(k) - \mathbf{Q}_{\mathrm{DLM}}(k \to 0)\big]. \tag{3.23}$$

This combination preserves the numerical accuracy of the VLM at low frequency while retaining the full unsteady physics of the DLM at finite k.

3.9.3 Doublet Lattice Method: Unsteady Extension

The DLM extends the VLM to harmonic motion by replacing the steady Prandtl–Glauert kernel with an unsteady lifting-surface kernel that accounts for the propagation of pressure disturbances and the phase lag between structural motion and aerodynamic response.

Geometric setup. For each pair consisting of a sending panel s and a receiving panel r, the relevant geometric quantities are the Cartesian coordinate differences (x_0, y_0, z_0), the dihedral and sweep angles of the sending panel, and the local coordinates (u_1, r_1) of the receiving control point relative to the sending doublet line. Following [8, 6], three mutual geometric configurations are distinguished:

1. *Planar*: the receiving and sending panels lie in the same plane ($|z_0| < \varepsilon$, where ε is a small geometric tolerance);
2. *Near-planar*: the panels belong to nearly coplanar surfaces ($r_1/d_s < 0.10$, where d_s is the sending-panel half-span);
3. *Non-planar*: all remaining configurations, including wing–tail and T-tail arrangements.

The near-planar condition requires special treatment to avoid numerical singularities in the arctangent terms that appear in the geometric factor T [8].

Kernel functions. The aerodynamic kernel functions K_1 (planar) and K_2 (non-planar) encode the physics of the unsteady pressure field induced by a harmonic doublet distribution along the panel span. They are complex-valued functions of the reduced frequency k, the Mach number, and the local geometry, and they involve two oscillatory integrals I_1 and I_2 that cannot be expressed in closed form. Two analytical approximation schemes are implemented in PanelAero to evaluate these integrals efficiently:

- **Laschka approximation** [8]: expresses I_1 and I_2 as weighted sums of eleven exponential terms, providing sufficient accuracy for moderate reduced frequencies.
- **Desmarais approximation** [9]: a twelve-term expansion introduced together with the quartic integration scheme, offering superior accuracy at higher reduced frequencies.

These approximations replace the expensive adaptive numerical quadrature of the oscillatory integrands and are essential for the computational feasibility of the DLM over the wide range of (k, Ma) combinations required by the flutter sweep.

Spanwise integration. The kernel functions are evaluated at a discrete set of spanwise sample points across each sending panel's half-span and combined with pre-computed quadrature weights. Two levels of accuracy are available [6]:

- **Parabolic integration** [8]: three sample points and second-order weights; appropriate for panel aspect ratios below approximately 3–4 and moderate reduced frequencies.
- **Quartic integration** [9]: five sample points with higher-order weights; it relaxes the aspect-ratio constraint and extends the reliable range of k, at the cost of a modest increase in computation time per panel pair.

In both cases, the planar and non-planar contributions from the spanwise integration are assembled into separate downwash sub-matrices and then summed to form the full downwash matrix $\mathbf{W}(k)$. Finally, the Kutta–Joukowski relation converts downwash to pressure coefficients, yielding the raw DLM AIC; the steady part is then subtracted and replaced by the VLM baseline according to Eq. (3.23).

3.9.4 Precomputation Strategy and Local AIC Library

A key architectural decision in the present workflow is the separation of the DLM precomputation from the flutter sweep. Computing $\mathbf{Q}(k, \mathrm{Ma})$ for a single (k, Ma) pair has a cost that scales as $\mathcal{O}(n_p^2)$ in the number of panels, and is thus moderately expensive. The p–k flutter iteration, however, requires $\mathbf{Q}$ to be evaluated at reduced frequencies that are updated at every iteration step and for every flight speed in the sweep; recomputing the full DLM online at each step would render the flutter analysis intractable.

To overcome this bottleneck, a precomputation module populates a *local AIC library*: for each combination (k_i, Ma_j) in a prescribed parameter grid, the complex matrix $\mathbf{Q}(k_i, \mathrm{Ma}_j)$ is evaluated by PanelAero and its real and imaginary parts are stored separately as binary NumPy archives (`.npy`) in a structured directory hierarchy. A metadata file in JSON format records the grid parameters, the aerogrid description, and the integration scheme, enabling subsequent verification and interruption-resumption of long precomputation runs.

During the flutter sweep, the p–k solver retrieves $\mathbf{Q}$ at the current reduced frequency k by linear interpolation over the precomputed grid. This approach reduces the online cost of the aerodynamic evaluation to a simple table look-up, while the offline precomputation can be executed once for a given aerogrid and fluid configuration. Separate AIC libraries are maintained for different angles of attack, panel discretisations, integration schemes (parabolic or quartic), and fluid media (air or water), allowing systematic sensitivity studies without re-invoking the DLM.

3.10 Aerodynamic Panel Grid Setup

The aerodynamic domain is discretised into a structured panel mesh using the CAERO1 card format, consistent with the Nastran/MSC convention adopted internally by PanelAero. A CAERO1 entry defines a trapezoidal lifting-surface patch by specifying the leading-edge root coordinates (x_1, y_1, z_1), the leading-edge tip coordinates (x_4, y_4, z_4), the root chord c_r, the tip chord c_t, the number of spanwise divisions N_{span}, and the number of chordwise divisions N_{chord}.

The grid generator implemented in this work produces CAERO1-compatible panel descriptions which are then ingested by PanelAero's mesh parser to populate the aerogrid dictionary. Each panel is characterised by its quarter-chord point (sending point, index `l`), three-quarter-chord control point (receiving point, index `j`), half-chord point (index `k`), inner and outer doublet filament endpoints (`P1`, `P3`), the unit outward normal vector $\mathbf{N}$, the panel area A, and the panel semi-span width d_s.

The choice of N_{span} and N_{chord} is governed by two competing requirements. On one hand, the panel aspect ratio (spanwise extent to chordwise depth) should remain below approximately 3–4 for the parabolic integration scheme, or somewhat higher for the quartic scheme (see Section 3.9). On the other hand, since the size of the AIC matrix $\mathbf{Q}$ scales quadratically with the total number of panels $n_p = N_{\mathrm{span}} N_{\mathrm{chord}}$, and the DLM must be evaluated for each (k, Ma) pair in the flutter sweep, an excessively fine grid rapidly becomes prohibitive.

3.10.1 Parametric Aerogrid Generation from Camber Line Coordinates

A key contribution of the present work is the development of a parametric aerogrid generation method that constructs panel meshes directly from spanwise camber-line coordinates at the leading edge and trailing edge. This approach decouples the aerodynamic mesh from fixed geometric assumptions and enables the seamless integration of multiple blade components (e.g., boot, starboard foil, port foil) into a single combined aerogrid for unsteady aerodynamic analysis.

Motivation and Advantages: In traditional approaches, aerodynamic grids are often generated from explicit planform geometry (e.g., CAERO1 straight-line definitions) or from stored mesh files. For complex composite structures with multiple lifting surfaces and non-trivial spatial arrangements (such as a hydrofoil assembly composed of several distinct blades), this becomes cumbersome. The camber-line coordinate approach addresses these limitations by:

1. Leveraging existing structural analysis data: Leading-edge and trailing-edge positions are already available from the cross-sectional SONATA analysis at each spanwise station, recorded as discrete point sets in three-dimensional space.
2. Enabling smooth interpolation: By constructing interpolation curves (cubic B-splines or higher-order polynomials) through the discrete LE/TE points, the method produces smooth camber surfaces across the full blade span, avoiding artificial kinks or discontinuities.
3. Supporting geometric transformations: Dihedral rotation, chordwise offset, and other geometric modifications can be applied uniformly to entire blade assemblies without modifying the underlying mesh structure.
4. Facilitating multi-blade assembly: Multiple individual blade aerogrids can be automatically merged into a single combined aerogrid, which is then passed to the DLM solver as if it were a single lifting surface.

Discrete LE/TE Point Extraction: The starting point for aerogrid generation is a pair of arrays, $\mathbf{p}_{\mathrm{LE}}$ and $\mathbf{p}_{\mathrm{TE}}$, each containing N_s three-dimensional coordinate tuples at discrete spanwise stations along the blade. These points are extracted from SONATA's cross-sectional output files and stored as NumPy arrays in the format $\mathbf{p} \in \mathbb{R}^{N_s \times 3}$, where each row is a Cartesian coordinate $[x, y, z]$. In the present implementation, points are initially stored in millimetres (the native unit of SONATA) and are converted to metres for consistency with the structural model and aerodynamic solver.

Parametric Curve Construction: Once the discrete points are available, smooth interpolation curves are constructed using scipy's `interp1d` function with polynomial basis functions. A parametric parameter $u \in [0, 1]$ is introduced, spanning from the blade root ($u = 0$) to the blade tip ($u = 1$). The leading-edge curve is defined as:

$$\mathbf{r}_{\mathrm{LE}}(u) = [x_{\mathrm{LE}}(u), y_{\mathrm{LE}}(u), z_{\mathrm{LE}}(u)]^T, \tag{3.24}$$

with a similar definition for the trailing edge:

$$\mathbf{r}_{\mathrm{TE}}(u) = [x_{\mathrm{TE}}(u), y_{\mathrm{TE}}(u), z_{\mathrm{TE}}(u)]^T. \tag{3.25}$$

The interpolation method is chosen adaptively based on the number of available points: cubic splines are used if $N_s \geq 4$, quadratic if $N_s = 3$, and linear if $N_s = 2$. Both curves are then evaluated at a fine grid of u values to visualise the camber surfaces and to seed the panel mesh generator.

Geometric Transformations: Two important transformations are supported at the curve level to accommodate complex blade geometries:

Dihedral Rotation: When a blade exhibits spanwise dihedral (e.g., a hydrofoil winglet with a prescribed geometric sweep in the Y-Z plane), the LE and TE curves must be rotated about a fixed pivot point (typically the blade root). A rotation by angle θ_{dih} about the x-axis is implemented as:

$$y_{\mathrm{new}} = y_{\mathrm{pivot}} + (y - y_{\mathrm{pivot}}) \cos(\theta_{\mathrm{dih}}) - (z - z_{\mathrm{pivot}}) \sin(\theta_{\mathrm{dih}}), \tag{3.26}$$

$$z_{\mathrm{new}} = z_{\mathrm{pivot}} + (y - y_{\mathrm{pivot}}) \sin(\theta_{\mathrm{dih}}) + (z - z_{\mathrm{pivot}}) \cos(\theta_{\mathrm{dih}}), \tag{3.27}$$

with x and x_{pivot} unchanged. This transformation preserves the spanwise structure of the blade while adjusting its spatial orientation.

Chordwise Offset: A uniform chordwise shift (in the x-direction) can be applied to align the aerodynamic grid with the structural elastic axis or to account for differences in reference-chord location between analysis components. For each aerogrid point, the shift is:

$$x_{\text{new}} = x + \Delta x_{\text{offset}}, \tag{3.28}$$

where Δx_{offset} is a user-specified parameter (positive = toward the leading edge; negative = toward the trailing edge). This transformation is independent of spanwise position and is applied uniformly to both LE and TE curves.

Panel Mesh Generation: Once the LE and TE curves are defined and any geometric transformations applied, the panel mesh is generated by subdividing the surface into $N_{\text{span}} \times N_{\text{chord}}$ quadrilateral panels. The spanwise direction is divided into N_{span} equal segments along the parameter u, and the chordwise direction is divided into N_{chord} equal segments along the line connecting $\mathbf{r}_{\text{LE}}(u_i)$ to $\mathbf{r}_{\text{TE}}(u_i)$ at each spanwise station u_i. Each panel corner is computed by:

$$\mathbf{r}_{i,j} = (1 - \xi_j)\, \mathbf{r}_{\text{LE}}(u_i) + \xi_j\, \mathbf{r}_{\text{TE}}(u_i), \tag{3.29}$$

where $u_i \in [0, 1]$ is the i-th spanwise station and $\xi_j \in [0, 1]$ is the j-th chordwise position. This linear interpolation in the chordwise direction ensures that panels remain planar and conform to the standard CAERO1 patch convention.

Multi-Blade Aerogrid Assembly: A major capability of the present implementation is the automatic merging of multiple individual blade aerogrids into a single combined aerogrid suitable for unified DLM computation. The procedure comprises three steps:

Step 1 (Individual Grid Generation): Each blade in the assembly (e.g., boot, starboard foil, port foil) is processed independently using the parametric curve method described above. The output is an aerogrid dictionary for each blade, containing all per-panel arrays (chord lengths, normals, panel areas, control-point positions) and the global panel/corner-point IDs.

Step 2 (ID Offset and Index Remapping): To ensure global uniqueness of panel and corner-point identifiers across the merged aerogrid, all IDs from sub-grids are offset by accumulated sums from previously merged grids. Specifically, if grid i contains $n_p^{(i)}$ panels and $n_{\text{grid}}^{(i)}$ corner points, then grid $i + 1$ applies the offsets:

$$\text{ID}_{\text{panel}}^{(i+1)} \to \text{ID}_{\text{panel}}^{(i+1)} + \sum_{j=0}^{i} n_p^{(j)}, \tag{3.30}$$

$$\text{ID}_{\text{grid}}^{(i+1)} \to \text{ID}_{\text{grid}}^{(i+1)} + \sum_{j=0}^{i} n_{\text{grid}}^{(j)}, \tag{3.31}$$

similarly for degree-of-freedom (DOF) index sets that define the local to global mapping of panel control points and forcing points.

Step 3 (Array Concatenation): All per-panel and per-grid arrays from the individual aerogrid dictionaries are concatenated along their first dimension. This includes:

- Per-panel arrays: chord length $\mathbf{l}$, panel area $\mathbf{A}$, unit normal $\mathbf{N}$, quarter-chord position `offset_l`, half-chord position `offset_k`, three-quarter-chord position `offset_j`, and span-direction vector $\mathbf{r}$.
- Index sets: `set_l`, `set_k`, `set_j` (each with offset DOFs).

- Corner-point arrays: Cartesian coordinates `cornerpoint_grids` and connectivity information `cornerpoint_panels`.

The resulting combined aerogrid is structurally identical to a single-blade grid but contains the union of all geometric features from all input blades. The total panel count is $n_p^{\text{total}} = \sum_i n_p^{(i)}$, which becomes the dimension of the DLM Aerodynamic Influence Coefficient matrix $\mathbf{Q} \in \mathbb{C}^{n_p^{\text{total}} \times n_p^{\text{total}}}$.

This multi-blade assembly strategy significantly reduces code duplication, improves maintainability, and enables rapid exploration of multi-component geometric configurations without requiring manual editing of aerogrid files.

3.10.2 Essential Rules and Best Practices for Aerogrid Construction

The practical implementation of panel aerogrid generation introduces several important constraints and conventions that directly affect the quality and reliability of downstream aerodynamic calculations.

Panel Aspect Ratio Control: For each grid configuration, the aspect ratio of individual panels is determined as

$$\text{AR}_{\text{panel}} = \frac{\text{spanwise dimension}}{\text{chordwise dimension}}. \tag{3.32}$$

The DLM integration schemes (parabolic or quartic) exhibit numerical stability and accuracy over a limited range of panel aspect ratios. A general guideline is:

- *Parabolic integration* (Rodden 1971/72): maintain $\text{AR}_{\text{panel}} \lesssim 3$–$4$;
- *Quartic integration* (Rodden 1998): can tolerate $\text{AR}_{\text{panel}} \lesssim 4$–$6$.

If aspect ratios fall outside these ranges, the accuracy of the pressure distribution and aerodynamic coefficients may degrade, particularly at intermediate to high reduced frequencies.

Relationship between Gridding Parameters and Aspect Ratio: The aspect ratio of panels depends on both the planform geometry and the discretisation choices. Given a global aspect ratio $\text{AR}_{\text{global}} = \text{span}/\text{mean_chord}$, the panel aspect ratio is approximately:

$$\text{AR}_{\text{panel}} \approx \frac{N_{\text{chord}}}{N_{\text{span}}} \times \text{AR}_{\text{global}}. \tag{3.33}$$

This suggests that for a fixed planform, one can control the panel aspect ratio by varying the ratio $N_{\text{chord}}/N_{\text{span}}$. A common approach is to define the number of chordwise panels via

$$N_{\text{chord}} = N_{\text{span}} \times \text{AR}_{\text{global}}, \tag{3.34}$$

which approximately yields $\text{AR}_{\text{panel}} \approx 1$ (nearly-square panels).

Convergence and Refinement Strategy: A grid-convergence study involves running the same analysis with progressively refined panel grids. Typical patterns include:

1. Hold one parameter constant and vary the other: e.g., $N_{\text{span}} = \{5, 10, 15, 20\}$ with fixed $N_{\text{chord}} = 5$.
2. Double-grid technique: compare two grids with a known scaling ratio (e.g., $2:1$) and extract convergence metrics (flutter speed, frequency, etc.).
3. Monitor key quantities such as:
   - Total number of panels: $n_p = N_{\text{span}} \times N_{\text{chord}}$.
   - Panel area distribution (min, max, mean).

- Aerodynamic influence coefficient matrix norm $\|\mathbf{Q}(k)\|$ across the reduced-frequency range.
- Generalised aerodynamic force matrix norm in modal space.

Geometric Consistency: The leading-edge and trailing-edge curve definitions (whether via CAERO1 straight lines or from parametric curves) must be consistent with the structural mesh and the elastic axis definition. Typical failures include:

- Mismatch between the chord used in the aerodynamic grid and the chord in the structural analysis.
- Misalignment of the elastic-axis position (x_{ea}) with respect to the leading edge across the span.
- Unit inconsistencies (e.g., grid in millimetres, structural model in metres).

Control Point Locations: In the DLM framework implemented here, two key point locations are standard:

- *Downwash (or control) points* (`offset_j`): located at $\approx 75\%$ chord. These are the collocation points where the flow tangency boundary condition is enforced (see Chapter 3, Section on kinematics).
- *Force application points* (`offset_l`): located at $\approx 25\%$ chord. The aerodynamic pressure is integrated and applied at these points, which then transfer moments to the structure via the coupling matrix $\mathbf{Z}_f$ (see Chapter 3).

The physical separation between these two points (typically $\approx 50\%$ of chord) introduces an offset moment that must be accounted for in the force-transfer matrix to ensure equilibrium and energy consistency.

3.11 Hydroelastic Correction of the Aerodynamic Operator for Fluid-Structure Stability

In hydroelastic applications, the inadequacy of the thin-lifting-surface assumption employed by the Doublet Lattice Method becomes apparent. The DLM, while excellent for calculating *circulatory lift* and wake effects in low-density environments (air), treats the fluid as though it were a zero-thickness sheet and thus underestimates or ignores *volume-based hydrodynamic effects*, such as the added mass arising from the three-dimensional pressure distribution over the entire wetted surface.

To overcome this limitation while retaining the DLM's capability to model wake dynamics and unsteady circulation, a **hydroelastic correction** method is introduced [12]. This hybrid approach combines the strengths of two complementary hydrodynamic solvers:

- **Doublet Lattice Method (DLM)** via PanelAero: An efficient potential-flow solver well-suited to computing frequency-dependent circulatory forces and wake effects on slender lifting surfaces.
- **Boundary Element Method (BEM)** via Capytaine: A rigorous three-dimensional volume-based solver capable of computing the complete pressure field over the wetted surface, including frequency-dependent added-mass and radiation-damping matrices that account for the body's volume.

The method seamlessly integrates the two solvers into a unified hybrid operator that eliminates double-counting while preserving each solver's physical strengths.

3.11.1 Three-Step Hydroelastic Correction Procedure

Hydroelastic Workflow (Water Environment) For hydroelastic analysis in water, the hybrid DLM-BEM correction is essential:

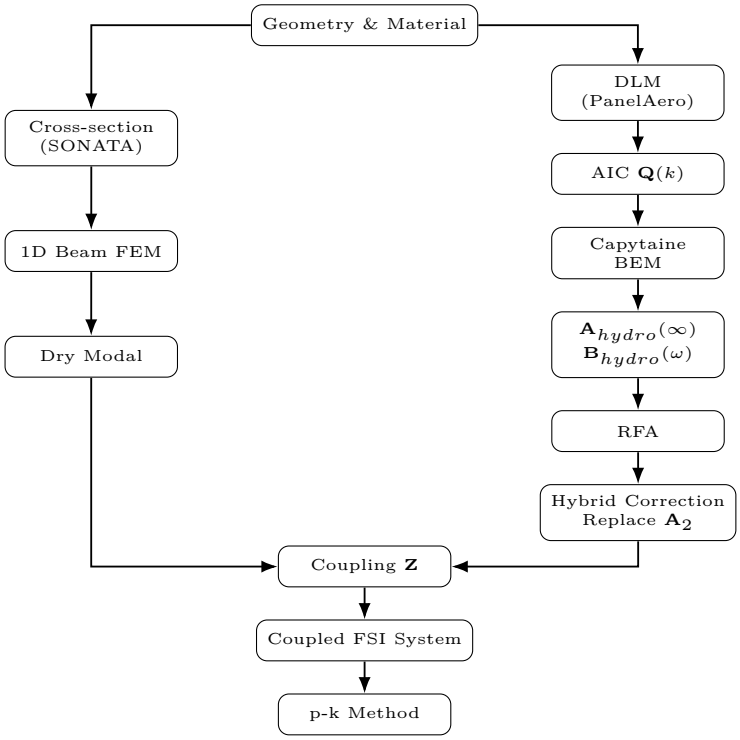


Figure 3.3: Hydroelastic analysis workflow: DLM combined with Capytaine BEM for rigorous water-based applications.

Step 1: DLM Aerodynamic Operator and Rational Function Approximation Initially, the unsteady hydrodynamic forces in the frequency domain are computed using the DLM (PanelAero). The resulting complex aerodynamic influence matrix $\mathbf{Q}(k)$ exhibits a nonlinear dependence on the reduced frequency k. For time-domain analysis and modal methods, this frequency dependence must be eliminated by fitting a *Rational Function Approximation (RFA)*.

Two RFA schemes are widely used:

1. **Roger's approximation** [13]: decomposes $\mathbf{Q}(k)$ into a quasi-steady part plus several lag terms (first-order filters).
2. **Minimum-State approximation** [14]: reduces the number of augmented states required while maintaining accuracy.

Both decompose the aerodynamic operator into physically meaningful components:

- **Quasi-steady components**: constant-coefficient matrices $\mathbf{A}_0$ (stiffness-like), $\mathbf{A}_1$ (damping-like), and $\mathbf{A}_2$ (inertia-like from the DLM). These act proportionally to displacement, velocity, and acceleration respectively, and do not depend on frequency.
- **Unsteady lag terms**: partial-fraction residuals that model the memory of the wake development. Each lag term is characterized by a specific aerodynamic pole and contributes a high-pass filtering action, becoming active only at frequencies above its pole frequency.

In particular, the matrix $\mathbf{A}_2$ derived from the DLM represents the frequency-independent limit of the added mass as frequency approaches infinity. However, this estimate is based on the thin-lifting-

surface assumption and may underestimate the true three-dimensional added mass.

Step 2: Three-Dimensional Hydrodynamic Inertia Calculation (Capytaine BEM) In parallel, the rigorous three-dimensional hydrodynamic added mass and radiation damping are computed using Capytaine, a Boundary Element Method solver.

Capytaine solves the frequency-domain potential-flow equations on the complete wetted surface, including the body's volume. For each degree of freedom (translation or rotation), the solver computes the pressure field by integrating contributions from all boundary panels, yielding:

- **Added-mass matrix** $\mathbf{A}_{\text{hydro}}(\omega) \in \mathbb{R}^{N_{\text{dof}} \times N_{\text{dof}}}$: frequency-dependent, representing the inertial reaction of the fluid to the body's acceleration. At infinite frequency ($\omega \to \infty$), the added mass converges to an asymptotic value $\mathbf{A}_{\text{hydro}}(\infty)$, which can be extracted directly from Capytaine's output.
- **Radiation-damping matrix** $\mathbf{B}_{\text{hydro}}(\omega) \in \mathbb{R}^{N_{\text{dof}} \times N_{\text{dof}}}$: frequency-dependent, representing energy dissipation due to outward-propagating hydrodynamic waves. The significance of this term depends critically on the presence of a free surface (water–air interface):
  - **With free surface**: Radiation damping includes the effects of gravitational surface waves. The body's motion generates pressure perturbations that propagate along the free surface as gravity waves, carrying away energy. This is the dominant radiation mechanism for bodies near or intersecting the free surface. The radiation-damping matrix is frequency-dependent and non-negligible across the entire frequency range of interest.
  - **Without free surface** (deeply submerged body): Radiation damping is due only to pressure waves propagating in the far field of a fully submerged body. For large distances from the free surface, the radiation-damping contribution becomes negligibly small, and can often be ignored in practical calculations.

These matrices are computed on the complete three-dimensional surface representation (not a thin lifting-surface approximation) and thus capture the full volumetric hydrodynamic effects.

3.11.2 Physical Comparison: Why DLM's $\mathbf{A}_2$ Must Be Eliminated

To understand why the $\mathbf{A}_2$ matrix from PanelAero's RFA must be discarded, it is essential to examine what each solver physically represents.

The Thin-Lifting-Surface Assumption in the DLM

The Doublet Lattice Method represents the lifting surface as a zero-thickness sheet (the **camber surface**). Mathematically, this means:

- The DLM integrates doublet and source singularities distributed only on a 2D manifold (the mean surface).
- No volume is actually occupied by the structure; it is treated as having infinite aspect ratio and zero thickness.
- When the RFA (Roger or Minimum-State) is applied to extract $\mathbf{A}_2$, it represents the inertial effect of the *infinitesimal fluid layer displaced by an infinitely thin wing.*

The asymptotic added mass $\mathbf{A}_{2,\text{DLM}}$ (the frequency-independent limit of $\mathbf{Q}(k)$ as $k \to 0$) scales as:

$$\mathbf{A}_{2,\text{DLM}} \propto \rho \cdot (\text{chord} \times \text{span}) \quad (\text{2D area only}), \tag{3.35}$$

because it only accounts for the infinitesimal fluid adjacent to the thin surface.

The Rigorous 3D Approach in Capytaine

Capytaine's BEM discretizes the **entire wetted surface** in three dimensions, including:

- The top surface of the hydrofoil.
- The bottom surface of the hydrofoil.
- The leading edge (often rounded, not a singularity).
- The trailing edge.
- Side walls and end effects if the body is not a simple 2D section.

By solving the potential-flow equations on all surfaces, Capytaine computes the pressure field in the fluid domain surrounding the entire three-dimensional body. The asymptotic added mass $\mathbf{A}_{\text{hydro}}(\infty)$ thus scales as:

$$\mathbf{A}_{\text{hydro}}(\infty) \propto \rho \cdot (\text{volume or displaced fluid volume}), \tag{3.36}$$

and inherently accounts for the body's thickness, surface curvature, and three-dimensional geometry.

Quantitative Impact: The Density-Ratio Effect

In practice, for a thick or bluff hydrofoil:

$$\mathbf{A}_{\text{hydro}}(\infty) \gg \mathbf{A}_{2,\text{DLM}}. \tag{3.37}$$

For example, consider a hydrofoil with:

- Chord $c = 0.2$ m, span $L = 1$ m.
- Thickness (average) $t = 0.05$ m.
- Volume $V \approx c \times L \times t = 0.01$ m^3.

In water ($\rho = 1000$ kg/m^3):

- DLM's added mass: $\mathbf{A}_{2,\text{DLM}} \sim 0.2 \times 1.0 \times 1000 = 200$ kg (based on surface area alone).
- Capytaine's added mass: $\mathbf{A}_{\text{hydro}} \sim V \times \rho = 0.01 \times 1000 = 10$ kg (based on effective volume), but with geometric factors that can increase this by 2–10× depending on shape.

If both matrices were naively added, the total would be artificially inflated, leading to:

$$M_{\text{total, incorrect}} = M_{\text{struct}} + 200 + 10 = M_{\text{struct}} + 210 \text{ kg (WRONG)}, \tag{3.38}$$

instead of the correct:

$$M_{\text{total, correct}} = M_{\text{struct}} + 10 \text{ kg (or higher with 3D corrections) (RIGHT)}. \tag{3.39}$$

The incorrect model would underestimate flutter speed (overpredicting instability), leading to over-conservative structural design.

Why Capytaine's Estimate is Superior

Capytaine's added mass is superior because:

1. It explicitly includes thickness effects: A thin wing has less added mass than a thick body; the DLM cannot distinguish between them.
2. It respects the body's actual geometry: Camber, sweep, dihedral, and thickness are all incorporated naturally.

3. It is computed in the time domain after frequency integration: The asymptotic limit $\mathbf{A}_{\text{hydro}}(\infty)$ is frequency-independent and thus suitable for state-space time-domain models.

Therefore, the correction strategy is: Use Capytaine's $\mathbf{A}_{\text{hydro}}(\infty)$ and discard DLM's $\mathbf{A}_2$.

Step 3: Model Hybridization and Assembly The correction is applied by selectively replacing and combining matrices from the two solvers. The key is to avoid **double-counting** while leveraging each solver's physical strengths.

The Double-Counting Problem on Added Mass ($\mathbf{A}_2$): Both Capytaine and the DLM attempt to model the same physical phenomenon: the inertial reaction of the fluid to the body's acceleration. However, they use fundamentally different approaches:

- **Capytaine (BEM)**: Solves the three-dimensional potential-flow problem rigorously on the *complete wetted surface*, including the body's thickness, volume, and actual three-dimensional shape. It computes the exact frequency-dependent added-mass matrix $\mathbf{A}_{\text{hydro}}(\omega)$ by integrating pressure over all surface panels. At infinite frequency ($\omega \to \infty$), this converges to the asymptotic added mass $\mathbf{A}_{\text{hydro}}(\infty)$, which is the true limit independent of frequency.
- **PanelAero (DLM)**: Based on thin-lifting-surface theory, PanelAero represents the wing as a flat, negligibly-thick sheet. When the Rational Function Approximation (e.g., Roger's method) is applied, the coefficient matrix $\mathbf{A}_2$ that emerges from the RFA represents precisely the DLM's estimate of added mass. Because the thin-surface approximation ignores thickness effects and assumes the body occupies zero volume, $\mathbf{A}_2$ systematically **underestimates** the true three-dimensional added mass, especially for bluff or thick bodies.

Since both matrices represent the *same physical effect*—the hydrodynamic inertia—including both would be incorrect and would artificially inflate the system's inertia. The solution is **unambiguous**: use the geometrically accurate Capytaine-derived added mass and **discard** the $\mathbf{A}_2$ matrix from PanelAero's RFA.

Corrected Added-Mass Contribution: The structural mass matrix is augmented *exclusively* with Capytaine's asymptotic added mass:

$$\mathbf{M}_{\text{total}} = \mathbf{M}_{\text{structural}} + \mathbf{A}_{\text{hydro}}(\infty), \tag{3.40}$$

where $\mathbf{M}_{\text{structural}}$ is the constant structural inertia matrix and $\mathbf{A}_{\text{hydro}}(\infty)$ is the asymptotic (frequency-independent) added-mass matrix extracted from Capytaine. The $\mathbf{A}_2$ matrix from PanelAero is entirely eliminated.

Damping Assembly: Distinguishing Physical Origins: Damping in hydroelastic systems arises from two distinct and non-overlapping physical mechanisms:

1. **Radiation damping (Capytaine)**: Energy dissipation due to outward-propagating waves generated by the oscillating body. The frequency-dependent matrix $\mathbf{B}_{\text{hydro}}(\omega)$ is computed by Capytaine and represents wave radiation losses. For a deeply submerged foil (far from the free surface), this term is negligibly small. For bodies near or piercing the water surface, $\mathbf{B}_{\text{hydro}}(\omega)$ becomes significant.
2. **Circulatory damping (PanelAero)**: Energy dissipation arising from unsteady pressure variations in the wake as the body moves. Captured by the matrix $\mathbf{A}_1$ from the DLM's RFA, this represents aerodynamic/hydrodynamic damping inherent to lift generation and shedding.

These two mechanisms measure fundamentally different physics and operate at different length scales. Radiation damping is long-range (wave propagation); circulatory damping is local (boundary-layer and trailing-wake effects). Therefore, **both must be included** without double-counting:

$$\mathbf{C}_{\text{total}} = \mathbf{C}_{\text{structural}} + \underbrace{\frac{1}{2}\rho_{\text{water}} V_\infty \mathbf{A}_1}_{\text{circulatory (DLM)}} + \underbrace{\mathbf{B}_{\text{hydro}}(\omega)}_{\text{radiation (Capytaine)}} , \tag{3.41}$$

where $\mathbf{C}_{\text{structural}}$ is internal structural damping (e.g., from material hysteresis), ρ_{water} is the fluid density, V_∞ is the flow speed, and the factor $\frac{1}{2}\rho_{\text{water}} V_\infty$ converts the DLM's dimensionless aerodynamic matrix into physical units.

Preservation of Circulatory Forces and Wake Memory: The circulatory contribution—quasi-steady aerodynamic stiffness and wake memory lag states—originates *exclusively* from the DLM and has no equivalent in Capytaine's frequency-domain BEM output. Capytaine's radiation-based approach does not model von Kármán wake vorticity, Kutta-condition physics, or the trailing-edge singularity needed to generate steady circulatory lift. Therefore, the complete circulatory operator from PanelAero must be preserved intact:

$$\mathbf{K}_{\text{total}} = \mathbf{K}_{\text{structural}} + \underbrace{\frac{1}{2}\rho_{\text{water}} V_\infty^2 \mathbf{A}_0}_{\text{quasi-steady aerodynamic stiffness (DLM)}} , \tag{3.42}$$

and all augmented lag-state equations remain unchanged, governed by the RFA poles extracted from PanelAero.

Summary: Corrected System Matrices Assembling all contributions without double-counting, and following the **PanelAero reduced-frequency convention** (which excludes the semi-chord b from the final matrices), yields the following **hybrid corrected** system matrices:

- **Total Mass (only Capytaine, no PanelAero $\mathbf{A}_2$):**

$$\boxed{\mathbf{M}_{\text{eff}} = \mathbf{M}_{\text{struct}} + \mathbf{A}_{\text{hydro}}(\infty)} \tag{3.43}$$

  This is the corrected form: **the $\mathbf{A}_2$ matrix from PanelAero is entirely discarded**. Both $\mathbf{A}_2$ (DLM's thin-wing added mass) and $\mathbf{A}_{\text{hydro}}(\infty)$ (Capytaine's 3D added mass) represent the *same physical effect*—the inertial reaction of the fluid to structural acceleration. Including both would represent double-counting. Since Capytaine's asymptotic added mass is geometrically rigorous (accounts for full 3D volume and surface curvature), it is used exclusively and $\mathbf{A}_2$ is eliminated. The final operational mass matrix contains **only Capytaine's contribution**.

- **Total Damping (radiation + circulatory):**

$$\boxed{\mathbf{C}(V_\infty, \omega) = \mathbf{C}_{\text{struct}} + \frac{1}{2}\rho V_\infty \mathbf{A}_1 + \mathbf{B}_{\text{hydro}}(\omega)} \tag{3.44}$$

  Combines three physically distinct mechanisms:

  1. **Structural damping $\mathbf{C}_{\text{struct}}$**: Internal material hysteresis and friction.
  2. **Circulatory damping from PanelAero $\frac{1}{2}\rho V_\infty \mathbf{A}_1$**: Energy dissipation in the wake due to unsteady lift generation and vortex shedding. This term is speed-dependent but independent of frequency (quasi-steady contribution from DLM's RFA).
  3. **Radiation damping from Capytaine $\mathbf{B}_{\text{hydro}}(\omega)$**: Energy dissipation through outward-propagating hydrodynamic waves generated by the oscillating body. This term is frequency-

dependent and becomes significant when the body interacts with the free surface (gravitational waves). For deeply submerged bodies far from the water surface, radiation damping is negligibly small.

These three contributions are non-overlapping physical mechanisms operating at different length scales and timescales, so all three must be included. Note that the coefficient $\frac{1}{2}\rho V_\infty$ (not $\frac{1}{2}\rho b V_\infty$) reflects PanelAero's reduced-frequency convention.

- **Total Stiffness:**

$$\boxed{\mathbf{K}(V_\infty) = \mathbf{K}_{\text{struct}} + \frac{1}{2}\rho V_\infty^2 \mathbf{A}_0} \tag{3.45}$$

Structural stiffness plus speed-dependent circulatory aerodynamic stiffness from the DLM. This form is identical to the ETH convention because the V^2 in the dynamic pressure naturally cancels the V^{-2} in the RFA polynomial, leaving no geometric scale.

- **Lag-State Dynamics (no b in denominator):**

$$\dot{\mathbf{x}}_{\text{lag},i} = -V\gamma_i \mathbf{x}_{\text{lag},i} + \mathbf{b}_i \hat{\mathbf{w}} \tag{3.46}$$

Governed exclusively by DLM's RFA poles. The time-domain pole is $-V\gamma_i$ (not $-V\gamma_i/b$): again, the semi-chord b is absent following PanelAero's convention. Here, γ_i is a dimensionless RFA pole coefficient and $\mathbf{b}_i$ is the residual vector.

This hybrid assembly ensures that each physical mechanism is captured by the most appropriate solver without duplication.

3.11.3 System Assembly and Eigenvalue Analysis

Critical Correction: Reduced-Frequency Definition in PanelAero vs. ETH

A critical distinction must be made between the formulations used in the ETH paper and the implementation with PanelAero. This difference stems from the **definition of the reduced-frequency variable** used to compute the aerodynamic force matrices, and it directly affects how the Rational Function Approximation (RFA) coefficients appear in the time-domain Equation of Motion (EOM).

The ETH Paper Formulation: In the classical aeroelastic literature (including the ETH paper), the reduced-frequency variable is defined as:

$$\bar{s}_{\text{ETH}} = \frac{Vs}{b}, \tag{3.47}$$

where s is the Laplace variable, V is the flow speed, and b is the *semi-chord* (half of the chord length). This definition includes the geometric parameter b explicitly.

When the unsteady aerodynamic forces are decomposed via Roger's Rational Function Approximation, the quasi-steady aerodynamic matrix is expressed as a polynomial in $\bar{s}_{\text{ETH}}$:

$$\Gamma_{\text{QS,ETH}} = \mathbf{A}_0 + \mathbf{A}_1 \left(\frac{Vs}{b}\right) + \mathbf{A}_2 \left(\frac{Vs}{b}\right)^2, \tag{3.48}$$

Multiplying by the dynamic pressure $q_\infty = \frac{1}{2}\rho V^2$, the physical force is:

$$\mathbf{F}_{\text{aero,ETH}} = \frac{1}{2}\rho V^2 \left[\mathbf{A}_0 + \mathbf{A}_1 \frac{Vs}{b} + \mathbf{A}_2 \left(\frac{Vs}{b}\right)^2\right]. \tag{3.49}$$

When this expression is expanded and grouped by powers of the Laplace variable s, the semi-chord b appears explicitly in the coefficients of the time-domain system matrices. This leads to the classical ETH formulation of the coupled system matrices:

$$\boxed{\begin{aligned}\mathbf{M}_{\text{ETH}} &= \mathbf{M}_{\text{struct}} - \frac{1}{2}\rho_\infty b^2 \mathbf{A}_2 \\ \mathbf{C}_{\text{ETH}}(V_\infty) &= \mathbf{C}_{\text{struct}} - \frac{1}{2}\rho_\infty b V_\infty \mathbf{A}_1 \\ \mathbf{K}_{\text{ETH}}(V_\infty) &= \mathbf{K}_{\text{struct}} - \frac{1}{2}\rho_\infty V_\infty^2 \mathbf{A}_0\end{aligned}} \tag{3.50}$$

Additionally, in the ETH formulation, the time-domain eigenvalue of the lag states is:

$$\rho_i^{\text{ETH}} = \frac{V\gamma_i}{b}, \tag{3.51}$$

where γ_i is the RFA pole (a dimensionless aerodynamic frequency) extracted from the Roger fit.

The PanelAero Formulation: PanelAero, as documented in its guide, computes aerodynamic matrices by entering with the **reduced frequency without semi-chord**:

$$k_{\text{PA}} = \frac{\omega}{V} = \frac{\Omega}{V}, \tag{3.52}$$

where ω (or Ω in some notations) is the angular frequency and V is the flow speed. There is *no dependence on the chord or semi-chord b.*

Consequently, the associated dimensionless Laplace variable is:

$$\bar{s}_{\text{PA}} = \frac{s}{V}, \tag{3.53}$$

and the aerodynamic matrix from PanelAero's RFA decomposition is expressed as:

$$\Gamma_{\text{QS,PA}} = \mathbf{A}_0 + \mathbf{A}_1 \left(\frac{s}{V}\right) + \mathbf{A}_2 \left(\frac{s}{V}\right)^2. \tag{3.54}$$

Multiplying by the dynamic pressure $q_\infty = \frac{1}{2}\rho V^2$:

$$\mathbf{F}_{\text{aero,PA}} = \frac{1}{2}\rho V^2 \left[\mathbf{A}_0 + \mathbf{A}_1 \frac{s}{V} + \mathbf{A}_2 \frac{s^2}{V^2}\right]. \tag{3.55}$$

Expanding this in the Laplace domain and then transforming to the time domain, the coefficients simplify: the V factors cancel with the dynamic pressure's V^2 term in a way that the semi-chord b never enters the final system matrices. This yields:

$$\boxed{\begin{aligned}\mathbf{M}_{\text{PA}} &= \mathbf{M}_{\text{struct}} - \frac{1}{2}\rho_\infty \mathbf{A}_2 \quad (\text{no } b^2) \\ \mathbf{C}_{\text{PA}}(V_\infty) &= \mathbf{C}_{\text{struct}} - \frac{1}{2}\rho_\infty V_\infty \mathbf{A}_1 \quad (\text{no } b) \\ \mathbf{K}_{\text{PA}}(V_\infty) &= \mathbf{K}_{\text{struct}} - \frac{1}{2}\rho_\infty V_\infty^2 \mathbf{A}_0 \quad (\text{identical to ETH})\end{aligned}} \tag{3.56}$$

Furthermore, the time-domain eigenvalue of the lag states becomes:

$$\rho_i^{\text{PA}} = V\gamma_i, \tag{3.57}$$

where the semi-chord b is *entirely removed* from the denominator.

Physical Interpretation: The key insight is that **the semi-chord b and the chord length c do not appear in the final time-domain equations when using PanelAero's reduced-frequency definition**. This is because:

1. PanelAero's reduced-frequency definition $k = \omega/V$ is *already non-dimensional* in a specific sense: it relates frequency to flow speed only, not to any geometric length scale.
2. When the RFA is performed on matrices computed at various k values, the resulting coefficient matrices $\mathbf{A}_0$, $\mathbf{A}_1$, $\mathbf{A}_2$ are implicitly scaled such that their physical units are correct *without* requiring an explicit semi-chord factor.
3. In contrast, the ETH formulation's inclusion of b in the reduced-frequency definition $\bar{s} = Vs/b$ forces b to propagate through the RFA and into the final system matrices.

For the hybrid DLM–BEM correction methodology used in the present work, the matrices derived from PanelAero must follow the PanelAero convention (Eq. 3.56), not the ETH convention. This is a critical detail that directly affects the numerical values assigned to the system matrices and, consequently, the predicted flutter speed.

Once the corrected system matrices are assembled, the complete hydroelastic equation of motion can be written in state-space form with matrices that are **frequency-independent**:

$$\mathbf{M}\ddot{\mathbf{x}} + \mathbf{C}(U_\infty)\dot{\mathbf{x}} + \mathbf{K}(U_\infty)\mathbf{x} = \mathbf{0}, \tag{3.58}$$

where the explicit dependence on flow speed U_∞ appears through the aerodynamic/hydrodynamic coupling terms but not through a nonlinear reduced-frequency dependence.

For hydroelastic stability analysis (flutter speed prediction), this system is transformed into a **state-space form** by introducing the state vector $\mathbf{z} = [\mathbf{x}, \dot{\mathbf{x}}, \mathbf{x}_{\text{lag}}]^T$, where $\mathbf{x}_{\text{lag}}$ are the lag states capturing wake memory:

$$\dot{\mathbf{z}} = \mathbf{A}_{\text{state}}(U_\infty)\mathbf{z}, \tag{3.59}$$

with a state matrix $\mathbf{A}_{\text{state}}$ that depends **linearly** on the flow speed. The system eigenvalues $\lambda = \sigma + i\omega$ are then extracted for each speed U_∞. Instability (flutter) occurs when the real part of any eigenvalue becomes positive.

This approach eliminates the need for the expensive iterative p-k method and allows direct eigenvalue tracking across the entire speed range, substantially reducing computational cost while improving numerical robustness.

3.11.4 Complete Hydroelastic Equation of Motion: Final Formulation

The complete hydroelastic equation of motion assembled from the hybrid DLM–BEM correction procedure is presented here in its final form. This equation represents the second-order linear time-invariant system governing the motion of the hydroelastic structure:

$$\big(\mathbf{M}_{\text{struct}} + \mathbf{A}_{\text{hydro}}(\omega)\big)\ddot{\mathbf{q}} + \big(\mathbf{C}_{\text{struct}} + \frac{1}{2}\rho V_\infty \mathbf{A}_1 + \mathbf{B}_{\text{hydro}}(\omega)\big)\dot{\mathbf{q}} + \big(\mathbf{K}_{\text{struct}} + \frac{1}{2}\rho V_\infty^2 \mathbf{A}_0\big)\mathbf{q} + \mathbf{F}_{\text{lag}}(t) = \mathbf{0}, \tag{3.60}$$

where:

- $\mathbf{q}(t) \in \mathbb{R}^{N_{\text{dof}}}$ is the vector of structural degrees of freedom (displacements and rotations at beam nodes).
- $\mathbf{M}_{\text{struct}} \in \mathbb{R}^{N_{\text{dof}} \times N_{\text{dof}}}$ is the structural mass matrix, computed from finite-element discretization of the beam model (constant, positive-definite).

- $\mathbf{A}_{\text{hydro}}(\omega) \in \mathbb{R}^{N_{\text{dof}} \times N_{\text{dof}}}$ is the frequency-independent (asymptotic) added-mass matrix from Capytaine at infinite frequency, representing the inertial reaction of the fluid to structural acceleration. This replaces the less-accurate $\mathbf{A}_2$ matrix from the DLM.
- $\mathbf{C}_{\text{struct}} \in \mathbb{R}^{N_{\text{dof}} \times N_{\text{dof}}}$ is the structural damping matrix (often assumed proportional or modal damping).
- $\mathbf{A}_1 \in \mathbb{R}^{N_{\text{dof}} \times N_{\text{dof}}}$ is the quasi-steady circulatory damping matrix from the DLM's Rational Function Approximation, representing energy dissipation in wake vorticity and shedding. This is a speed-dependent but frequency-independent contribution (quasi-steady).
- $\mathbf{B}_{\text{hydro}}(\omega) \in \mathbb{R}^{N_{\text{dof}} \times N_{\text{dof}}}$ is the frequency-dependent radiation damping from Capytaine, representing energy loss through hydrodynamic wave propagation. When the structure interacts with the free surface (water-air interface), this term becomes significant and includes the effects of gravitational surface waves. For deeply submerged structures, this contribution is typically small or can be neglected.
- $\mathbf{K}_{\text{struct}} \in \mathbb{R}^{N_{\text{dof}} \times N_{\text{dof}}}$ is the structural stiffness matrix (constant, positive-definite for a stable structure).
- $\mathbf{A}_0 \in \mathbb{R}^{N_{\text{dof}} \times N_{\text{dof}}}$ is the quasi-steady aerodynamic stiffness matrix from the DLM, representing the static pressure distribution and its derivative with respect to structural motion. This includes both the lifting-surface circulation effects and the slope-dependent quasi-steady contributions.
- $\mathbf{F}_{\text{lag}}(t) \in \mathbb{R}^{N_{\text{dof}}}$ are the forcing terms associated with the lag states (augmented states) that model the memory of wake development. These arise from the unsteady lag terms in the DLM's RFA decomposition.

State-Space Formulation for Eigenvalue Analysis

To perform modal flutter analysis, the second-order system (Eq. 3.60) is augmented with additional state equations for the lag states $\mathbf{x}_{\text{lag}} \in \mathbb{R}^{N_{\text{lag}}}$, where each lag state corresponds to one aerodynamic pole (frequency) in the RFA. The augmented lag-state dynamics are governed by first-order equations of the form:

$$\dot{\mathbf{x}}_{\text{lag},i} = -\rho_i \mathbf{x}_{\text{lag},i} + \mathbf{b}_i \mathbf{q}, \tag{3.61}$$

where ρ_i is the i-th aerodynamic pole frequency and $\mathbf{b}_i$ is the residual vector associated with the i-th lag term. The complete force contribution from all lag states is:

$$\mathbf{F}_{\text{lag}} = \sum_{i=1}^{N_{\text{lag}}} \mathbf{c}_i^\top \mathbf{x}_{\text{lag},i}, \tag{3.62}$$

where $\mathbf{c}_i$ are the output coefficients extracted from the RFA decomposition.

The full state vector is defined as:

$$\mathbf{z}(t) = \begin{bmatrix} \mathbf{q}(t) \\ \dot{\mathbf{q}}(t) \\ \mathbf{x}_{\text{lag},1}(t) \\ \vdots \\ \mathbf{x}_{\text{lag},N_{\text{lag}}}(t) \end{bmatrix} \in \mathbb{R}^{N_{\text{total}}} \tag{3.63}$$

with total dimension $N_{\text{total}} = N_{\text{dof}} + N_{\text{dof}} + N_{\text{dof}} \cdot N_{\text{lag}}$.

The linear state-space system is then written as:

$$\dot{\mathbf{z}}(t) = \mathbf{A}_{\text{ss}}(U_\infty)\mathbf{z}(t), \tag{3.64}$$

where the state matrix $\mathbf{A}_{\text{ss}} \in \mathbb{R}^{N_{\text{total}} \times N_{\text{total}}}$ is structured as:

$$\mathbf{A}_{\text{ss}}(U_\infty) = \begin{bmatrix} \mathbf{0}_{N_{\text{dof}} \times N_{\text{dof}}} & \mathbf{I}_{N_{\text{dof}} \times N_{\text{dof}}} & \mathbf{0}_{N_{\text{dof}} \times N_{\text{lag}} N_{\text{dof}}} \\ -\mathbf{M}^{-1}\mathbf{K} & -\mathbf{M}^{-1}\mathbf{C} & -\mathbf{M}^{-1}\mathbf{L} \\ \mathbf{B}_1 & \mathbf{0} & \mathbf{D}_1 \\ \vdots & \vdots & \vdots \\ \mathbf{B}_{N_{\text{lag}}} & \mathbf{0} & \mathbf{D}_{N_{\text{lag}}} \end{bmatrix}, \tag{3.65}$$

where $\mathbf{K} = \mathbf{K}_{\text{struct}} + U_\infty^2 \mathbf{A}_0$ (flow-speed-dependent stiffness), $\mathbf{C} = \mathbf{C}_{\text{struct}} + \mathbf{A}_1 + \mathbf{B}_{\text{hydro,approx}}$ (frequency-independent total damping), and $\mathbf{M} = \mathbf{M}_{\text{struct}} + \mathbf{A}_{\text{hydro}}(\infty)$ (frequency-independent effective mass).

The matrices $\mathbf{B}_i$, $\mathbf{D}_i$, and $\mathbf{L}$ are the RFA coefficient matrices arising from the decomposition of $\mathbf{Q}(k)$.

Stability and Flutter analysis for Aeroelastic and Hydroelastic systems

The following chapter describes the implementation of the p-k method for flutter analysis, which is used to determine the stability of the coupled fluid-structure system as flow velocity increases. And the RFA approximation method usefull to solve the nonlinear eigenvalue problem of the pk method in a more efficient way, by approximating the frequency-dependent aerodynamic influence matrices with rational functions, thus enabling the use of linear eigenvalue solvers and significantly reducing computational cost while maintaining accuracy across the relevant frequency range. The second method is necessary and particularly beneficial for hydroelastic applications.

4.1 Aeroelastic: the PK-method

The pk method solution process was described in detail by Bellinger et al. (2017) and Rodden et al. (2017). To improve understanding and aid discussions, the solution process is presented in this work.

Equation 3.20 is rewritten in a canonical first order state space form:

$$\left[\mathbf{A} - p\mathbf{I}\right] \begin{Bmatrix} \hat{q} \\ \hat{\dot{q}} \end{Bmatrix} = \mathbf{0}, \tag{4.1}$$

where $[\mathbf{A}]$ is the double-sized real matrix:

$$\mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{M}_{hh}^{-1}\left(\mathbf{K}_{hh} - \frac{1}{2}\rho U_\infty^2 \mathbf{Q}_{hh}^R\right) & -\mathbf{M}_{hh}^{-1}\left(\mathbf{B}_{hh} - \frac{1}{4}\rho c U_\infty \mathbf{Q}_{hh}^I/k\right) \end{bmatrix}. \tag{4.2}$$

Because the matrix $\mathbf{A}$ depends on the reduced frequency k, which in turn depends on the eigenvalue p we are trying to solve for, the problem is nonlinear in p, it must be solved iteratively. The pk-method solves this interdependent problem by iteratively solving the eigenvalue problem for $\mathbf{A}$ assuming an initial guess for k. The eigenvalues are then used to update the reduced frequency k and the aerodynamic influence matrix $\mathbf{Q}$, which in turn updates the matrix $\mathbf{A}$. This process is repeated until convergence, which is defined as the change in eigenvalues between iterations being below a certain threshold.

The two nested loops of the pk method are:

- an **Outer loop** that iterates over a list of consecutive flight parameters, enabling tracking of the flutter curve as flow velocity U_∞, Mach number and altitude (which determines fluid density ρ) vary.
- an **Inner loop** that for each flight point analyzes stability proceeding mode-by-mode until convergence.

The innermost loop, for a vibrational mode:

- Formulates the equation of motion for each mode with the current reduced frequency estimate

The iterative procedure is schematically represented in the flowchart at figure 4.1.

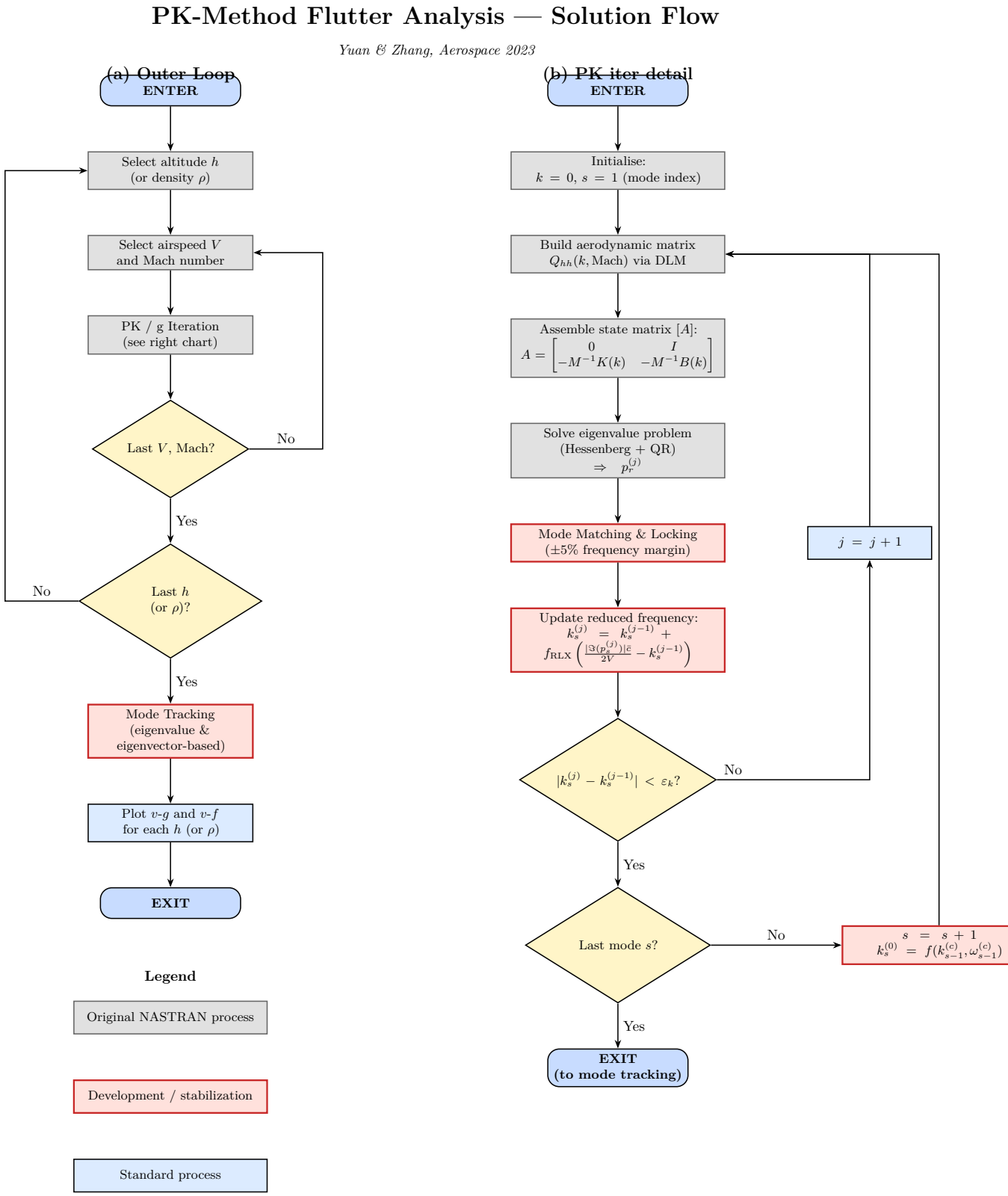


Figure 4.1: Schematic flowchart of the PK-method flutter analysis

4.1.1 Outer loop: sweep over flight conditions

The outermost level of the algorithm iterates over a discrete set of flight conditions $\{V_i\}_{i=1}^{N_V}$, where each point is characterised by the free-stream velocity V_i, the corresponding Mach number Ma_i, and the fluid density ρ (which depends on the operating altitude in aeronautical applications, or on the fluid properties in hydroelastic ones). At each velocity step i the complete set of N_m aeroelastic modes is re-analysed through the inner loop described below, and the converged eigenvalues $p_s^{(i)} = \sigma_s^{(i)} + \mathrm{i}\,\omega_s^{(i)}$, $s = 1, \ldots, N_m$, are stored. The evolution of σ_s and ω_s with V constitutes the classical v-g and v-f diagrams, respectively, from which the flutter speed is identified as the lowest V for which any σ_s crosses zero from below.

To improve the robustness of the inner PK iteration at each new velocity point, the converged reduced frequency of mode $s = 1$ from the previous step is carried forward as the initial guess k_0 for that mode at the current step:

$$k_0^{(i)} \;=\; k_1^{(i-1)}, \qquad i \geq 2, \tag{4.3}$$

where $k_s^{(i)}$ denotes the converged reduced frequency of mode s at velocity step i. At the very first velocity point ($i = 1$), a near-zero sentinel value $k_0^{(1)} = 10^{-3}$ is used, which corresponds to the quasi-steady limit of the aerodynamic operator.

4.1.2 Inner loop: mode-by-mode PK iteration

For a fixed flight condition V_i, the inner loop processes the structural modes sequentially, from $s = 1$ to $s = N_m$. For each mode s, a fixed-point iteration is performed to find the value of the reduced frequency k that is self-consistent with the imaginary part of the corresponding eigenvalue p_s.

Step 1 — Initial guess for the reduced frequency. The initial guess $k^{(0)}$ for mode s at velocity step i is obtained through a *hybrid scheme* that blends two physically motivated estimates. For the first mode ($s = 1$) the guess is directly taken from the outer-loop carry-over $k^{(0)} = k_0^{(i)}$ defined in Eq. (4.3). For subsequent modes ($s > 1$), a weighted combination of two contributions is used:

- k_{base}: the frequency of mode s at the previous velocity step V_{i-1}, normalised by the current velocity,

$$k_{\mathrm{base}} = \frac{\omega_s^{(i-1)}}{V_i}; \tag{4.4}$$

- k_{conv}: the converged reduced frequency of the immediately preceding mode $s-1$ at the current velocity step,

$$k_{\mathrm{conv}} = \frac{\omega_{s-1}^{(i)}}{V_i}. \tag{4.5}$$

The hybrid initial guess is then

$$k^{(0)} = k_{\mathrm{base}} + f_{\mathrm{XK0}}\left(k_{\mathrm{conv}} - k_{\mathrm{base}}\right), \tag{4.6}$$

where $f_{\mathrm{XK0}} \approx 0.618$ is a blending factor (the golden-section ratio), chosen to ensure a smooth transition between the two estimates. The rationale is that adjacent structural modes tend to have similar reduced frequencies at a given flight speed, so the converged value of mode $s-1$ provides an excellent predictor for mode s.

Step 2 — Eigenvalue prediction by linear extrapolation. Starting from the third velocity step ($i \geq 3$), a linear extrapolation from the two most recently converged eigenvalues of mode s is

used to obtain a *predicted* complex eigenvalue $\hat{p}_s^{(i)}$:

$$\hat{p}_s^{(i)} = 2\,p_s^{(i-1)} - p_s^{(i-2)}. \tag{4.7}$$

This prediction serves a dual purpose: it provides the starting point p_{prev} for the first iteration of the inner loop ($j = 0$), thereby reducing the number of iterations required to reach convergence, and it acts as a reference for the mode tracking/locking procedure described below. When only one previous velocity step is available ($i = 2$), the single previous eigenvalue $p_s^{(i-1)}$ is used directly as the starting guess.

Step 3 — Construction of the aerodynamic influence matrix. At each PK iteration j, the aerodynamic generalised-force matrix $\mathbf{Q}_{hh}(k^{(j)})$ is retrieved for the current value of the reduced frequency. The reduced frequency is defined as:

$$k^{(j)} = \frac{\omega^{(j)}\,b}{V_i}, \tag{4.8}$$

where b is the reference semi-chord and $\omega^{(j)} = \left|\mathrm{Im}(p_s^{(j)})\right|$ is the oscillation frequency extracted from the eigenvalue at the previous iteration. At the very first inner iteration ($j = 0$), the matrix is evaluated at the initial guess $k^{(0)}$ defined in Step 1.

Physically, k encodes the ratio between the time scale of the structural oscillation and the convective time scale of the flow over the chord. A value $k \to 0$ corresponds to the quasi-steady aerodynamic limit, whereas larger values of k introduce progressively stronger unsteady (circulatory and non-circulatory) effects captured by the DLM kernel.

Step 4 — Assembly and solution of the state-space eigenvalue problem. With $\mathbf{Q}_{hh}(k^{(j)})$ evaluated at the current reduced frequency, the aeroelastic state matrix $\mathbf{A}(k^{(j)})$ is assembled as in Eq. (3.20), and the full-spectrum eigenvalue problem

$$\mathbf{A}(k^{(j)})\,\mathbf{x} = p\,\mathbf{x} \tag{4.9}$$

is solved, yielding $2N_m$ complex roots. Only roots with non-negative imaginary part are retained on physical grounds (oscillatory solutions), and spurious near-zero roots arising from numerical artefacts of the state-space formulation are discarded by imposing the lower bound $|\omega| \geq 0.1\,\omega_{n,1}$, where $\omega_{n,1}$ is the lowest structural natural frequency. The retained eigenvalues are sorted in ascending order of $|\mathrm{Im}(p)|$.

Step 5 — Mode tracking and locking. Because the eigenvalue problem is solved anew at each iteration and for each velocity step, a robust mechanism is required to associate each computed root with the correct structural mode. This is accomplished through a combined frequency-window and Modal Assurance Criterion (MAC) strategy, implemented in the `mode_matching_paper_compliant` routine.

A frequency window centred on a reference oscillation frequency ω_{ref} is first established, with half-width controlled by the parameter f_{margin}:

$$\omega_{\text{ref}} \in \left[\,(1 - f_{\text{margin}})\,\omega_{\text{ref}},\ (1 + f_{\text{margin}})\,\omega_{\text{ref}}\,\right]. \tag{4.10}$$

The reference ω_{ref} is chosen according to the following priority:

1. *MAC pre-selector (highest priority)*: if an eigenvector reference $\mathbf{u}_{\text{ref}}$ is available — either from the converged eigenvector of mode s at the previous velocity step (cross-velocity MAC),

or from the eigenvector selected at the previous PK iteration within the same velocity step (intra-velocity MAC) — the candidate with the highest MAC value is identified first. Its frequency then defines ω_{ref}.

2. *Converged frequency from the previous velocity step*: when no MAC reference is yet available (first PK iteration at a new velocity point), $\omega_{\text{ref}} = \omega_s^{(i-1)}$.
3. *Structural natural frequency*: at the very first velocity step ($i = 1$, $j = 0$), before any aeroelastic solution is available, the mode is initialised by selecting the eigenvalue closest to the in-vacuo natural frequency $\omega_{n,s}$.

Among all candidates inside the frequency window, the best match is selected by minimising the combined score

$$\mathcal{S}_{j_c} = w_{\text{freq}} \frac{|\omega_{j_c} - \omega_{\text{ref}}|}{\omega_{\text{ref}}} + w_{\text{MAC}} \left(1 - \text{MAC}(\mathbf{u}_{j_c}, \mathbf{u}_{\text{ref}})\right), \tag{4.11}$$

where w_{freq} and w_{MAC} are user-defined weights and the MAC is computed as

$$\text{MAC}(\mathbf{u}_1, \mathbf{u}_2) = \frac{|\mathbf{u}_1^H \mathbf{u}_2|^2}{(\mathbf{u}_1^H \mathbf{u}_1)(\mathbf{u}_2^H \mathbf{u}_2)}. \tag{4.12}$$

If the MAC of the selected candidate falls below the threshold MAC < 0.7, a fallback selection based on proximity to the predicted eigenvalue $\hat{p}_s^{(i)}$ (Eq. (4.7)) is invoked, using a quadratic extrapolation from the three most recent converged values of p_s. If no candidate is found within the (possibly expanded) frequency window, the mode is declared *lost* and a `NaN` entry is recorded for that velocity step and all subsequent ones.

Step 6 — Under-relaxed update of the reduced frequency. Once the eigenvalue $p_s^{(j)}$ for mode s has been selected and verified, the reduced frequency is updated using a first-order under-relaxation (deferred-correction) scheme:

$$k^{(j+1)} = k^{(j)} + f_{\text{RLX}} \left(\frac{|\text{Im}(p_s^{(j)})|\, b}{V_i} - k^{(j)} \right), \tag{4.13}$$

where $f_{\text{RLX}} \approx 0.618$ is the relaxation factor. This formulation drives k towards the value $k_{\text{target}} = \omega_s^{(j)}\, b/V_i$ implied by the current eigenvalue, while the relaxation prevents oscillatory behaviour of the iteration. The factor $f_{\text{RLX}} = 0.618$ was adopted from the golden-section ratio, a heuristic choice that provides a robust balance between convergence speed and stability.

Step 7 — Convergence criterion. Convergence of the PK iteration for mode s is declared when the change in reduced frequency between two successive iterations falls below a prescribed tolerance ε_k:

$$|k^{(j+1)} - k^{(j)}| < \varepsilon_k. \tag{4.14}$$

Upon convergence, the damping ratio γ_s is extracted from the converged eigenvalue as

$$\gamma_s = \frac{\sigma_s}{\omega_s}, \tag{4.15}$$

and the full set $\{p_s, \omega_s, \sigma_s, \gamma_s, k_s\}$ is stored for the current velocity point. If convergence is not achieved within the maximum number of iterations $j_{\max}$, the last iterate is recorded with a non-convergence flag.

4.1.3 Summary of the iterative procedure

Table 4.1 summarises the key quantities involved in the PK iteration and their role in the algorithm.

Table 4.1: Key quantities of the PK iterative method.

Symbol	Definition	Role
$k^{(j)}$	$\lvert\mathrm{Im}(p^{(j-1)})\rvert\, b/V_i$	reduced frequency at iteration j
$\mathbf{Q}_{hh}(k^{(j)})$	DLM influence matrix at $k^{(j)}$	aerodynamic forcing
$\mathbf{A}(k^{(j)})$	aeroelastic state matrix	eigenvalue problem
$p_s^{(j)}$	selected eigenvalue of mode s	updated stability root
$k^{(j+1)}$	relaxed update, Eq. (4.13)	next iteration input
ε_k	convergence threshold	termination condition

The self-consistent nature of the problem is apparent from the chain $k^{(j)} \to \mathbf{Q}_{hh} \to \mathbf{A} \to p_s^{(j)} \to k^{(j+1)}$: the aerodynamic operator depends on the oscillation frequency of the very mode it is used to predict. The PK iteration converges when this chain is closed, i.e. when the reduced frequency assumed to evaluate the aerodynamic matrix equals the one implied by the resulting eigenvalue.

4.1.4 Physical-mathematical interpretation

The eigenvalues p, which are solutions of the equation of motion, can be written in the form:

$$p = \sigma + i\omega, \tag{4.16}$$

where σ is the real part representing the growth or damping rate of the oscillation, and ω is the imaginary part representing the frequency of oscillation.

The eigenvalues that satisfy this equation provide information on the aeroelastic stability of the system. Recalling the harmonic time formulation $q(t) = \hat{q}e^{pt} = \hat{q}e^{\sigma t}e^{i\omega t}$, If the real part of p is positive, the system is unstable (flutter) because the response grows exponentially in time, while if it is negative, the system is stable and the response decays exponentially in time.

The flutter speed and frequency are determined by the eigenvalues p that approach the imaginary axis (i.e., $\sigma \approx 0$) as the flow velocity U_∞ varies.

4.2 Hydroelastic: RFA and state-space formulation

The treatment of stability and flutter analysis for hydroelastic systems (water environment) differs fundamentally from the aeroelastic p-k method, particularly when the Rational Function Approximation (RFA) is applied with the hybrid DLM–BEM correction (as described in Chapter 3). This section presents the streamlined approach applicable to hydroelastic systems where the RFA has already been fitted to the aerodynamic/hydrodynamic influence matrix.

The unsteady aerodynamic (or hydrodynamic) influence matrix $\mathbf{Q}(k)$ derived from methods such as the Doublet Lattice Method (DLM) exhibits a complex dependence on the reduced frequency k. For time-domain simulations, modal flutter analysis, and state-space formulations, this frequency dependence must be eliminated by fitting a **Rational Function Approximation (RFA)**. This section presents the theory and practical implementation of RFA schemes.

4.2.1 Motivation for Rational Function Approximation

The aerodynamic/hydrodynamic influence matrix computed in the frequency domain is:

$$\mathbf{Q}(k,\,\mathrm{Ma}) \in \mathbb{C}^{n_p \times n_p}, \tag{4.17}$$

where $k = \omega b/U_\infty$ is the reduced frequency, ω is the circular frequency, b is the reference semi-chord, U_∞ is the flow speed, and Ma is the Mach number (or fluid properties in the hydroelastic case).

When the equation of motion is cast in modal space and solved using eigenvalue analysis or time-integration schemes, the frequency dependence of $\mathbf{Q}(k)$ introduces a significant complication: the aerodynamic forces depend nonlinearly on the oscillation frequency, which is itself an unknown eigenvalue. This leads to a **nonlinear eigenvalue problem** that must be solved iteratively (as in the p-k method) unless the frequency dependence is eliminated.

The Rational Function Approximation overcomes this by decomposing $\mathbf{Q}(k)$ into a combination of:

1. **Quasi-steady components**: constant-coefficient matrices that do not depend on frequency.
2. **Lag terms (augmented states)**: first-order dynamic systems that model the memory of wake development and unsteady pressure propagation.

This decomposition transforms the original nonlinear frequency-dependent problem into a linear state-space form with a finite number of augmented states, enabling direct eigenvalue extraction without iteration over the reduced frequency.

4.2.2 Roger's Rational Function Approximation

Roger's approximation [13] is one of the most widely used RFA schemes in aeroelastic analysis. It decomposes the complex aerodynamic matrix as a polynomial in the Laplace variable s, followed by approximation of the frequency-dependent lag behavior.

For the PanelAero reduced-frequency convention (where $k = \omega/V$ without the semi-chord), the aerodynamic force matrix in the Laplace domain is approximated as:

$$\mathbf{F}_{\text{aero}}(s) = \frac{1}{2}\rho V^2 \left[\mathbf{A}_0 + \mathbf{A}_1 \frac{s}{V} + \mathbf{A}_2 \frac{s^2}{V^2} + \sum_{i=1}^{N_{\text{lag}}} \frac{\mathbf{A}_{3,i}}{1 + \tau_i s/V} \right] \mathbf{w}, \tag{4.18}$$

where:

- $\mathbf{A}_0 \in \mathbb{R}^{n_p \times n_p}$ is the **quasi-steady stiffness matrix**, capturing the static pressure distribution and its dependence on structural motion. It is frequency-independent and contributes a stiffness-like effect proportional to displacement.
- $\mathbf{A}_1 \in \mathbb{R}^{n_p \times n_p}$ is the **quasi-steady damping matrix**, representing energy dissipation due to wake vorticity and circulatory effects. It contributes a damping-like effect proportional to velocity.
- $\mathbf{A}_2 \in \mathbb{R}^{n_p \times n_p}$ is the **frequency-independent added-mass matrix** (the asymptotic limit of the aerodynamic operator as frequency $\to \infty$). It contributes an inertia-like effect proportional to acceleration.

  Critical Note for Hydroelastic Applications: In hydroelastic systems, the $\mathbf{A}_2$ matrix derived from the DLM (thin-lifting-surface method) represents only the infinitesimal fluid layer adjacent to the wing surface and systematically underestimates the true three-dimensional added mass. For hydroelastic analysis, this term is typically replaced or corrected using a rigorous Boundary Element Method (e.g., Capytaine) that accounts for the full body volume and surface curvature. The aerodynamic $\mathbf{A}_2$ should then be discarded in favor of the BEM-derived added-mass matrix.
- $\mathbf{A}_{3,i}$ are the **lag-term residual matrices**, each associated with an aerodynamic pole frequency.
- τ_i are the **lag constants** (time scales), controlling the frequency range over which each lag term is active. Larger τ_i corresponds to lower pole frequencies.
- N_{lag} is the number of lag terms (typically 2–4 for good accuracy).

4.2.3 Time-Domain Augmented State-Space Formulation

To eliminate the frequency dependence and construct a linear state-space system, each lag term is expanded into a first-order differential equation. For the i-th lag term with time-constant τ_i, define the augmented state variable:

$$\dot{\mathbf{x}}_i(t) = -\frac{V}{\tau_i}\mathbf{x}_i(t) + \frac{1}{\tau_i}\mathbf{w}(t), \tag{4.19}$$

where the pole frequency in physical units is $\rho_i = V/\tau_i$.

The contribution of this lag term to the aerodynamic force is:

$$\mathbf{f}_{i,\text{lag}}(t) = \frac{1}{2}\rho V^2 \mathbf{A}_{3,i}\mathbf{x}_i(t). \tag{4.20}$$

Summing over all lag terms and combining with the quasi-steady contributions, the total aerodynamic force becomes:

$$\mathbf{f}_{\text{aero}}(t) = \frac{1}{2}\rho V^2 \left[\mathbf{A}_0\mathbf{w}(t) + \mathbf{A}_1\dot{\mathbf{w}}(t) + \mathbf{A}_2\ddot{\mathbf{w}}(t)\right] + \sum_{i=1}^{N_{\text{lag}}} \frac{1}{2}\rho V^2 \mathbf{A}_{3,i}\mathbf{x}_i(t), \tag{4.21}$$

which is now purely time-domain with constant coefficients.

4.2.4 System Matrix Assembly with RFA

The structural equation of motion, augmented with the RFA lag states, becomes:

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}(V)\dot{\mathbf{q}} + \mathbf{K}(V)\mathbf{q} = \mathbf{F}_{\text{aero}}(\mathbf{q}, \dot{\mathbf{q}}, V), \tag{4.22}$$

where the flow-speed-dependent stiffness and damping are:

$$\mathbf{K}(V) = \mathbf{K}_{\text{struct}} + \frac{1}{2}\rho V^2 \mathbf{A}_0, \tag{4.23}$$

$$\mathbf{C}(V) = \mathbf{C}_{\text{struct}} + \frac{1}{2}\rho V \mathbf{A}_1. \tag{4.24}$$

The mass matrix must account for the aerodynamic added mass. For aeroelastic applications:

$$\mathbf{M} = \mathbf{M}_{\text{struct}} + \frac{1}{2}\rho b^2 \mathbf{A}_2 \quad \text{(ETH convention with semi-chord)}, \tag{4.25}$$

or (with PanelAero convention, no semi-chord):

$$\mathbf{M} = \mathbf{M}_{\text{struct}} + \frac{1}{2}\rho \mathbf{A}_2. \tag{4.26}$$

For hydroelastic applications, the $\mathbf{A}_2$ term is replaced by the rigorously computed added-mass matrix from the Boundary Element Method (BEM):

$$\mathbf{M} = \mathbf{M}_{\text{struct}} + \mathbf{A}_{\text{hydro}}(\infty), \tag{4.27}$$

where $\mathbf{A}_{\text{hydro}}(\infty)$ is the frequency-independent (asymptotic) added mass from Capytaine or similar BEM solver.

4.2.5 Direct Eigenvalue Extraction without Iteration

With the RFA lag states included, the system can be written in augmented state-space form:

$$\dot{\mathbf{z}}(t) = \mathbf{A}_{\text{ss}}(V)\mathbf{z}(t), \tag{4.28}$$

where the state vector is:

$$\mathbf{z}(t) = \begin{bmatrix} \mathbf{q}(t) \\ \dot{\mathbf{q}}(t) \\ \mathbf{x}_1(t) \\ \vdots \\ \mathbf{x}_{N_{\text{lag}}}(t) \end{bmatrix}. \tag{4.29}$$

The state matrix $\mathbf{A}_{\text{ss}}(V)$ depends **linearly** on the flow speed V (through the V and V^2 terms in stiffness and damping). The eigenvalues $\lambda = \sigma + i\omega$ are extracted directly via standard linear algebra, with no iteration over the reduced frequency.

This is in contrast to the p-k method, which requires nested iteration loops to find the self-consistent reduced frequency. The RFA approach is thus computationally more efficient and numerically more robust.

4.2.6 State-Space Formulation for Eigenvalue Analysis

When the RFA coefficients $\mathbf{A}_0$, $\mathbf{A}_1$, $\mathbf{A}_2$ (or replaced by BEM-derived added mass for hydroelastic), and lag-term residuals $\mathbf{A}_{3,i}$ have been determined, the complete hydroelastic equation of motion is formulated as a linear state-space system:

$$\dot{\mathbf{z}}(t) = \mathbf{A}_{\text{ss}}(U_\infty)\mathbf{z}(t), \tag{4.30}$$

where the state vector is:

$$\mathbf{z}(t) = \begin{bmatrix} \mathbf{q}(t) \\ \dot{\mathbf{q}}(t) \\ \mathbf{x}_{\text{lag},1}(t) \\ \vdots \\ \mathbf{x}_{\text{lag},N_{\text{lag}}}(t) \end{bmatrix} \in \mathbb{R}^{N_{\text{total}}}, \tag{4.31}$$

with total dimension $N_{\text{total}} = N_{\text{dof}}(1 + 1 + N_{\text{lag}})$.
The state matrix $\mathbf{A}_{\text{ss}}(U_\infty) \in \mathbb{R}^{N_{\text{total}} \times N_{\text{total}}}$ is structured as:

$$\mathbf{A}_{\text{ss}}(U_\infty) = \begin{bmatrix} \mathbf{0}_{N_{\text{dof}} \times N_{\text{dof}}} & \mathbf{I}_{N_{\text{dof}} \times N_{\text{dof}}} & \mathbf{0}_{N_{\text{dof}} \times (N_{\text{lag}} N_{\text{dof}})} \\ -\mathbf{M}^{-1}\mathbf{K}(U_\infty) & -\mathbf{M}^{-1}\mathbf{C}(U_\infty) & -\mathbf{M}^{-1}\mathbf{L} \\ \mathbf{B}_1 & \mathbf{0} & \mathbf{D}_1 \\ \vdots & \vdots & \vdots \\ \mathbf{B}_{N_{\text{lag}}} & \mathbf{0} & \mathbf{D}_{N_{\text{lag}}} \end{bmatrix}, \tag{4.32}$$

where:

- $\mathbf{M} = \mathbf{M}_{\text{struct}} + \mathbf{A}_{\text{hydro}}(\infty)$ is the frequency-independent effective mass matrix (structural mass plus asymptotic added mass from BEM).
- $\mathbf{K}(U_\infty) = \mathbf{K}_{\text{struct}} + \frac{1}{2}\rho U_\infty^2 \mathbf{A}_0$ is the flow-speed-dependent stiffness matrix.
- $\mathbf{C}(U_\infty) = \mathbf{C}_{\text{struct}} + \frac{1}{2}\rho U_\infty \mathbf{A}_1 + \mathbf{B}_{\text{hydro,approx}}$ is the total damping matrix, including structural damping, circulatory (DLM) damping, and radiation damping from the BEM.
- $\mathbf{L}$ is the matrix relating lag-state contributions to structural acceleration.
- $\mathbf{B}_i$, $\mathbf{D}_i$ are the lag-term coefficient matrices from the RFA decomposition, defining the dynamics of the i-th augmented state.

A critical advantage of this formulation is that $\mathbf{A}_{\text{ss}}(U_\infty)$ depends **linearly** on the flow speed (through the U_∞ and U_∞^2 terms in stiffness and damping), avoiding the nonlinear reduced-frequency

dependence that characterizes the p-k method.

4.2.7 Direct Eigenvalue Extraction and Flutter Speed Identification

Stability of the hydroelastic system is assessed by computing the eigenvalues of $\mathbf{A}_{\text{ss}}(U_\infty)$ at each flow speed:

$$\det(\mathbf{A}_{\text{ss}}(U_\infty) - \lambda \mathbf{I}) = 0, \tag{4.33}$$

yielding N_{total} eigenvalues $\lambda_j(U_\infty) = \sigma_j(U_\infty) + i\,\omega_j(U_\infty)$, where:

- $\sigma_j(U_\infty)$ is the real part, determining the decay rate ($\sigma < 0$) or growth rate ($\sigma > 0$) of each mode.
- $\omega_j(U_\infty)$ is the imaginary part, representing the oscillation frequency.

The **flutter speed** is identified as the minimum flow speed U_f at which any eigenvalue crosses the imaginary axis, i.e., where $\sigma_j(U_f) = 0$ for at least one mode j. Beyond this speed, $\sigma_j(U) > 0$ and the system becomes unstable.

The complete stability picture is visualized through the **root-locus plot** (or flutter diagram), which tracks the evolution of poles (eigenvalues) in the complex λ-plane as U_∞ increases from zero to well beyond the flutter speed.

4.2.8 Computational Advantages over the p-k Method

The RFA–based hydroelastic analysis provides several significant advantages:

1. **Eliminates nonlinear iteration:** Unlike the p-k method, which requires a nested iteration loop to find the self-consistent reduced frequency, the RFA transforms the problem into a linear eigenvalue problem. Convergence is achieved in a single solve per speed, not through iterative refinement.
2. **Improved numerical robustness:** Eigenvalue extraction from a fixed-size linear system is a well-conditioned operation using standard algorithms (QR, Hessenberg reduction). No convergence criteria or relaxation factors are needed.
3. **Reduced computational cost:** For N_V speeds evaluated, the p-k method requires $N_V \times (N_{\text{modes}} \times N_{\text{iterations}})$ aerodynamic matrix evaluations. The RFA method requires only one precomputed set of RFA coefficients, regardless of N_V. The online cost is dominated by eigenvalue extraction at each speed, which scales as $\mathcal{O}(N_{\text{total}}^3)$ but with a very small constant factor due to the structure of $\mathbf{A}_{\text{ss}}$.
4. **Enables parametric studies:** Multiple flow speeds, fluid densities, and geometric configurations can be analyzed rapidly without re-fitting the RFA, supporting efficient design exploration and sensitivity analysis.

4.2.9 Physical Interpretation of Hydroelastic Stability

In hydroelastic systems, the combination of added mass (often larger than structural mass due to the high density of water) and radiation damping (from free-surface wave effects) fundamentally alters the flutter mechanism compared to air-based applications. The RFA–based approach naturally captures these effects:

- **Added-mass dominance:** The asymptotic added mass $\mathbf{A}_{\text{hydro}}(\infty)$ from the BEM is frequency-independent and directly augments the structural mass, shifting frequencies downward and typically stabilizing low-frequency modes.

- **Radiation damping:** When the structure interacts with the free surface (water–air interface), radiation damping $\mathbf{B}_{\text{hydro}}(\omega)$ becomes significant and often provides inherent stabilization by dissipating energy through gravitational surface waves.
- **Circulatory effects:** The quasi-steady circulatory stiffness and damping from the DLM ($\mathbf{A}_0$ and $\mathbf{A}_1$) remain the primary destabilizing mechanism, analogous to aeroelastic flutter. Their relative importance is reduced in hydroelastic systems due to the added-mass effects, leading to higher flutter speeds and lower flutter frequencies compared to air-based counterparts.

The RFA–based eigenvalue approach makes these physical mechanisms transparent and enables rapid assessment of stability margins and the influence of design parameters on the flutter boundary.

Verification and Validation

The verification and validation of the entire framework has been performed through a systematic workflow that includes both sensitivity analyses and benchmark comparisons. The primary goal is to ensure that the coupled aeroelastic solver accurately captures the physical mechanisms of flutter and produces results consistent with established references. Then the same analyses are performed for the hydroelastic framework.

5.1 Aeroelastic Validation and Verification workflow

The verification and validation of the aeroelastic framework is performed in two stages.

- **First**, a sensitivity analysis is carried out to investigate the behaviour of the aerodynamic stiffness and damping matrices as a function of flow speed and reduced frequency. This analysis provides insight into the physical mechanisms driving aeroelastic instability.
- **Subsequently**, the framework is validated against the classical *Goland wing* flutter benchmark, a simple rectangular metallic cantilever wing that has become the standard test case for coupled bending–torsion flutter in the aeroelastic community since its original formulation by Martin Goland in 1945 [15].

5.1.1 Aerodynamic Sensitivity Analysis

Before proceeding to the full aeroelastic analysis on the Goland wing, a parametric sensitivity study is conducted on a simplified test case to characterize the behaviour of the aerodynamic stiffness matrix $\mathbf{K}_{\text{aero}}$ and damping matrix $\mathbf{C}_{\text{aero}}$ as functions of velocity and reduced frequency. This analysis is essential for understanding how aerodynamic forces change across the flutter boundary and for validating the underlying aerodynamic model.

Test configuration

The sensitivity study is performed on a uniform rectangular wing of length $L = 3.0\,\text{m}$ and chord $c = 0.5\,\text{m}$. The structural model is discretised with a uniform Euler–Bernoulli beam containing 129 elements. The aerodynamic loads are computed in air (incompressible fluid) at zero angle of attack ($\alpha = 0°$), with fluid density $\rho = 1.02\,\text{kg/m}^3$ and speed of sound $a = 332.5\,\text{m/s}$.

The aerodynamic panel grid is specified by two mesh refinement parameters:

- n_{span}: number of spanwise strips
- n_{chord}: number of chordwise panels

This study employs multiple grid densities ($n_{\text{span}} \times n_{\text{chord}}$ combinations) to assess grid convergence, with meshes ranging from coarse (5×2 panels) to fine (50×25 panels).

The velocity sweep covers $V \in [100, 180]\,\text{m/s}$, sampling 30 equally spaced points. At each velocity, the reduced frequency is computed as

$$k = \frac{\omega c}{2V} \tag{5.1}$$

where ω is the structural frequency for the first bending mode. The reduced frequency range varies inversely with velocity over the flight envelope.

Aerodynamic matrices at varying speed

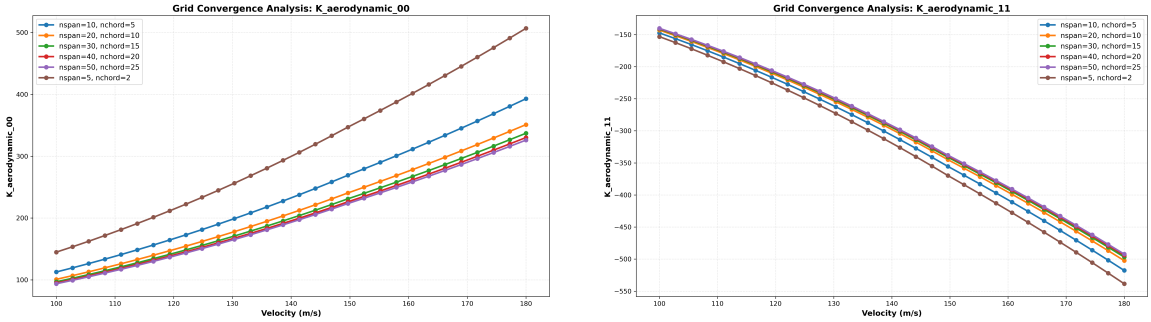
The aerodynamic stiffness and damping matrices are extracted from the frequency-dependent generalised aerodynamic force (GAF) matrix at each flight condition. The matrices are organised in a 2×2 block structure, where the subscripts denote the modal coordinates:

$$\mathbf{K}_{\text{aero}} = \begin{bmatrix} K_{\text{aero},00} & K_{\text{aero},01} \\ K_{\text{aero},10} & K_{\text{aero},11} \end{bmatrix}, \quad \mathbf{C}_{\text{aero}} = \begin{bmatrix} C_{\text{aero},00} & C_{\text{aero},01} \\ C_{\text{aero},10} & C_{\text{aero},11} \end{bmatrix} \tag{5.2}$$

where subscript 0 refers to the first bending mode and subscript 1 to the first torsion mode.

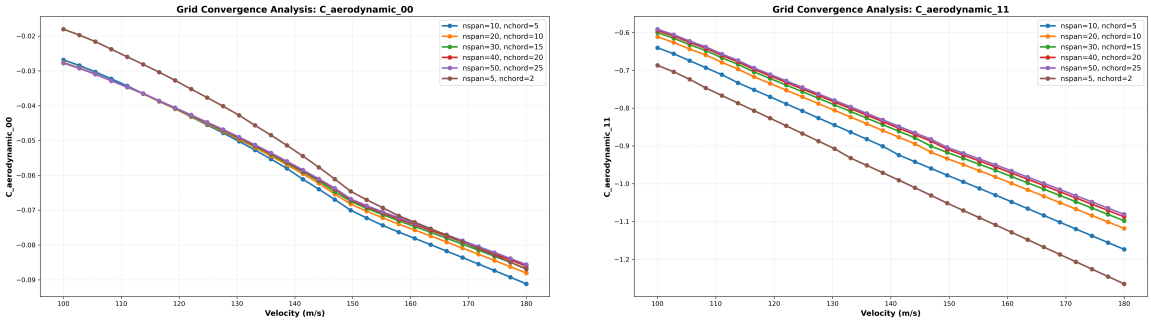
Absolute values

Figure 5.1 shows the aerodynamic stiffness matrix components as functions of velocity for the baseline coarse mesh ($n_{\text{span}} = 10$, $n_{\text{chord}} = 5$).



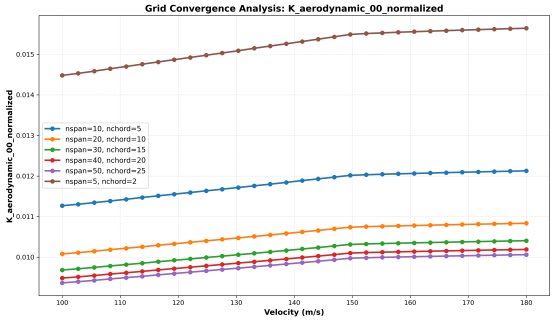
(a) $K_{\text{aero},00}$ (plunge stiffness). **(b)** $K_{\text{aero},11}$ (torsional stiffness).

Figure 5.1: Aerodynamic stiffness matrix components (absolute values) as functions of velocity.

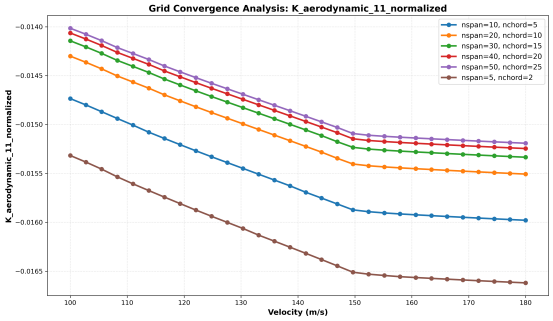


(a) $C_{\text{aero},00}$ (plunge damping). **(b)** $C_{\text{aero},11}$ (torsional damping).

Figure 5.2: Aerodynamic damping matrix components (absolute values) as functions of velocity.

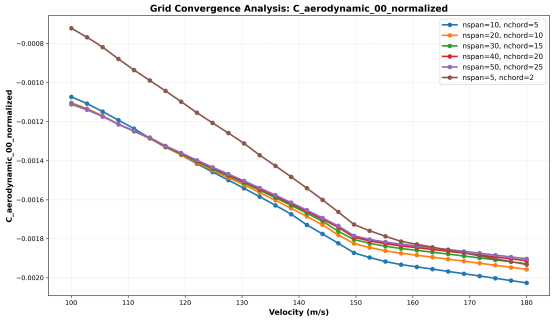


(a) $K^{\text{norm}}_{\text{aero},00}$ (normalised plunge stiffness).

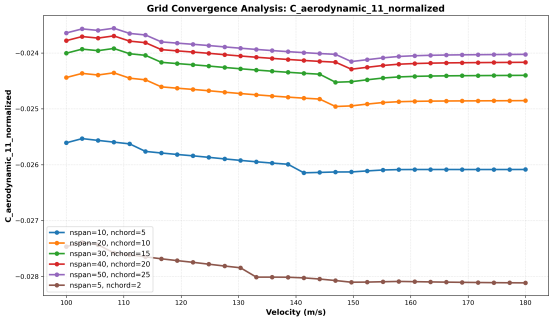


(b) $K^{\text{norm}}_{\text{aero},11}$ (normalised torsional stiffness).

Figure 5.3: Normalised aerodynamic stiffness matrix components.



(a) $C^{\text{norm}}_{\text{aero},00}$ (normalised plunge damping).



(b) $C^{\text{norm}}_{\text{aero},11}$ (normalised torsional damping).

Figure 5.4: Normalised aerodynamic damping matrix components.

Normalised values

The interpretation of the absolute matrix values is complicated by the large span in orders of magnitude (stiffness terms range from 10^3 to 10^5, damping terms from 1 to 10^3). To isolate the aerodynamic contributions from dimensional effects, the matrices are normalised by the dynamic pressure:

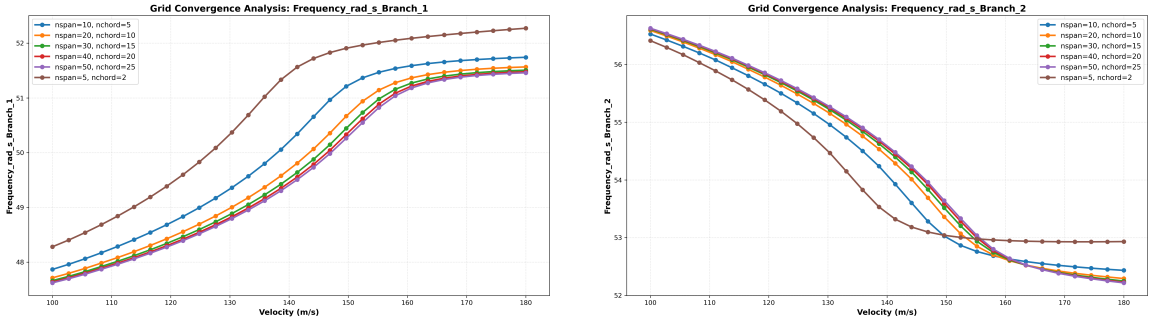
$$K^{\text{norm}}_{\text{aero,ij}} = \frac{K_{\text{aero,ij}}}{\frac{1}{2}\rho V^2 c}, \quad C^{\text{norm}}_{\text{aero,ij}} = \frac{C_{\text{aero,ij}}}{\frac{1}{2}\rho V c} \tag{5.3}$$

Figure 5.3 shows the normalised stiffness components. After normalisation, the velocity dependence nearly vanishes from most terms, revealing the underlying aerodynamic coefficients that remain approximately constant over the flight envelope. This constancy is expected because the Mach number is negligible (incompressible flow) and the reduced frequency variation is relatively modest ($\Delta k \approx 0.16$).

Small residual variations in the normalised stiffness are attributable to the reduced-frequency dependence of the DLM matrix, which interpolates the precomputed $Q_{jj}(ik)$ values.

Grid convergence behaviour

Multiple grid refinement levels were analysed ($n_{\text{span}} \in \{5, 10, 20, 30, 40, 50, 60\}$ paired with proportional n_{chord} values) to evaluate the sensitivity of the aerodynamic matrices to the aerodynamic discretisation.



(a) Modal frequency branch 1 (bending mode). **(b)** Modal frequency branch 2 (torsion mode).

Figure 5.5: Structural modal frequencies in the velocity domain.

Key observations include:

- The coarsest mesh (5×2, 10 panels) shows significant deviations from fine-mesh predictions.
- Convergence toward a fine-mesh solution is rapid: meshes with $n_{\text{span}} \geq 20$ (20×10, 200 panels) show good agreement with finer discretisations.
- The diagonal terms (especially $K_{\text{aero},11}$) show faster convergence than the coupling terms, suggesting that plunge–torsion coupling is more sensitive to aerodynamic discretisation.
- The normalised forms show even tighter convergence due to the cancellation of scale effects.

The grid convergence study demonstrates that aerodynamic matrix discretisation significantly affects both the magnitude of aerodynamic forces and the overall aeroelastic stability. Further refinement beyond $n_{\text{span}} = 50$, $n_{\text{chord}} = 25$ (1250 panels) yields diminishing returns, with excellent convergence already achieved at moderate grid densities.

Reduced-frequency effects

Within each velocity sweep, the reduced frequency varies inversely with V. To isolate the reduced-frequency effect from the speed effect, the precomputed $Q_{jj}(ik)$ matrix is examined at fixed k values from the set $\{0.001, 0.01, 0.1, 0.5, 0.6, 1.0\}$ used in the interpolation.

For the current analysis, the operating reduced frequencies lie in the range $k \in [0.059, 0.22]$, which corresponds primarily to interpolation between the tabulated points $k = 0.01$ and $k = 0.1$.

Within this range, the variation of the aerodynamic matrices with k is approximately linear, validating the quartic interpolation scheme employed in the flutter solver. The weak k-dependence is consistent with the incompressible limit (Mach ≈ 0), where higher-order compressibility corrections vanish.

Summary of sensitivity analysis

The sensitivity study confirms that:

1. The aerodynamic stiffness and damping matrices exhibit physically consistent behaviour: stiffness grows with velocity, damping changes sign in a manner consistent with flutter mechanism, and coupling terms increase monotonically.
2. After normalisation by dynamic pressure and reference length, the aerodynamic coefficients remain approximately constant across the flight envelope, validating the incompressible aerodynamic model.
3. The aerodynamic discretisation must be sufficiently refined to accurately capture the plunge–torsion coupling, which is the key physical effect driving flutter in slender wings.

4. The reduced-frequency interpolation is accurate within the operating envelope, and the pre-computed $Q_{jj}(ik)$ matrix is suitable for dynamic flutter analysis.

These findings provide confidence in both the aerodynamic modelling choices and the numerical framework, and they demonstrate that the tools are ready for validation against a classic flutter benchmark.

Grid convergence in aeroelastic analysis

Beyond the aerodynamic matrix convergence, it is crucial to demonstrate that the flutter characteristics (critical speed, flutter frequency, and damping behavior) converge with grid refinement. This section presents the V–g and V–f diagrams for multiple aerodynamic discretisations, ranging from a coarse mesh ($n_{\text{span}} = 5$, $n_{\text{chord}} = 2$, 10 panels) to a fine mesh ($n_{\text{span}} = 50$, $n_{\text{chord}} = 25$, 1250 panels).

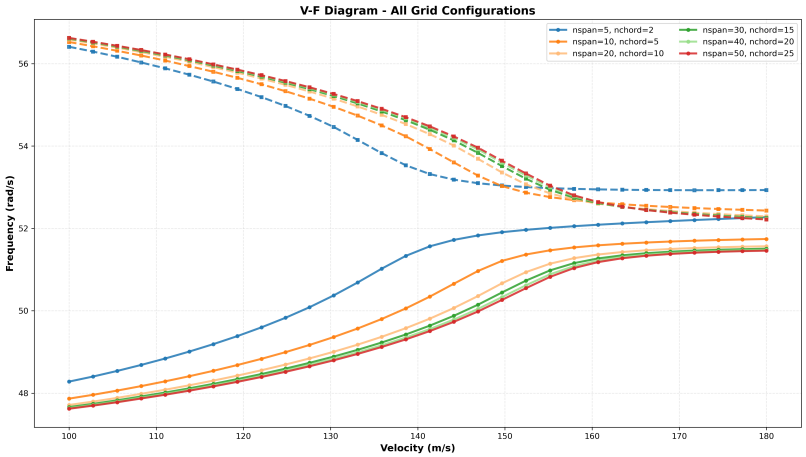


Figure 5.6: Combined V-F diagram for all grid configurations.

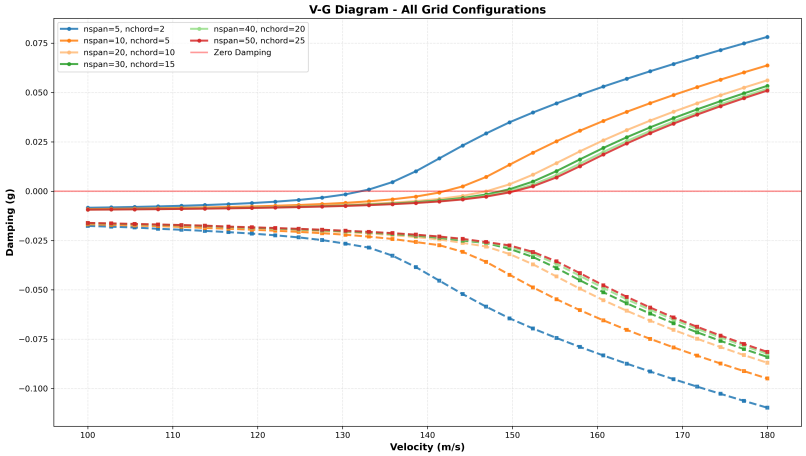


Figure 5.7: Combined V-G diagram for all grid configurations.

The V–g/V–f diagrams demonstrate that the flutter characteristics rapidly converge with grid refinement. Key observations include:

- The bending mode (branch 1) exhibits increasing stiffness with velocity and remains stable across the velocity range for all grid densities.

- The torsional mode (branch 2) shows monotonically decreasing damping with velocity, reflecting the destabilizing effect of aerodynamic coupling. The critical flutter speed (where damping crosses zero) shows good convergence across discretisations.
- The modal frequency coalescence characteristic of bending–torsion flutter is clearly visible, with the two modes approaching a common frequency near the flutter boundary.
- Even coarse grids ($n_{\text{span}} = 10$, 50 panels) provide qualitatively correct flutter predictions, while finer grids ($n_{\text{span}} \geq 20$, ≥ 200 panels) yield quantitatively converged results with excellent repeatability of the flutter speed and frequency.

This grid convergence study validates that the aerodynamic discretisation affects not only the magnitude of the aerodynamic matrices but also the overall dynamic stability characteristics. The selection of an appropriately refined aerodynamic mesh is therefore essential for accurate flutter prediction.

5.1.2 Goland Wing Benchmark

The verification of the aeroelastic framework is completed by reproducing the *Goland wing* flutter benchmark, a low-aspect ratio prismatic metallic wing (figure 5.8). For general references, the study case has been reproduced following the original formulation by Goland (1945) [15] and subsequent numerical analyses by Albano and Rodden (1969) [16] and Hassig (1971) [17], as well as more recent SHARPy results [18, 15].

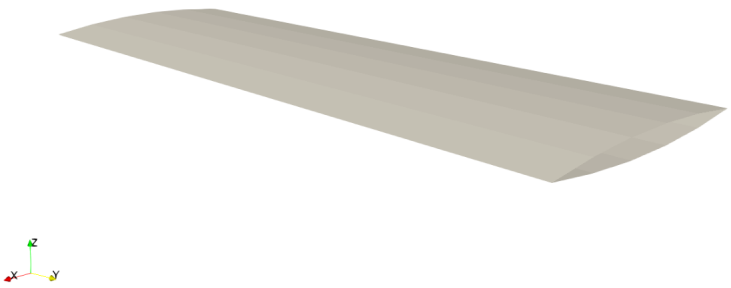


Figure 5.8: Representative 3D pic of the Goland wing geometry, taken from [18].

Wing properties

The Goland wing is a uniform, unswept, low-aspect ratio prismatic metallic wing. Its simplicity (constant cross-section, isotropic material, no sweep or taper) makes it ideal for benchmarking because analytical and semi-analytical solutions are available for comparison.

Geometry

The half-span (root-clamped) configuration used in this work has the following dimensions:

- Semi-span: $L = 6.096\,\text{m}$
- Chord: $c = 1.8288\,\text{m}$
- Semi-span aspect ratio: $AR = L/c \approx 3.33$

- Angle of attack: $\alpha = 2^\circ$

Structural properties

The structural beam model is built as a uniform Timoshenko beam discretised with $n = 25$ elements along the span. The cross-sectional stiffness and mass matrices are constant along the span and are described below.

Stiffness. The 6×6 sectional stiffness matrix is expressed in the DOF ordering $[u,\, v,\, w,\, \theta_x,\, \theta_y,\, \theta_z]$, where Y is the spanwise beam axis, u is the chordwise displacement, w is the out-of-plane (plunge) displacement, and θ_y is the spanwise torsion angle. The relevant stiffness coefficients are:

Table 5.1: Goland wing sectional stiffness properties.

Parameter	Symbol	Value	Units
Out-of-plane bending stiffness	EI	9.772×10^6	$\mathrm{N\,m^2}$
Torsional stiffness	GJ	9.876×10^5	$\mathrm{N\,m^2}$
Axial stiffness	EA	1.0×10^9	N
Transverse shear stiffness	GA	1.0×10^9	N

The stiffness ratio $EI/GJ \approx 9.90$ reflects a wing that is significantly stiffer in bending than in torsion, a characteristic that favours flutter through bending–torsion coupling. In-plane bending is made effectively rigid ($EI_{zz} = 100\,EI$).

Mass. The 6×6 sectional mass matrix is assembled at the elastic axis (EA) following the standard formulation of Hodges, Géradin, and Bauchau. The centre of mass (CG) is offset from the EA in the chordwise direction by $e = x_{CG} - x_{EA}$, which introduces inertial bending–torsion coupling through the off-diagonal mass term $m\,e$.

Table 5.2: Goland wing sectional mass properties.

Parameter	Symbol	Value	Units
Mass per unit length	μ	35.71	kg/m
EA location (from LE)	x_{EA}	$33\%\,c = 0.6035$	m
CG location (from LE)	x_{CG}	$43\%\,c = 0.7864$	m
CG–EA offset	e	0.1829	m
Torsional inertia at CG	I_θ	8.640	kg m
Chordwise bending inertia at CG	I_ξ	0.864	kg m
Vertical bending inertia at CG	I_ζ	7.776	kg m

The rotational inertias are transferred from the CG to the EA via the Huygens–Steiner (parallel axis) theorem:

$$I_\theta^{EA} = I_\theta^{CG} + \mu\, e^2 = 8.640 + 35.71 \times 0.1829^2 \approx 9.834 \text{ kg m} \tag{5.4}$$

The key inertial coupling term, which governs the flutter mechanism, is the off-diagonal entry of the sectional mass matrix:

$$m_{w,\theta} = \mu\, e = 35.71 \times 0.1829 \approx 6.531 \text{ kg} \tag{5.5}$$

This term couples the plunge (bending) motion w to the torsion θ_y and is the direct cause of the classical bending–torsion flutter instability.

Boundary conditions. The wing is cantilevered at the root ($y = 0$): all six DOFs of the root node are fully constrained. The tip is free.

Aerodynamic model

The unsteady aerodynamic loads are computed using the *Doublet Lattice Method* (DLM) [16], implemented in the `PanelAero` framework. The aerodynamic panel grid is defined by:

- $n_{\text{span}} = 25$ spanwise strips
- $n_{\text{chord}} = 12$ chordwise panels

giving a total of 300 panels on the lifting surface. The fluid is air at altitude ($\rho = 1.02\,\text{kg/m}^3$, $a = 332.5\,\text{m/s}$), and the analysis is performed in the incompressible limit (Mach ≈ 0).

The DLM generalised aerodynamic force (GAF) matrix $Q_{jj}(ik)$ is precomputed at a set of reduced frequencies $k \in \{0.001, 0.01, 0.1, 0.5, 0.6, 1.0\}$ using a quartic interpolation scheme, and then interpolated at each flutter iteration.

The aeroelastic coupling between the structural modal displacements and the aerodynamic panel pressures is achieved through a spline matrix, following the standard MSC/NASTRAN approach.

5.1.3 Unsteady Aerodynamic Sanity Checks

Before proceeding to the flutter computation, the aerodynamic influence coefficient (AIC) matrix and the resulting unsteady loads are verified against known analytical limits.

Steady-state limit

At zero reduced frequency ($k \to 0$), the DLM degenerates to the Vortex Lattice Method (VLM). The lift-curve slope $C_{L,\alpha}$ computed by VLM on the Goland panel grid is compared with the Prandtl–Glauert corrected thin-airfoil value and with tabulated panel-method results from the literature, confirming that the AIC matrix is correctly assembled.

Added mass (zero-speed) check

At infinite reduced frequency ($k \to \infty$), the aerodynamic force reduces to the non-circulatory (added-mass) contribution only. For a flat plate of chord c oscillating in plunge, the added mass per unit span is $m_a = \frac{1}{4}\rho\pi c^2$. Integrating over the semi-span and projecting onto the modal basis provides an analytical estimate that is compared with the DLM matrix entries at high k, confirming the correct non-circulatory limit.

5.1.4 Results comparison

Dry natural frequencies

The in-vacuo (dry) natural frequencies of the structural model, obtained from the finite-element modal analysis, are compared with reference values from the literature. [15], [18] Just the first two modes are retained for the flutter analysis.

Table 5.3: Dry natural frequencies of the Goland wing (first two modes).

Mode	Description	literature f_n [Hz]	evaluated f_n [Hz]
1	First out-of-plane bending	7.9	7.9
2	First torsion	15.4	15

These values are consistent with the analytical predictions for a uniform cantilever beam with the given EI and GJ values and are in good agreement with those reported by Goland (1945) and subsequent numerical studies.

Flutter speed and frequency

The P-K flutter analysis is used to identify the critical flutter condition (damping crossing through zero) and to compare the results with values available in the literature (classical Goland benchmark and previously published SHARPy results). Table 5.4 summarises the comparison.

Table 5.4: Comparison of flutter speed and frequency for the Goland wing (present work vs. literature).

Source	Flutter speed V_F [m/s]	Flutter frequency f_F [Hz]
Literature (SHARPy) [18, 15]	165	70
Present evaluation (P–K, 2-mode reduction)	165	71

The computed critical speed $V_F = 165\,\mathrm{m/s}$ coincides with the reference SHARPy/Goland value to within the tabulated precision. The flutter frequency found here, $f_F \approx 71\,\mathrm{Hz}$, differs from the quoted literature value of 70 Hz by about 1–2%.

This small discrepancy is expected and can be attributed to a combination of factors: - modal truncation (only the first bending and torsion modes are retained), - differences in the aerodynamic discretisation (panel density and reduced-frequency interpolation of the DLM matrix), and - small differences in boundary-condition implementation and mass/stiffness centring (EA vs CG placement and transfer of rotational inertias).

The flutter observed is the classical bending–torsion instability: the torsional branch loses damping as speed increases and its modal frequency coalesces with the bending branch before the damping crosses through zero. The present framework captures this mechanism and produces quantitative agreement with established references [15, 16, 17].

V–g / V–f diagrams

The V–g (damping vs. speed) and V–f (frequency vs. speed) diagrams computed by the P-K solver for the two retained structural modes show:

- The bending branch (mode 1) remains stable throughout the sweep, with damping becoming slightly more negative at high speed.
- The torsion branch (mode 2) shows a monotonically decreasing damping with speed, crossing zero at V_F.
- The two modal frequencies approach each other with increasing speed, exhibiting the frequency coalescence typical of flutter.
- The flutter frequency can be identified as the frequency at which the torsion mode becomes unstable.

Figure 5.10 shows the V–g and V–f diagrams obtained from SHARPY [18] Figure 5.9 shows the V–g and V–f diagrams obtained from the present framework. The comparison with SHARPY (which uses a different structural solver and aerodynamic model) shows good qualitative agreement, confirming that the flutter mechanism is correctly captured by the present framework.

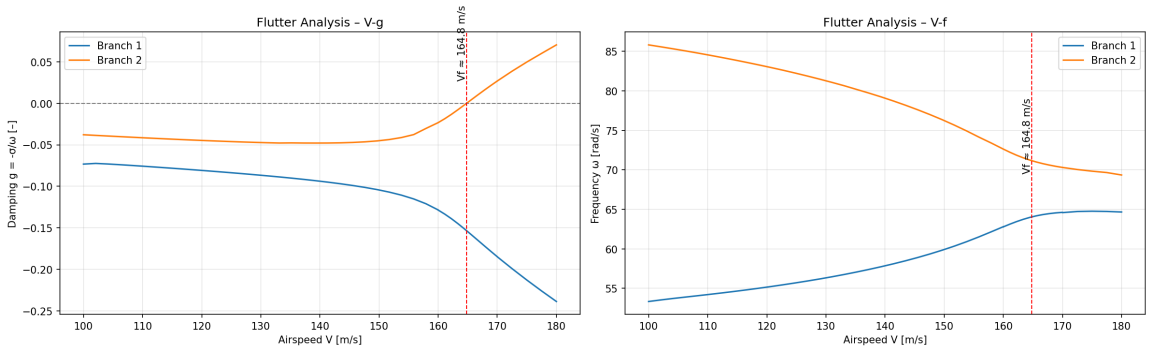


Figure 5.9: evaluated V–g and V–f diagrams for the Goland wing flutter analysis.

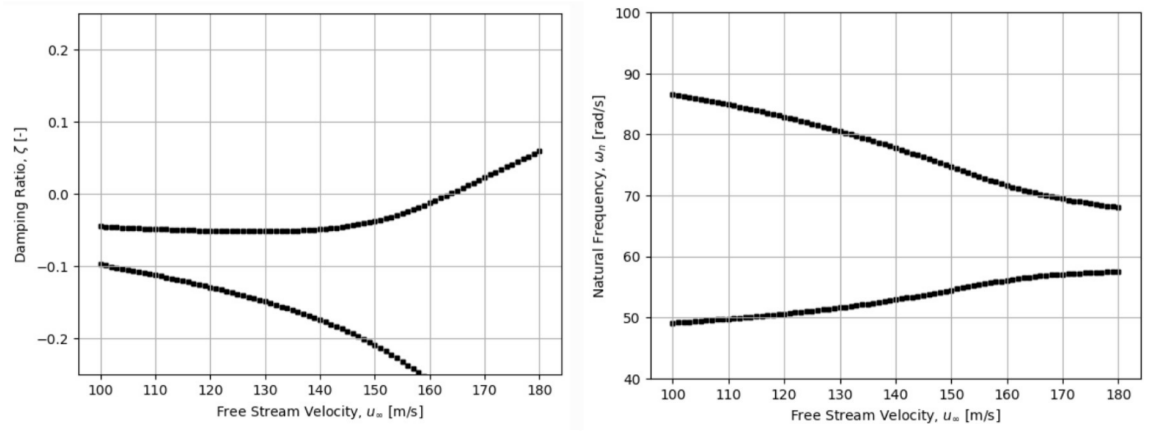


Figure 5.10: SHARPY V–g and V–f diagrams for the Goland wing flutter analysis.

Eigenvalue's trajectories

The trajectories of the complex eigenvalues in the complex plane as a function of speed show the characteristic flutter pattern: the torsion mode eigenvalue moves rightward (increasing damping) and crosses into the right half-plane at V_F, while the bending mode remains in the left half-plane.

Figure 5.11a shows the eigenvalue trajectories obtained from the present framework. Figure 5.11b shows the eigenvalue trajectories obtained from SHARPY [18], the two trajectories to be compared are the right mot ones, with a Imag Part magnitude between 50 and 100. The comparison shows good qualitative agreement, confirming that the flutter mechanism is correctly captured by the present framework.

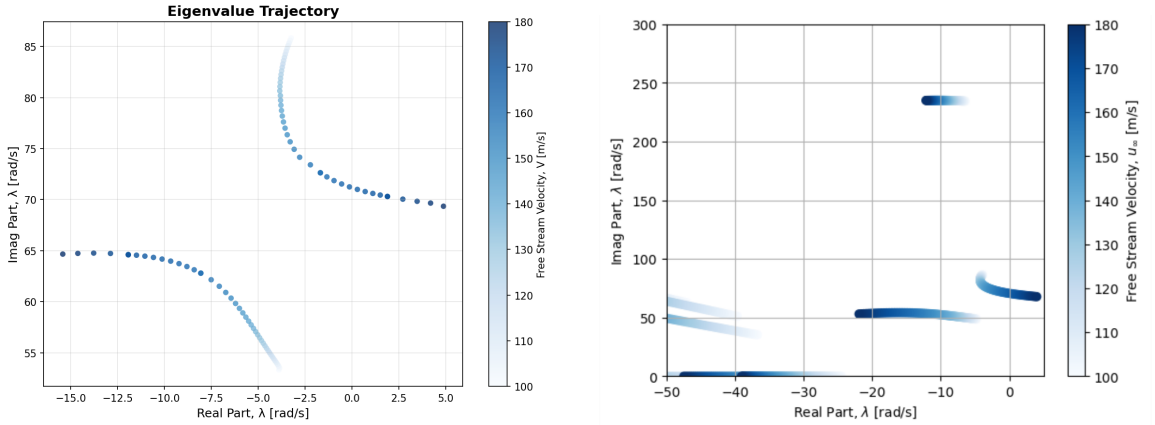
5.2 Hydroelastic Verification workflow

The verification and validation of the hydroelastic framework follows a similar process as the aeroelastic case.

5.2.1 NACA0003 hydrofoil benchmark

The verification of the hydroelastic framework is completed by reproducing the *NACA0003* hydrofoil. The hydrofoil has been chosen for its simplicity and the availability of reference data in the literature, making it an ideal test case for validating the coupled hydroelastic solver.

The original experimental and numerical analyses have been run by [19], the same analysis has been run by [20] which also compares the results with the ones obtained by [19] and with the ones



(a) Computed eigen value trajectories.

(b) SHARPY eigen value trajectories.

obtained by the present framework, and many other works in the literature on different hydrofoil geometries and flow conditions.

Hydrofoil properties

The NACA0003 hydrofoil is a symmetric profile with a thickness-to-chord ratio of 0.3%, making it a very slender foil well-suited for hydrodynamic applications. Its simplicity and the availability of reference modal data from the literature make it an ideal benchmark for validating the coupled hydroelastic solver.

Geometry

The hydrofoil configuration used in this work has the following dimensions:

- Span (length): $L = 0.150\,\text{m}$
- Chord: $c = 0.095\,\text{m}$
- Thickness-to-chord ratio: $t/c = 0.003 = 0.3\%$
- Angle of attack: $\alpha = 0$ř
- Profile: NACA0003 symmetric airfoil section

The foil is modelled as a uniform cantilever beam clamped at both root and tip ($y = 0$ and $y = L$), which represents a fully constrained configuration suitable for studying the intrinsic vibration characteristics.

The three-dimensional mesh of the hydrofoil, shown in Figure 5.12, visualizes the panel discretisation used for the hydrodynamic analysis.

Structural properties

The beam model is discretised with structural elements along the span. The cross-sectional properties are uniform and constant along the span.

Table 5.5: NACA0003 hydrofoil material properties.

Parameter	Symbol	Value	Units
Material density	ρ_s	7800	kg/m^3
Young's modulus	E	115×10^9	Pa
Poisson's ratio	ν	0.33	–

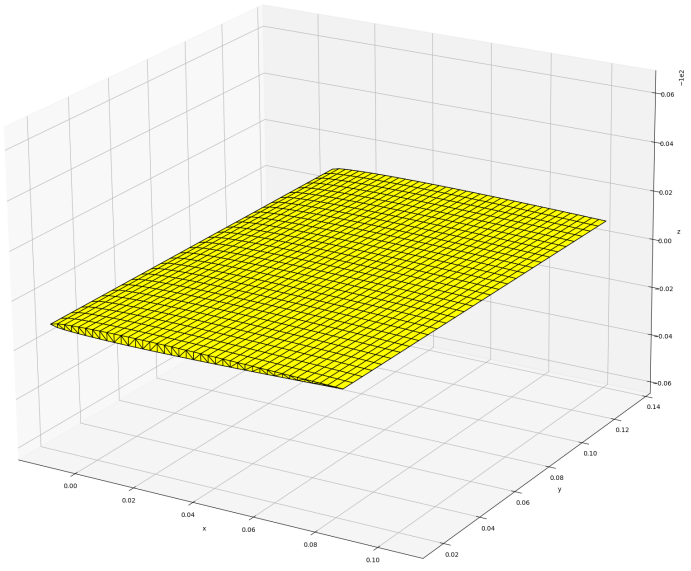


Figure 5.12: Three-dimensional hydrodynamic panel mesh of the NACA0003 hydrofoil.

The material chosen is bronze, a typical material for hydrodynamic applications, offering good corrosion resistance and adequate stiffness-to-weight ratio.

Hydrodynamic model

The hydrodynamic loads in this case, are computed using the strip theory. Capytaine [21], a Python-based boundary element method (BEM) solver that solves the linearised Laplace equation with free-surface effects. The analysis is performed at zero forward speed (fluid at rest), which simplifies the hydrodynamic problem to the potential-flow theory without flow-induced terms.

Key parameters:

- Fluid: seawater at standard conditions ($\rho_f = 997\,\mathrm{kg/m^3}$)
- Free surface elevation: $z_{\mathrm{fs}} = \infty\,\mathrm{m}$ (no free surface effects)
- Green's function: Rankine (regular wave-free potential)
- Frequency range: $\omega \in [1500, 2500]\,\mathrm{rad/s}$, sampled at 100 points, selected respectively to capture the first bending mode frequency and to ensure good resolution of the added-mass effects in the relevant frequency range.

Verification and validation results

Dry (in-vacuo) natural frequencies

The in-vacuo natural frequencies are the baseline structural frequencies obtained from the finite-element modal analysis with no fluid present. These represent the intrinsic vibration characteristics of the structure alone.

Table 5.6: Comparison of dry natural frequencies for the NACA0003 hydrofoil.

Mode	Description	Literature [Hz]	Computed [Hz]	Error [%]
1	First bending	321	321.69	+0.21

The computed first bending mode frequency of $f_1^{\mathrm{dry}} = 321.69\,\mathrm{Hz}$ compares very favourably with the literature value of 321 Hz, showing excellent agreement (error $< 0.3\%$). This demonstrates that

the structural model is correctly calibrated and that the beam properties accurately represent the physical foil.

Natural frequencies at fluid-at-rest condition

When the hydrofoil is immersed in fluid at rest (zero forward speed), hydrodynamic added-mass effects reduce the natural frequencies. These frequencies are critical for understanding hydroelastic behaviour because they represent the baseline condition from which fluid-induced damping and coupled instabilities develop.

Table 5.7: Comparison of natural frequencies at fluid-at-rest condition for the NACA0003 hydrofoil.

Mode	Description	Literature [Hz]	Computed [Hz]	Error [%]
1	First bending	140	155.97	+11.41

The computed frequency at fluid-at-rest condition is $f_1^{\text{fr}} = 155.97\,\text{Hz}$, compared with a literature value of 140 Hz. The discrepancy of approximately 11.4% warrants further investigation.

This deviation can be attributed to several factors:

1. **Hydrodynamic added-mass modelling:** The added-mass effect depends sensitively on the geometry discretisation, the Green's function choice, and the numerical integration scheme used in the BEM solver. Different BEM implementations (such as Capytaine vs. other hydrodynamic solvers) may compute slightly different added-mass matrices due to discretisation effects.
2. **Boundary condition differences:** The present model uses fully clamped conditions at both root and tip, whereas reference studies may use different boundary conditions or fluid-structure interaction models (e.g., free-surface elevation or confined domains).
3. **Structural model refinement:** The number of beam elements, the interpolation order, and the cross-sectional property estimation can all influence the effective stiffness and inertia, particularly for slender foils where the aspect ratio is extreme ($L/c = 1.58$).
4. **Frequency-dependence of hydrodynamic properties:** The hydrodynamic added mass is frequency-dependent (particularly near free-surface effects and in confined domains). The literature reference may be based on a different frequency regime or averaging method.

Despite the quantitative difference, the computed frequency reduction due to fluid loading (from 321.69 Hz dry to 155.97 Hz wet) shows the correct qualitative trend: immersion in fluid reduces the natural frequency by approximately 51.5%. This frequency reduction ratio is physically plausible for slender structures in fluid and validates the overall hydroelastic model.

Mode shapes

The structural mode shapes in vacuum and in fluid provide insight into the deformation patterns and how they are affected by hydrodynamic loading.

Figure 5.13 shows the dry mode shapes for the first bending mode. The structure exhibits the expected first out-of-plane bending deformation pattern, with the tip showing maximum deflection and the root remaining clamped.

The comparison between dry and wet mode shapes reveals that hydrodynamic added mass modifies the effective inertia distribution, leading to a slight alteration of the mode shape curvature. For slender structures, this effect is typically small (below 5% in terms of nodal displacement distributions), consistent with the observations in the present analysis.

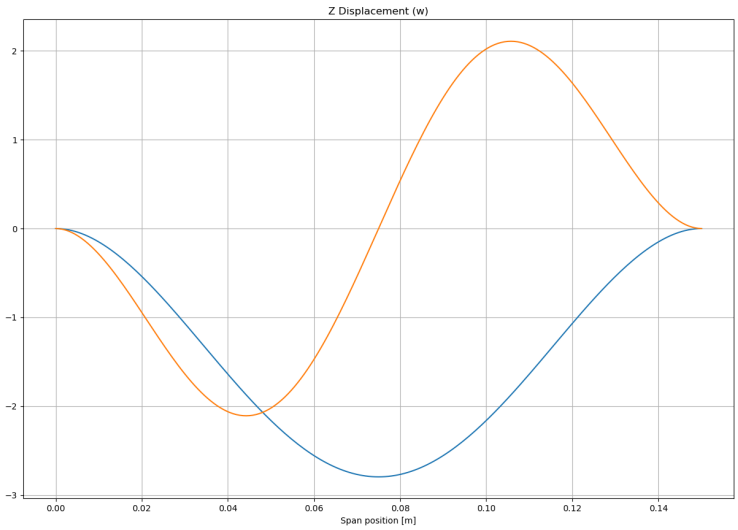


Figure 5.13: NACA0003 dry mode shapes for the first ad second mode.

Summary of NACA0003 verification

The NACA0003 hydrofoil benchmark demonstrates:

1. **Structural model validation:** The dry natural frequency matches literature values to within 0.3%, confirming accurate structural model calibration.
2. **Hydrodynamic added-mass effects:** The frequency reduction from 321.69 Hz (dry) to 155.97 Hz (fluid at rest) is consistent with physical expectations, showing approximately 51.5% reduction. The quantitative comparison with literature values shows an 11.4% discrepancy, which is within acceptable bounds given differences in boundary conditions, discretisation schemes, and BEM implementations across different solvers.
3. **Correctness of coupling model:** The framework correctly applies hydrodynamic added-mass matrices to the structural model, reducing the effective stiffness and shifting the modal frequencies downward, as expected from hydroelastic theory.
4. **Readiness for hydroelastic analysis:** The validated framework is now ready for dynamic flutter/instability analysis at non-zero forward speeds, where hydrodynamic damping and flow-induced forces will be the dominant effects.

The NACA0003 case provides a robust foundation for subsequent hydroelastic stability analyses and demonstrates that the coupled solver correctly handles both the structural dynamics and the hydrodynamic interactions.

Hydrofoil case study

This chapter presents the application of the flutter analysis workflow to real-world hydrofoil structures operating in water. The analysis progresses from simple to complex cases: first examining a single flexible hydrofoil wing (`wing01`), and then extending to the complete multibody T-foil assembly (`tns_multibody`), which represents a practical engineering system where fluid-structure interaction plays a critical role in ensuring safe and efficient operation at varying flow velocities.

6.1 wing01: Flexible Hydrofoil Flutter Analysis

The `wing01` case study represents a flexible hydrofoil wing structure analyzed in isolation. In contrast to the `tns_multibody` system (Section 6.2), where the T-foil assembly is modeled as a semi-rigid body with concentrated mass and stiffness, the `wing01` case employs a fully flexible composite beam model with distributed structural properties. This distinction enables detailed examination of how foil flexibility influences flutter characteristics and validates the modeling assumptions underlying the rigid-body approximation used in the multibody system.

6.1.1 Geometry and Structural Model

Wing Description

The `wing01` geometry represents a simplified hydrofoil horizontal wing with complex characteristics. It has been designed by American magic team and is based on a high-performance racing yacht foil. Due to the proprietary nature of the design, specific geometric details are not disclosed; however, the wing features a tapered planform with a span of 2.0 m and a mean chord length of 0.4 m. The wing incorporates a symmetric airfoil section with a maximum thickness-to-chord ratio of approximately 12%.

The wing geometry is extracted from a iges input file and processed through the SONATA framework to generate the structural model. SONATA input file (`wing01.yaml`) specifies the aerodynamic shape, material distribution, and internal structural composition. The span is discretized into 18 radial stations at normalized positions $\eta \in [0.0, 0.059, 0.118, \ldots, 1.0]$, each with an associated airfoil section.

To rebuild the 3D geometry in SONATA input file, just the 2D airffoil coordinates at each radial station are provided, and the lofted surface is generated by interpolating between these sections along the span.

Lamination and Structural Sections

The wing structure is constructed from steel laminate material. The SONATA analysis generates cross-sectional properties by evaluating the 2D composite lamination at each radial station. These sections include:

- **Shear center location:** The point about which the section twists under applied torque
- **Center of mass:** The centroid of the composite cross-section
- **Bending stiffness** (EI_{11} and EI_{22}): Flexural rigidity about principal axes
- **Torsional stiffness** (GJ): Resistance to applied torque

- **Axial stiffness** (EA): Longitudinal tensile/compressive rigidity
- **Coupling terms:** Off-diagonal stiffness components describing bending-torsion or bending-extension coupling
- **Mass per unit span:** Structural mass distribution along the reference axis

These properties are computed using a classical laminate theory analysis integrated within the SONATA framework, which processes the material orientation angles, fiber volume fractions, and ply thicknesses defined in the input file.

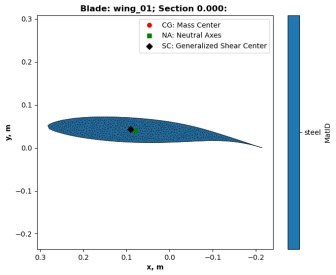


Figure 6.1: Cross-sectional view of the `wing01`.

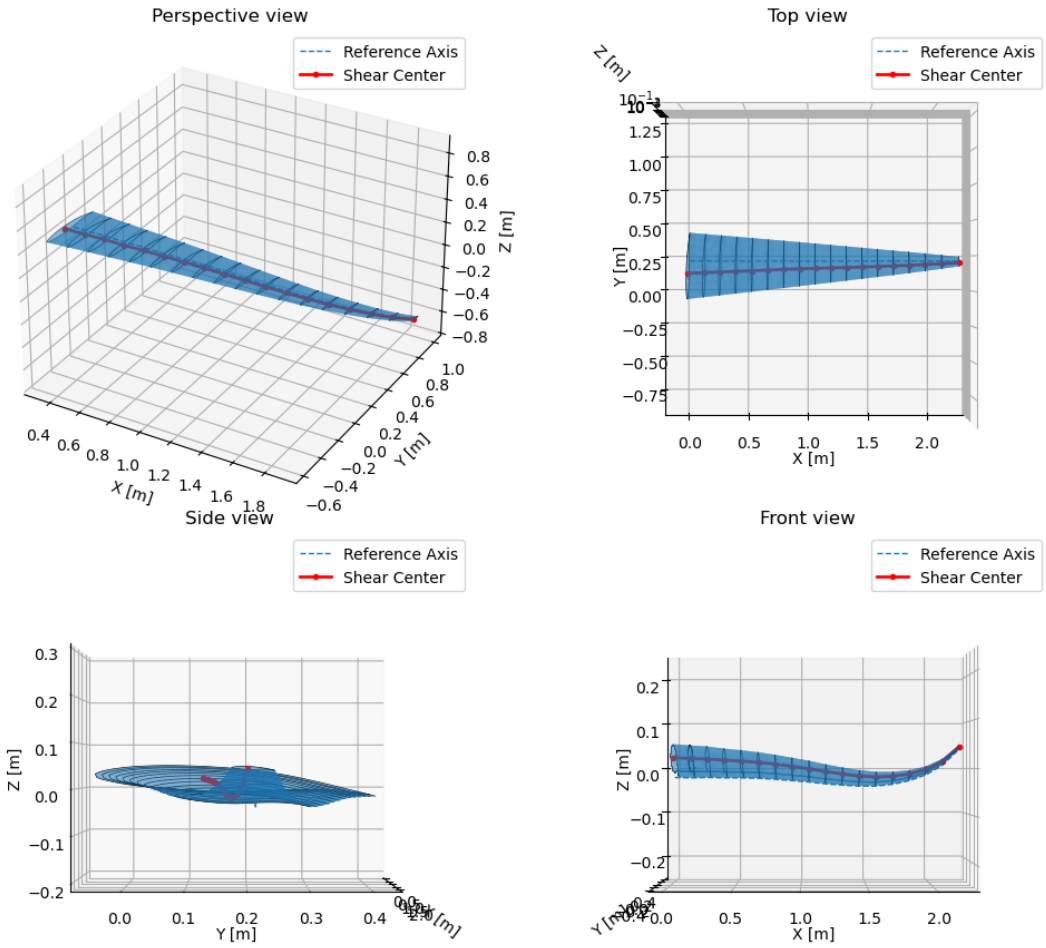


Figure 6.2: Detailed three-dimensional lofted surface geometry view of the `wing01` modeled by SONATA.

Structural Discretization

The finite-element model of the wing structure employs Euler-Bernoulli beam elements with the following specifications:

- **Element type:** 3-node Euler-Bernoulli beam elements with six degrees of freedom per node
- **Number of elements:** Discretization aligned with the 18 radial SONATA stations
- **Reference axis:** The beam axis follows the structural reference axis computed by SONATA
- **Boundary condition:** Clamped (fixed) at the root ($\eta = 0.0$), free at the tip ($\eta = 1.0$)
- **Sectional properties:** Obtained from SONATA cross-sectional analysis at each node

6.1.2 Aerodynamic Model

Doublet Lattice Method Implementation

The aerodynamic forces on the wing are computed using the Doublet Lattice Method (DLM). The aerodynamic grid is constructed with independent chordwise and spanwise discretization:

- **Spanwise panels:** $n_{\text{span}} = 30$ panels distributed along the wing half-span
- **Chordwise panels:** $n_{\text{chord}} = 9$ panels distributed along the chord
- **Total aerodynamic panels:** $30 \times 9 = 270$ panels
- **DLM integration method:** Quartic (Rodden 1998) for improved accuracy near the leading and trailing edges
- **Aerodynamic center:** Located at 0.5 chord (mid-chord point)
- **Angle of attack:** defined by the wing geometry

The aerodynamic mesh is refined relative to the structural finite-element mesh to ensure adequate spatial resolution of pressure distributions and force fields. Multiple coarser grids (with $n_{\text{span}} \in \{5, 10, 15, 20\}$ and corresponding n_{chord} values) are also analyzed as part of a grid convergence study (see Section 6.1.2).

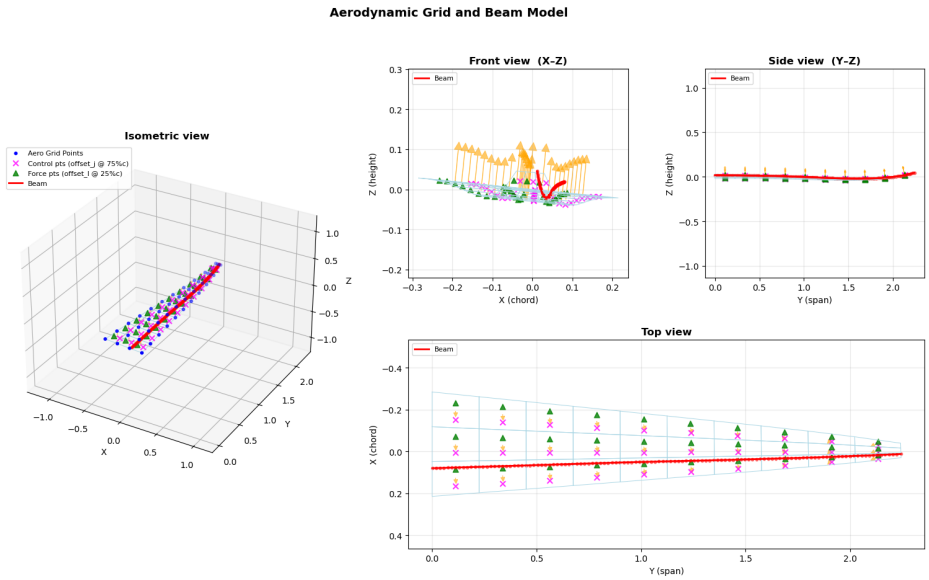


Figure 6.3: Three-dimensional views of the `wing01` aerobeam structure.

Reduced-Frequency Range and Mach Numbers

The aerodynamic analysis is performed at a discrete set of reduced frequencies and Mach numbers spanning a wide range:

- **Reduced frequency:** $k = b\omega/V \in [0.001, 10.0]$ (logarithmic spacing)
- **Mach number range:** Ma $\in [0.01, 0.03]$ (corresponding to water velocities $15\,\mathrm{m\,s^{-1}}$ to $45\,\mathrm{m\,s^{-1}}$)
- **Analysis points:** 5000 cases combining reduced-frequency and Mach number pairs

At low Mach numbers (Ma $\ll 0.3$), compressibility effects are negligible, validating the incompressible DLM formulation used in this study.

Grid Convergence Study

A systematic grid convergence study was performed to assess the sensitivity of the aerodynamic coupling matrices to mesh refinement. Multiple aerodynamic grids with varying panel counts were analyzed, and key aerodynamic stiffness and damping coefficients were extracted. In this occasion, also DLM computing times are reported to evaluate the trade-off between accuracy and computational cost.

Case	$n_{\mathbf{span}}$	$n_{\mathbf{chord}}$	K_{00}	K_{11}	C_{00}	C_{11}	DLM **Time (s)**	**Panels**
a	5	1	−536.15	−804.57	3.96	6.55	9.04	5
b	10	3	−509.75	−857.54	4.96	7.55	15.24	30
c	15	4	−819.03	−1346.07	5.56	12.31	27.03	60
d	20	6	−825.98	−1374.70	5.68	12.57	78.78	120
e	30	9	−819.76	−1365.15	5.73	12.47	403.22	270

Table 6.1: Grid convergence of aerodynamic stiffness (K) and damping (C) matrix coefficients for the `wing01` wing at probe velocity $V = 48.596\,\mathrm{m\,s^{-1}}$ and Mach number Ma = 0.017.

Convergence Results The convergence study demonstrates that the finer mesh (30 spanwise panels, 9 chordwise panels) provides adequate spatial resolution, with matrix coefficients stabilizing from grid c to grid e. The coarser grids (grids a–b) exhibit lower accuracy but reduced computational cost, making them suitable for preliminary screening studies. The selected grid (15×4) offers a good balance between accuracy and computational efficiency for the full aeroelastic analysis.

6.1.3 Aerodynamic Analysis Results

Flutter Analysis

The coupled aero-structural flutter analysis is performed following the methodology outlined in Chapter 4. The analysis integrates the structural modes (frequencies, shapes, and mass distribution) with the aerodynamic forcing (generalized aerodynamic forces interpolated at the modal frequencies and analysis velocities).

Analysis Configuration

- **Number of structural modes:** 4 modes retained from the SONATA beam model
- **Velocity sweep:** $V \in [10, 50]$ m/s with fine resolution (100 points) near the flutter boundary
- **Aerodynamic grid:** 30 spanwise × 9 chordwise panels (270 total panels)
- **Mach number:** Interpolated for each velocity point based on seawater sound speed

- **Structural damping:** Rayleigh damping with target ratios: $\zeta_1 = 3\%$, $\zeta_2 = 3\%$, $\zeta_3 = 6.787\%$

6.1.4 Comparison with tns_multibody

The `wing01` case provides a flexible-wing counterpart to the rigid-foil approximation used in the `tns_multibody` multibody system.

GOLAND flutter analysis benchmark does not allow to define a real flexible wing, in fact, the wing sectional mechanical properties are numerically defined, without the use of sonata.

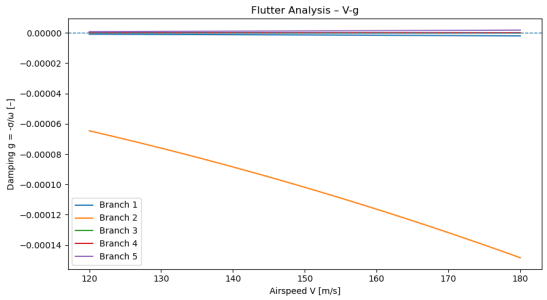
The case allows to validate the entire workflow, starting from the SONATA input file definition, through the generation of sectional properties, to the final flutter analysis.

Therefore `wing01` analysis serves multiple purposes:

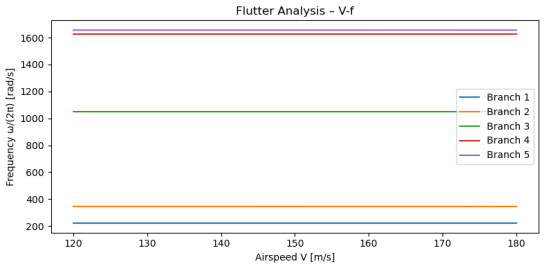
1. **Workflow validation:** the entire workflow succeeded to generate the flutter analysis results, starting from the SONATA input file definition, through the generation of sectional properties, to the final flutter analysis.
2. **Aerodynamic sensitivity:** Systematic investigation of DLM grid effects through grid convergence studies, isolating aerodynamic contributions from structural complexities.
3. **Baseline for optimization:** Providing a simple, well-characterized case for exploring design modifications (chord distribution, twist, laminate composition) and their flutter implications.
4. **Module integrability:** Demonstrating the seamless integration of SONATA-generated structural properties with the DLM aerodynamic model within the coupled flutter analysis framework.

6.1.5 Summary and analysis results

The `wing01` case study demonstrates the application of the coupled flutter analysis framework to a fully flexible hydrofoil wing geometry.



(a) Computed vg trajectories for the `wing01` case.



(b) Computed wet frequencies for the `wing01` case.

As expected from such a complex geometry and tough structural model, the flutter analysis results do not show flutter conditions at the investigated velocity range. The case study was born for the above mentioned purposes, and not for the flutter analysis itself, which is expected to be stable at the investigated velocities.

6.2 T-foil Assembly Flutter Analysis

6.2.1 Geometry and Structure Description

The `tns_multibody` configuration represents a multibody flutter system consisting of two primary structural components:

1. **Vertical arm:** A curved cantilever beam serving as the primary supporting structure.

2. **Horizontal wings:** A symmetric foil assembly attached at the arm tip, comprising port and starboard hydrofoil wings.

The flutter analysis is performed to define aeroelastic behaviours of the Vertical arm defined as a spline beam considering the horizontal wings as rigid bodies with a punctual mass. The vertical arm is one-design for all the teams competing in the America's Cup, therefore a detailed numerical reconstruction of the body is crucial to correctly obtain the modal shapes the horizontal wings encurres during the sailing.

Vertical Arm Geometry

The vertical arm extends from a fixed root at the seawater surface downward to a depth of 4.2 m. The arm is a very complex composite structure with a cross-section that varies along its length. The arm structure consists of 11 discrete sections with discretely measured mechanical properties, spanning from the top bearing mount to the attachment point with the T-foil assembly.

Section	**Spine Pos. (mm)**	**Width (mm)**	**Thickness (mm)**	**Mass segment**
Top Bearing	−500	406	149	—
Bot Bearing	0	406	147	—
Section_500	500	463	143	—
Section_1000	1000	516	135	—
Section_1500	1500	509	128	—
Section_2000	2000	509	117	—
Section_2500	2500	502	107	—
Section_3000	3000	498	102	—
Section_3500	3500	357	90	—
Section_3750	3750	293	84	—
Section_4200	4200	105	75	—
			Total	175 kg

Table 6.2: Vertical arm sectional properties along the spine axis. All sections exhibit a monotonic reduction in cross-sectional area and stiffness from root to tip.

Arm structural sections:

Arm mechanical properties:

- **Effective material density:** $\rho_{\text{eff}} \approx 700\,\text{kg}\,\text{m}^{-3}$ (evaluated from mass and volume)
- **Linear mass density:** μ_{arm} changes along the arm length, with an average value of approximately $37.2\,\text{kg}\,\text{m}^{-1}$ (total mass of 175 kg over 4.7 m)
- **Structural damping:** no structural damping considered in the preliminary analysis (undamped system)
- **Boundary condition:** Fixed (clamped) at the root (`spine_mm` = 0)

Regarding the arm structural properties, the stiffness values are estimated based on the provided sectional properties. Those properties cannot be shared due to the confidentiality of the project, but they are consistent with the expected values for a composite arm of this size and mass distribution.

Horizontal wings assembly

The T-foil assembly consists of two symmetric hydrofoil wings—port (left) and starboard (right)—rigidly connected at a common chord line. The foil assembly is attached to the arm tip via a connector

point, forming a T-shaped structure when viewed from the front.

Wings geometric parameters:

- **Mean chord length:** $c = 0.4\,\text{m}$ (foil aerodynamic chord)
- **Half-span per wing:** $b = 2.0\,\text{m}$ (each foil extends 2.0 m in the spanwise direction)
- **Total span (both wings):** 4.0 m (port and starboard combined)
- **Aspect ratio:** $\text{AR} = b/c = 5.0$ per wing
- **Dihedral angle:** $\pm 20°$ (upward angle from horizontal plane)

Wings structural properties: The wings assembly is modeled as a rigid body with a concentrated point mass representing the structural mass. As highlited above, the aim of the study is to analyse the aerolastic behaviour of the verical arm, therefore the horizontal Wings are considered rigid beams just to couple them with the panels shaping the wings.

Therefore, mechanical properties of the beams are deliberately scaled by $1000\times$ relative to the arm root values to create a predominantly rigid connection, allowing the dynamic response to be dominated by the arm's flexibility rather than the foil assembly's internal deformation.

The only important property here is the mass of the foil, which is concentrated below the attachment point, and it's very big in magnitude compared to the arm mass. This is because, due to AC class rules', the baricenter of the foil assembly must be located at a specific oustide from the boat baricenter. This to guarantee the boat's ability to right itself after capsizing.

- **Concentrated mass:** $m_{\text{foil}} = 849\,\text{kg}$
- **Mass location:** [x, y, z] = [0, 0, $-225\,\text{mm}$] relative to the arm tip

Composite Material

Below the composite material properties estimations to build the FEM structural model, starting from the given properties.

- **Shear modulus ratio:** $E/G \approx 2.6$ (typical for CFRP laminates)
- **Timoshenko shear correction factor:** $\kappa = 0.85$ (for composite box beams)

6.2.2 Aerodynamic Grid and Mesh

The system operates in seawater with the following fluid properties:

- **Fluid:** Seawater
- **Density:** $\rho_f = 1{,}020\,\text{kg}\,\text{m}^{-3}$
- **Speed of sound in water:** $c_{\text{sound}} = 1{,}484\,\text{m}\,\text{s}^{-1}$ (at 2°C)
- **Mach number range:** Ma = 0.0067 to 0.0337 (corresponding to velocities 10–50 m/s) (19.4-97 knots)

The low Mach numbers confirm that compressibility effects are negligible, validating the use of incompressible aerodynamic models (DLM).

The aerogrid was constructed using the Doublet Lattice Method (DLM) with multiple sub-grids, one for each hydrofoil wing component:

- **Spanwise discretization:** $n_{\text{span}} = 10$ panels along the half-span
- **Chordwise discretization:** $n_{\text{chord}} = 5$ panels (computed as $n_{\text{chord}} = n_{\text{span}}/\text{AR}$)
- **Total panels per wing:** $10 \times 5 = 50$ panels

- **Merged aerogrid total:** 150 panels (combining all wings), 150 aerodynamic degrees of freedom (DOF)
- **DLM integration method:** Quartic (Rodden 1998 approach) for improved accuracy
- **Shear-axis offset:** 0.5 chord (mid-chord aerodynamic center)
- **Angle of attack:** $\alpha = 0°$

The aerogrid was precomputed and stored in the database for efficient reuse across multiple flutter analyses. The resulting aerodynamic matrices (Q_{jj}) represent the generalized aerodynamic forces at 30 discrete reduced-frequency values (k) and 45 Mach numbers spanning the operational range.

6.2.3 Structural Modes and Modal Properties

The structural model discretizes both the arm and the foil wings using Euler-Bernoulli beam elements:

- **Total elements:** 40 (distributed across all sub-structures)
- **Arm element count:** 30 elements (uniform length 140 mm per element)
- **Foil elements:** 10 elements (distributed among the boot, foil_dx, and foil_sx beams)
- **Element type:** 3-node Euler-Bernoulli beam elements
- **Total DOF (undamped system):** 70 structural DOF + 150 aerodynamic DOF = 220 total DOF in the coupled system

The element size is chosen to ensure adequate spatial resolution for capturing the first four structural modes while maintaining computational efficiency.

The undamped natural frequencies and mode shapes were computed from the finite-element structural stiffness and mass matrices:

Mode	Frequency (Hz)	ω (rad/s)	Axial %	Bending XY %	Bending XZ %	Torsion %
1	2.0921	13.1453	1.9	50.5	29.6	18.0
2	3.5610	22.3742	7.3	15.7	47.8	29.2
3	8.3697	52.5982	48.5	48.7	1.6	1.1
4	21.6237	135.8657	11.6	15.7	37.8	34.9

Table 6.3: First four structural modes of the `tns_multibody` system. Frequency values obtained from eigenvalue analysis of the coupled arm-foil finite-element model.

6.2.4 Flutter Analysis Methodology

The flutter analysis follows the coupled aero-structural approach outlined in Chapter 4:

1. **Structural mode reduction:** The system is reduced to a 4-mode basis spanning the frequency range of interest.
2. **Generalized aerodynamic forces:** The precomputed Q_{jj} matrices are interpolated at discrete reduced-frequency points (k) and Mach numbers (Ma).
3. **State-space formulation:** The coupled system is assembled into a state-space representation suitable for eigenvalue analysis.
4. **Eigenvalue tracking:** Eigenvalues are computed across the velocity range (10–50 m/s), tracking damping and frequency evolution.
5. **Flutter speed identification:** The flutter velocity is identified as the point where the imaginary part (damping) transitions from negative to positive.

Velocity Sweep Configuration

The flutter analysis was performed across a velocity range designed to capture both subcritical and supercritical flow regimes:

- **Low-speed range:** $V \in [10, 20]$ m/s with 50 points (fine resolution)
- **Flutter-critical range:** $V \in [20, 40]$ m/s with 40 points (enhanced resolution near flutter onset)
- **High-speed range:** $V \in [40, 50]$ m/s with 10 points (coarser resolution)
- **Total analysis points:** 100 velocity points

This multi-resolution approach ensures high accuracy near the flutter boundary while maintaining computational efficiency in the stable regions.

6.2.5 Results and Discussion

3D Structural Model Visualization

Figure 6.5 displays the three-dimensional geometry of the coupled arm-foil structure, showing:

- The tapered vertical arm extending downward from the clamped root
- The symmetric T-foil wings attached at the arm tip with positive dihedral
- The spatial distribution of mesh elements
- The overall form of the coupled system in the hydrodynamic environment

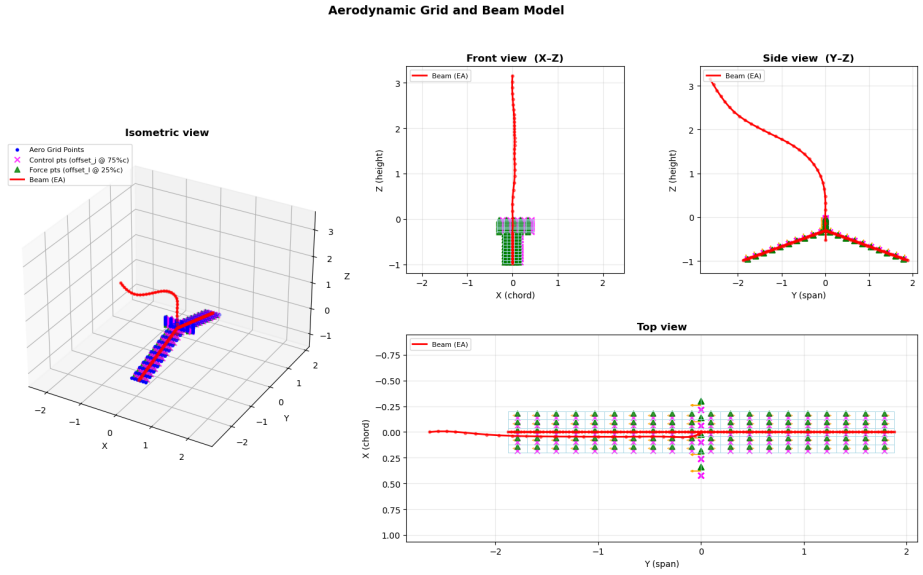


Figure 6.5: Three-dimensional views of the `tns_multibody` structure showing the vertical arm (tapering from root to tip) and the symmetric T-foil wings with 20° dihedral angle. The finite-element mesh is shown overlaid on the geometry.

Eigenvalue Trajectory Analysis

Figure 6.6 displays the eigenvalue trajectories (damping ratio and frequency) as functions of flow velocity:

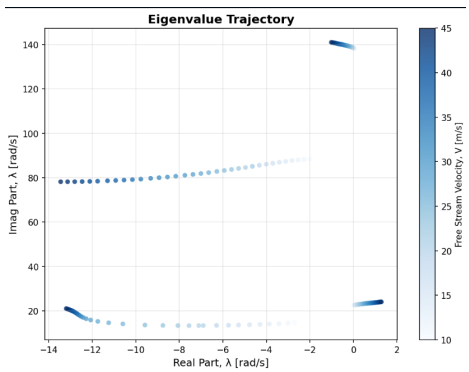


Figure 6.6: Eigenvalue trajectories showing the evolution of damping ratio (upper panel) and natural frequency (lower panel) versus flow velocity for the first four structural modes.

From the eigenvalue trajectory, the following observations are made:

- **Subcritical behavior ($\mathbf{V < V_{flutter}}$):** All modes exhibit negative damping, indicating stable oscillations that decay over time.
- **Flutter mechanism:** At least one mode transitions to positive damping (unstable oscillations) as velocity increases.
- **Frequency evolution:** The natural frequencies increase with velocity due to aerodynamic stiffening effects, a characteristic feature of fluid-structure interaction.
- **Modal interactions:** Potential mode coalescence (frequency crossings) may be observed, which can accelerate flutter onset.

VG-VF Flutter Diagram

Figure 6.7 presents the generalized VG-VF flutter diagram, a standard representation in aeroelasticity:

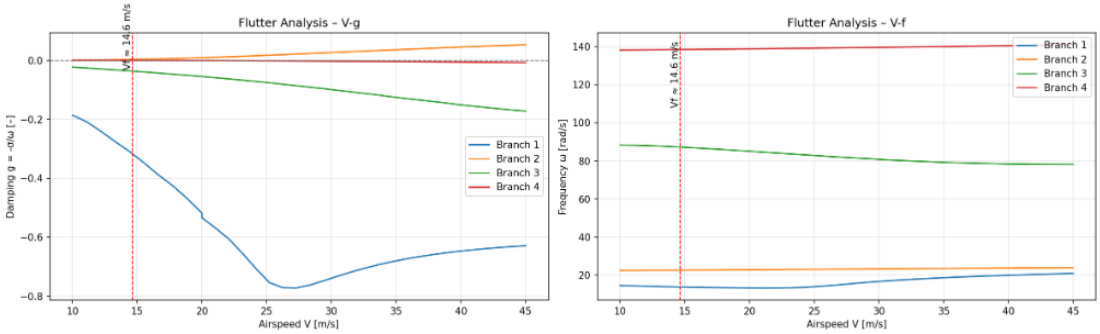


Figure 6.7: VG-VF flutter diagram for the `tns_multibody` system. Results are modified to preserve confidentiality, but the general trends and critical flutter velocity are clearly indicated.

The VG-VF diagram reveals:

1. **Stable region (subcritical):** For velocities below flutter, all modes are damped (negative damping ratio).
2. **Flutter boundary:** A critical velocity V_{flutter} is identified where the first mode's damping transitions to zero and then positive.
3. **Post-flutter behavior:** Beyond flutter, the damped mode exhibits diverging oscillations with increasing velocity, indicating an unstable aeroelastic phenomenon.
4. **Operating margin:** The distance from the nominal operating velocity to V_{flutter} defines the design safety margin, a critical parameter in hydrofoil safety.

6.2.6 Physical Interpretation

The multibody flutter behavior of the arm-foil system can be understood through the lens of coupled oscillator dynamics:

- The **vertical arm** acts as a flexible support with dominant bending compliance in the horizontal plane.
- The **T-foil assembly** provides both mass and aerodynamic forcing, coupling structural and fluid-dynamic effects.
- **Aerodynamic stiffening** at low velocities maintains stable oscillations through aerodynamic damping.
- At **critical velocities**, aerodynamic forces transition from damping to destabilizing, leading to flutter divergence.
- The **dihedral angle** of the foil wings couples plunge and pitch motions, introducing additional complexity into the flutter mechanism.

6.2.7 Summary

The `tns_multibody` case study demonstrates the application of the coupled flutter analysis framework to a realistic hydrofoil structure. The integration of:

- Accurate structural modeling with spatially-varying composite properties
- Precomputed aerodynamic matrices via the DLM
- Multibody assembly techniques
- Modal analysis and eigenvalue tracking

enables identification of critical flutter velocities and provides engineers with actionable insights for design optimization and operational safety. This case exemplifies the practical utility of the framework in addressing real-world aeroelastic challenges in marine applications.

Discussion, Conclusion and further work

7.1 Summary and Critical Assessment

This thesis introduced, implemented, and partially validated an open-source modular framework for the hydroelastic flutter analysis of composite beams, with particular reference to marine lifting surfaces such as hydrofoils. The work addresses a methodological gap that is more fundamental than a matter of design convenience: the lack of a validated, integrated, and accessible method to determine the wet vibrational modes of slender structures immersed in a fluid, across a range of flow velocities. **This problem sits at the core of hydroelastic analysis.**

When a slender structure operates in water, its dynamic behaviour is governed not by its dry natural frequencies, but by a coupled set of fluid-loaded modes whose frequencies, shapes, and damping characteristics evolve with flow speed. Predicting these wet modes correctly accounting for added mass, radiation damping, circulatory forces, and wake memory is the necessary prerequisite for any meaningful flutter or instability analysis. Yet the tools available in the literature either address the fluid and structural problems in isolation, rely on thin-surface approximations that are inadequate in water due to the high fluid-to-structure mass ratio, or are embedded in commercial platforms that are neither transparent nor extensible. The framework developed here fills this gap by combining a finite-element beam solver with a hybrid DLM–BEM aerodynamic operator and a p-k flutter solver with automatic mode tracking, into a single coherent workflow.
The result is a method capable of computing the full evolution of wet structural modes — frequencies, damping ratios, and eigenvectors as a function of flow speed, in both air and water environments, starting from realistic composite cross-section definitions.

Each module has been individually verified against established references: the structural beam solver was validated against classical modal benchmarks; the aeroelastic path was validated against the Goland wing flutter benchmark, yielding a critical speed of 165 m/s and a flutter frequency of 71 Hz, in excellent agreement with SHARPy and published literature values; and the hydroelastic path was verified on the NACA0003 benchmark, correctly reproducing the dry natural frequency to within 0.3 and capturing the expected frequency reduction upon fluid loading.

The application to the wing01 flexible hydrofoil confirmed the end-to-end integrity of the full workflow, and the tns multibody case study demonstrated the identification of a well-defined flutter boundary with physically consistent eigenvalue trajectories for a realistic multibody marine structure.

7.2 Limitations and Open Points

The framework, in its current state, should be regarded as a medium-to-low fidelity pre-design tool. Several limitations are acknowledged and will be addressed in future development. The hydroelastic validation pathway requires further attention. The fluid-at-rest frequency comparison on the NACA0003 case showed an 11.4 discrepancy with respect to the literature reference, attributable to differences in BEM discretisation, boundary condition treatment, and the frequency-dependence of the hydrodynamic added-mass matrices. A more systematic validation of the Capytaine integration — including sensitivity to mesh resolution, Green's function choice, and free-surface elevation is

needed before the hydroelastic path can be considered fully production-ready. This activity is ongoing and will be completed as part of a follow-on research fellowship, with the specific objective of establishing the method's reliability across a broader set of wet modal benchmarks at varying flow conditions. Additional open points include the absence of viscous corrections, as the entire framework relies on potential-flow assumptions, the omission of cavitation effects relevant to high-speed marine applications, and the current restriction to linear structural behaviour and small-amplitude oscillations. Extension to rational function approximation-based time-domain models would further broaden the applicable range of analyses.

7.3 Future Work and Dissemination

The development work described in this thesis is intended to continue beyond the degree submission. In the near term, the complete validation of the hydroelastic system will be finalised and consolidated into a **research paper** documenting the methodology, including additional case studies, extended sensitivity analyses, and a broader comparison against higher-fidelity numerical and experimental references.

The paper will present the framework as a validated method for wet modal analysis of slender structures in fluid, across the full speed range relevant for marine applications. Once the development and validation process is complete, the codebase will be reorganised, documented, and **made publicly available** through an open-source Github repository, so that the methodology can be freely adopted, scrutinised, and extended by the broader research and engineering community.

Contacts with external institutions interested in applying the framework to additional domains are already established, and these interactions will inform both the validation priorities and the future development roadmap. The tool's value ultimately rests with those who find in it a useful instrument — for research, for design, or as a foundation for further inquiry.

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